

Check_In

2025-09-25

```
# set working directory
setwd("/Users/jmelb/Projects/HDS-Foundations_of_Data_Science/Final_Project/")

# 1.a.i read in data
counts <- read.csv("data/counts.csv", row.names = 1, check.names = FALSE, stringsAsFactors = FALSE)
meta_data <- read.csv("data/meta_data.csv")
meta_data <- data.frame(meta_data)
```

For Gene 1: I selected the gene ENSG00000003402.20 (CFLAR) because it is implicated in apoptosis and immune response regulation. In cancer, elevated expression of CFLAR is correlated with resistance to therapy (Zhang et al., J Inflamm Res, 2025)

For Gene 2: I selected the gene ENSG00000000003.15 (TSPAN6) because it is involved in extracellular signalling to recruit immune cells the tumor microenvionrment (Molostvov et al., Cell Rep, 2023)

```
# 2.a.
# prep for figure generation
library(ggplot2)

# make data frame with both genes
gene_data <- data.frame(
  barcode = colnames(counts),
  gene1 = as.numeric(counts['ENSG00000003402.20', ]),
  gene2 = as.numeric(counts['ENSG00000000003.15', ]))

# pull in Hunter's Cooler code
coolor_code <- cool_palette <- c(
  "#E3E9EA",
  "#B8C4C2",
  "#7A9E9F",
  "#527C83",
  "#314D58",
  "#585858"
)

# set themes for subsequent plots
theme_set(
  theme_minimal(base_family = "Helvetica", base_size = 12) +
  theme(
    # backgrounds
    plot.background = element_rect(fill = coolor_code[1], color = NA),
    panel.background = element_rect(fill = coolor_code[1], color = NA),

    # grid lines
```

```

panel.grid.major = element_line(color = cooler_code[2], linewidth = 0.4),
panel.grid.minor = element_line(color = cooler_code[2], linewidth = 0.2, linetype = "dotted"),

# text and titles
plot.title = element_text(
  hjust = 0.5,
  face = "bold",
  size = 14,
  color = cooler_code[5]
),
axis.title = element_text(face = "bold", color = cooler_code[5]),
axis.text = element_text(color = cooler_code[5]),

)
)

```

```

# 3.b. Generate summary statistics
print(summary(gene_data$gene1))

```

```

##      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      664   3346   4799   5442   6720   21182

```

```

print(sd(gene_data$gene1))

```

```

## [1] 3040.001

```

```

print(summary(gene_data$gene2))

```

```

##      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##       69   1562   2700   3208   4144   40780

```

```

print(sd(gene_data$gene2))

```

```

## [1] 2510.336

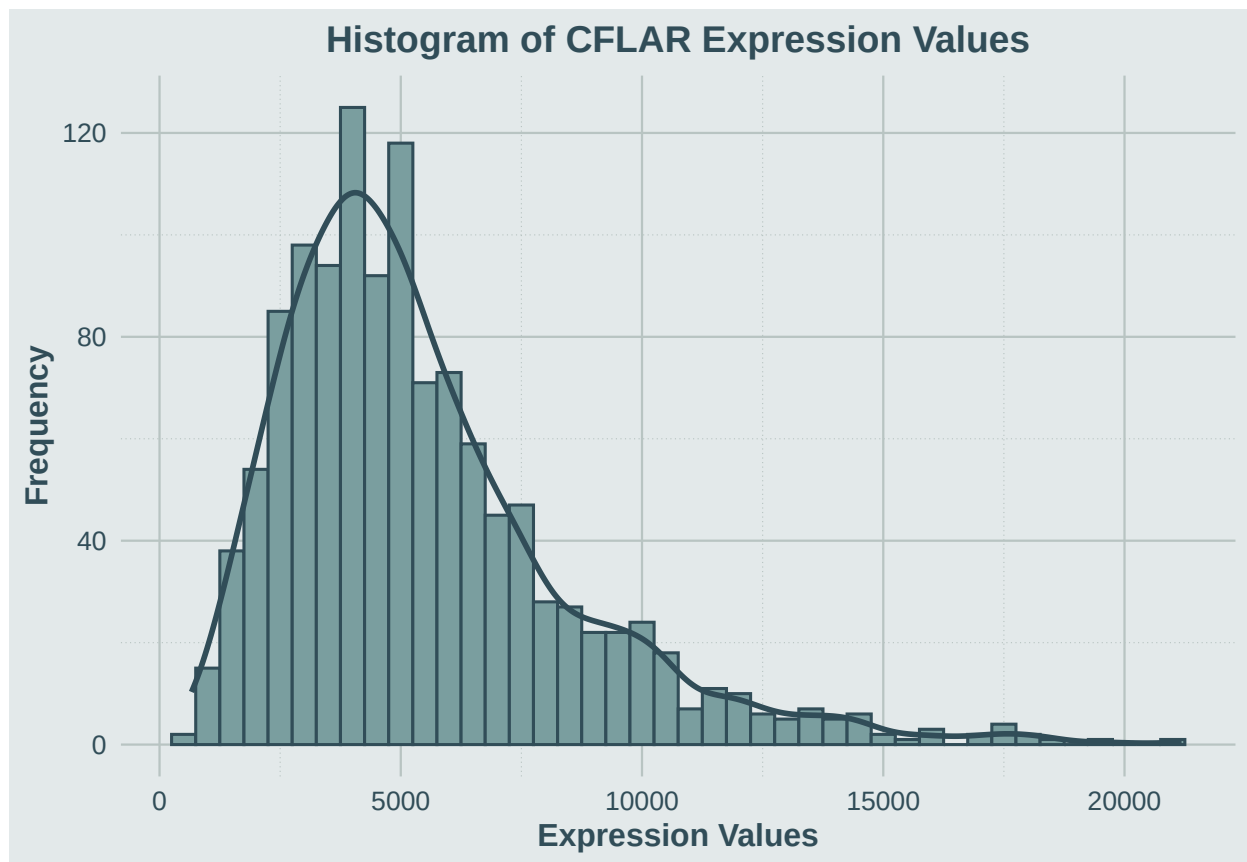
```

```

# 3. Visualization
# 3.a.- Histogram
p <- ggplot(gene_data, aes(x = gene1)) +
  geom_histogram(aes(y = after_stat(count)), binwidth = 500,
    fill = cooler_code[3], color = cooler_code[5]) +
  geom_density(aes(y = after_stat(count) * 500), color = cooler_code[5],
    linewidth = 1, alpha = .5) +
  labs(
    title = "Histogram of CFLAR Expression Values",
    x = "Expression Values",
    y = "Frequency"
  )

print(p)

```



```
png("/Users/jmelb/Projects/HDS-Foundations_of_Data_Science/Final_Project/figures_tables/hist.png", width=1000, height=500)
print(p)
dev.off()
```

```
## pdf
## 2
```

```
# 3.b. - Scatter plot
library(ggpubr)

# plot gene1 against gene2
p <- ggplot(gene_data, aes(x = gene1, y = gene2)) +
  # add scatter plot
  geom_point(
    fill = cooler_code[3],
    color = cooler_code[5],
    shape = 21,
    size = 4,
    stroke = .3,
    alpha = 0.65
  ) +
  # add regression line with cooler code colors
  geom_smooth(
    method = "lm",
```

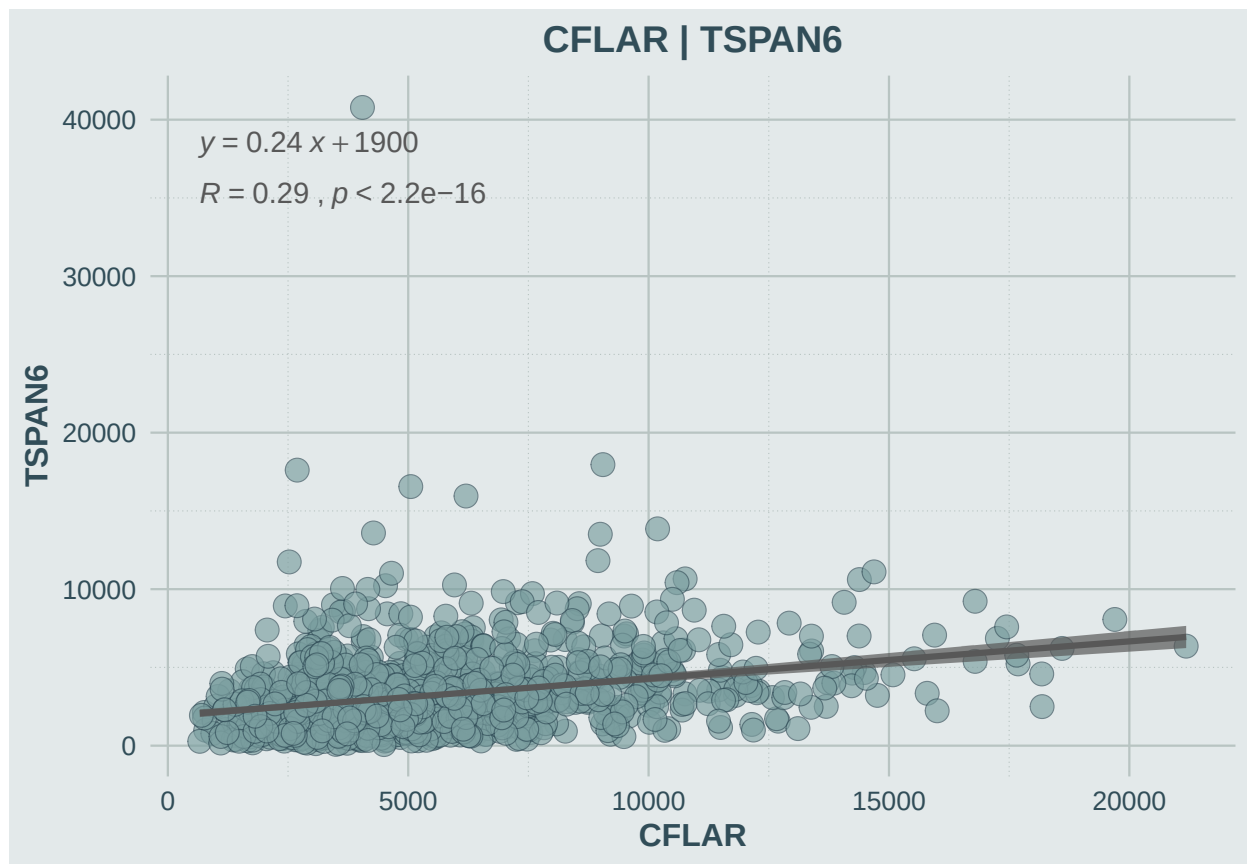
```

    se = TRUE, # add confidence interval
    color = coolor_code[6],
    fill = coolor_code[6],
    alpha = 0.7,
    linewidth = 1.1
  ) +
  labs(
    title = "CFLAR | TSPAN6",
    x = "CFLAR",
    y = "TSPAN6") +
  # add regression line equation
  stat_regline_equation(
    aes(label = paste(after_stat(eq.label), sep = "~~~")),
    label.x.npc = "left", # x position. - leftcorner
    label.y.npc = 0.95, # y position - upper corner
    color = coolor_code[6],
    size = 4
  ) +
  # add correlation coefficient and p value
  stat_cor(
    aes(label = paste(after_stat(r.label), after_stat(p.label), sep = "~``,`~")),
    label.x.npc = "left", # same x position
    label.y.npc = 0.87, # slightly below other y
    color = coolor_code[6],
    size = 4
  )
)

print(p)

```

```
## 'geom_smooth()' using formula = 'y ~ x'
```



```
png("/Users/jmelb/Projects/HDS-Foundations_of_Data_Science/Final_Project/figures_tables/scatter.png", w
print(p)
```

```
## 'geom_smooth()' using formula = 'y ~ x'
```

```
dev.off()
```

```
## pdf
## 2
```

```
# 3.c. - Violin plot
# define data frame for gene of interest
meta_col <- "tissue_type" # <-- change this to any column in meta (e.g., "definition")

# merge gene and meta data
gene_meta <- merge(gene_data[, c("barcode", "gene1")], meta_data[, c("barcode", meta_col)], by = "barcode")

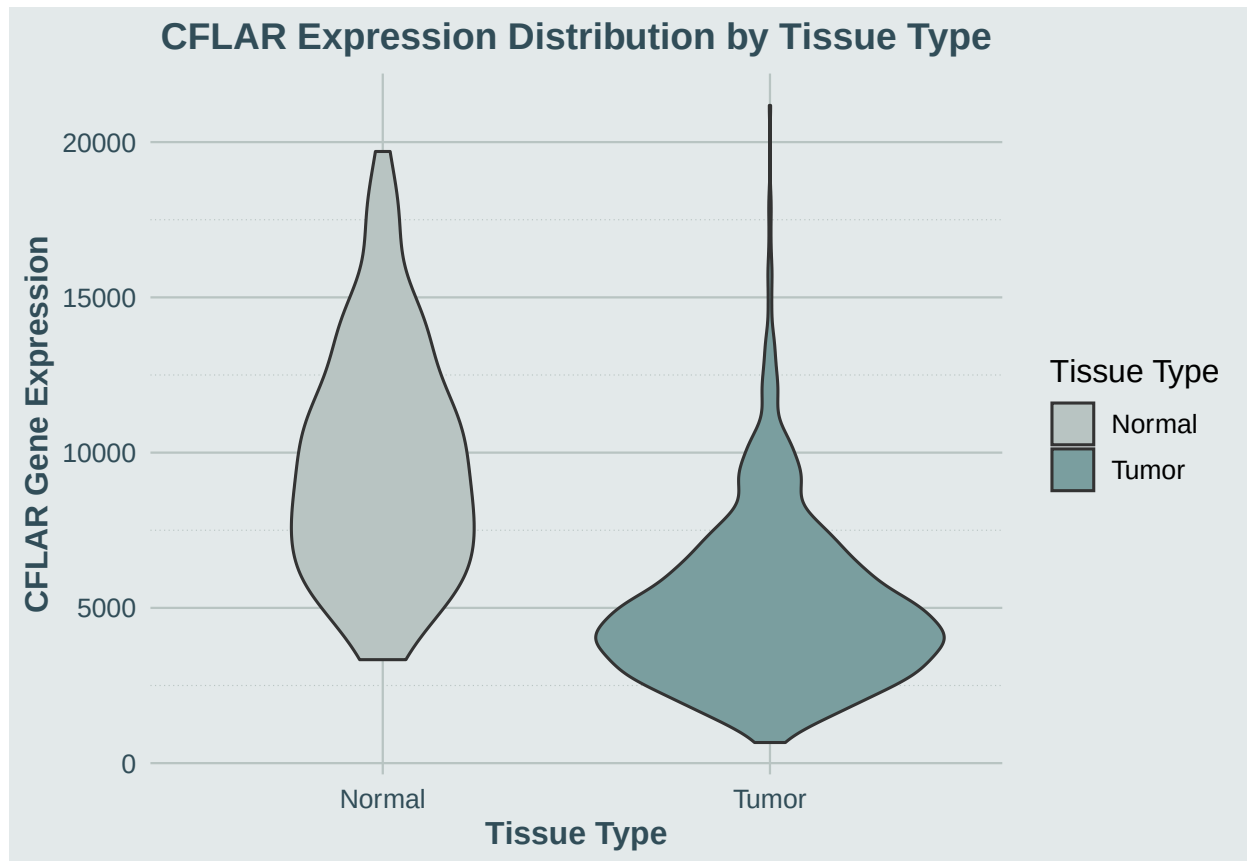
# make sure tissue_type is a factor not a continuous variable
gene_meta$tissue_type <- as.factor(gene_meta$tissue_type)

library(ggplot2)
# make violin plot
p <- ggplot(gene_meta, aes(x = tissue_type, y = gene1, fill = tissue_type)) +
  geom_violin() +
```

```

scale_fill_manual(values = cooler_code[2:5]) +
labs(
  x = "Tissue Type",
  y = "CFLAR Gene Expression",
  fill = "Tissue Type", # reformat fill legend
  title = "CFLAR Expression Distribution by Tissue Type"
)
print(p)

```



```

png("/Users/jmelb/Projects/HDS-Foundations_of_Data_Science/Final_Project/figures_tables/violin.png", width = 1000, height = 500)
print(p)
dev.off()

```

```

## pdf
## 2

```

Heatmap Analysis: Select 10 genes: Choose a set of 10 different genes from the count matrix for your heatmap. Generate a Heatmap: Use the ComplexHeatmap package in R to create a heatmap of the count data for the selected genes. Add an Annotation Bar: Include an annotation bar reflecting your chosen covariate for further context and interpretation of the data.

```

# implement Chayce's method for changing ensemble ID ==> Gene Names
library(biomaRt)

```

```

# find top ten most variable genes
gene_vars <- apply(counts, 1, var)
genes_to_plot <- names(sort(gene_vars, decreasing = TRUE)[1:10])

# subset genes
counts_subset <- counts[genes_to_plot, ]

# Remove version suffix from ensembl IDs
genes_to_plot <- sub("\\.\\d+$", "", genes_to_plot)
rownames(counts_subset) <- sub("\\.\\d+$", "", rownames(counts_subset))

# Connect to ensembl with mirror
ensembl_df <- useEnsembl(
  biomaRt = "genes",
  dataset = "hsapiens_gene_ensembl", # human genes
  mirror = "useast" # use mirror because original ensembl not available
)

# get ensembl gene names for genes_to_plot values
gene_names <- getBM(
  attributes = c("ensembl_gene_id", "external_gene_name"),
  filters = "ensembl_gene_id",
  values = genes_to_plot,
  mart = ensembl_df
)

# make rownames ensembl_gene_id
rownames(gene_names) <- gene_names$ensembl_gene_id

# remove ensembl_gene_id
gene_names$ensembl_gene_id <- NULL

# check
head(gene_names)

```

```

##              external_gene_name
## ENSG00000108821      COL1A1
## ENSG00000115414        FN1
## ENSG00000153002      CPB1
## ENSG00000164692      COL1A2
## ENSG00000168542      COL3A1
## ENSG00000198804      MT-CO1

```

```

#merge external names with counts_subset
subset_merged <- merge(gene_names, counts_subset, by = "row.names", all = TRUE)

# rinse and repeat
# make rownames external_gene_name
rownames(subset_merged) <- subset_merged$external_gene_name

```

```
# remove ensembl_gene_id
subset_merged$external_gene_name <- NULL
subset_merged$Row.names <- NULL
```

```
head(subset_merged)
```

```
##          TCGA-GM-A2DL-01A-11R-A18M-07 TCGA-AC-A2QI-01A-12R-A19W-07
## COL1A1          460584          848283
## FN1            583910          226903
## CPB1              10           1321
## COL1A2          379843          534876
## COL3A1          330889          419949
## MT-CO1          539156          408607
##          TCGA-A8-A06R-01A-11R-A00Z-07 TCGA-EW-A1PD-01A-11R-A144-07
## COL1A1          476371          700718
## FN1            677816          468828
## CPB1             32           10035
## COL1A2          437803          517547
## COL3A1          396071          506666
## MT-CO1          805821          436080
##          TCGA-A0-A12D-01A-11R-A115-07 TCGA-AR-A24N-01A-11R-A169-07
## COL1A1          404508          557915
## FN1            399899          627614
## CPB1             35             22
## COL1A2          324008          522520
## COL3A1          255682          421486
## MT-CO1          312220          307334
##          TCGA-AR-A24U-01A-11R-A169-07 TCGA-D8-A1JU-01A-11R-A13Q-07
## COL1A1          274604          1225089
## FN1            654020          428313
## CPB1            137           7500
## COL1A2          251577          1169868
## COL3A1          244193          1288139
## MT-CO1          1080765          321115
##          TCGA-A8-A0AD-01A-11R-A056-07 TCGA-E2-A3DX-01A-21R-A213-07
## COL1A1          633123          1078573
## FN1            453082          326573
## CPB1            1428           223
## COL1A2          596956          777717
## COL3A1          451812          876169
## MT-CO1          363438          389556
##          TCGA-BH-A18J-11A-31R-A12D-07 TCGA-A2-A0YL-01A-21R-A109-07
## COL1A1          20797           553656
## FN1            11719          238833
## CPB1             377           216
## COL1A2          29392          447256
## COL3A1          21272          427683
## MT-CO1          519386          460405
##          TCGA-C8-A12P-01A-11R-A115-07 TCGA-A0-A0JA-01A-11R-A056-07
## COL1A1          691677          833292
## FN1            489609          682823
## CPB1              7           1512
```


##	COL1A2	576979	659729
##	COL3A1	470638	593976
##	MT-CO1	582612	570263
##	TCGA-EW-A10V-01A-11R-A144-07	TCGA-5L-AATO-01A-12R-A41B-07	
##	COL1A1	195570	442529
##	FN1	103010	194163
##	CPB1	14	429
##	COL1A2	126579	292992
##	COL3A1	132573	263625
##	MT-CO1	741152	316058
##	TCGA-BH-A18J-01A-11R-A12D-07	TCGA-D8-A27V-01A-12R-A17B-07	
##	COL1A1	603383	556349
##	FN1	690033	134029
##	CPB1	162	10
##	COL1A2	477795	553245
##	COL3A1	310462	815335
##	MT-CO1	522415	806143
##	TCGA-A7-A26I-01B-06R-A220-07	TCGA-A8-A096-01A-11R-A00Z-07	
##	COL1A1	166430	1004036
##	FN1	382526	1103787
##	CPB1	0	113
##	COL1A2	130254	867991
##	COL3A1	129631	627200
##	MT-CO1	138578	432019
##	TCGA-AR-A256-01A-11R-A169-07	TCGA-A7-A426-01A-22R-A24H-07	
##	COL1A1	128539	580989
##	FN1	228035	76009
##	CPB1	16	50975
##	COL1A2	101679	321630
##	COL3A1	91268	355143
##	MT-CO1	542895	595460
##	TCGA-E2-A1BC-11A-32R-A12P-07	TCGA-E2-A1LH-11A-22R-A14D-07	
##	COL1A1	14534	2623
##	FN1	8448	3522
##	CPB1	1021	975
##	COL1A2	22084	4087
##	COL3A1	17685	4290
##	MT-CO1	761255	293680
##	TCGA-E2-A1LH-01A-11R-A14D-07	TCGA-E9-A1N8-01A-11R-A144-07	
##	COL1A1	83518	55058
##	FN1	47783	69034
##	CPB1	18	4
##	COL1A2	47052	46222
##	COL3A1	42717	51163
##	MT-CO1	1107811	127783
##	TCGA-BH-A0HK-11A-11R-A089-07	TCGA-EW-A2FR-01A-11R-A21T-07	
##	COL1A1	65051	1936562
##	FN1	53161	1878689
##	CPB1	17417	13
##	COL1A2	132972	1772686
##	COL3A1	56214	1688885
##	MT-CO1	408523	505454
##	TCGA-E2-A154-01A-11R-A115-07	TCGA-A2-A0T4-01A-31R-A084-07	
##	COL1A1	344930	554331

##	FN1	337717	193830
##	CPB1	8	228462
##	COL1A2	255725	404094
##	COL3A1	189223	356409
##	MT-CO1	339806	380036
##	TCGA-E9-A1NH-01A-11R-A14D-07	TCGA-AR-A2LO-01A-31R-A18M-07	
##	COL1A1	611637	305188
##	FN1	710056	101313
##	CPB1	22	10299
##	COL1A2	518295	208630
##	COL3A1	392729	220960
##	MT-CO1	341607	292377
##	TCGA-Z7-A8R6-01A-11R-A41B-07	TCGA-AC-A3YJ-01A-11R-A22U-07	
##	COL1A1	263585	51990
##	FN1	198540	26943
##	CPB1	1482	321624
##	COL1A2	144688	32806
##	COL3A1	113008	23100
##	MT-CO1	603938	270076
##	TCGA-A2-A04U-01A-11R-A115-07	TCGA-A7-A26I-01A-11R-A169-07	
##	COL1A1	185847	246135
##	FN1	93627	1051569
##	CPB1	10	19
##	COL1A2	108296	466069
##	COL3A1	81879	557045
##	MT-CO1	225840	50999
##	TCGA-A8-A09A-01A-11R-A00Z-07	TCGA-A0-A126-01A-11R-A10J-07	
##	COL1A1	329976	1250326
##	FN1	497717	197717
##	CPB1	1631	2531
##	COL1A2	302138	699041
##	COL3A1	324221	426780
##	MT-CO1	716197	902231
##	TCGA-EW-A1J6-01A-11R-A13Q-07	TCGA-LQ-A4E4-01A-11R-A266-07	
##	COL1A1	44245	222637
##	FN1	22041	48939
##	CPB1	7385	10
##	COL1A2	27551	118994
##	COL3A1	26583	104151
##	MT-CO1	3478847	820287
##	TCGA-E2-A1IL-01A-11R-A14D-07	TCGA-D8-A1XW-01A-11R-A14M-07	
##	COL1A1	474705	140585
##	FN1	229177	194188
##	CPB1	168	68
##	COL1A2	320996	136353
##	COL3A1	279886	132571
##	MT-CO1	217039	878123
##	TCGA-A8-A07C-01A-11R-A034-07	TCGA-EW-A2FV-01A-11R-A17B-07	
##	COL1A1	97773	178338
##	FN1	282896	197558
##	CPB1	2	17
##	COL1A2	105063	161772
##	COL3A1	124284	149989
##	MT-CO1	403518	847441

##	TCGA-E2-A1AZ-01A-11R-A12P-07	TCGA-AC-A2FM-11B-32R-A19W-07
##	COL1A1	211641 19038
##	FN1	150163 7274
##	CPB1	8 2099
##	COL1A2	144220 26188
##	COL3A1	154506 15229
##	MT-CO1	125911 264944
##	TCGA-BH-A1FJ-01A-11R-A13Q-07	TCGA-EW-A1J1-01A-11R-A13Q-07
##	COL1A1	159431 150449
##	FN1	163075 123630
##	CPB1	13 404056
##	COL1A2	148540 133148
##	COL3A1	119932 110654
##	MT-CO1	225241 489938
##	TCGA-A7-AODB-11A-33R-A089-07	TCGA-A7-AODB-01A-11R-A277-07
##	COL1A1	357041 174169
##	FN1	152127 327252
##	CPB1	323 9
##	COL1A2	321103 225334
##	COL3A1	429740 290072
##	MT-CO1	798100 62840
##	TCGA-A7-A56D-01A-11R-A27Q-07	TCGA-AC-A30D-01A-11R-A21T-07
##	COL1A1	621023 45444
##	FN1	130425 72333
##	CPB1	58 2260
##	COL1A2	318423 96801
##	COL3A1	125933 146202
##	MT-CO1	64983 24576
##	TCGA-E2-A109-01A-11R-A10J-07	TCGA-BH-A0DK-11A-13R-A089-07
##	COL1A1	498812 61297
##	FN1	321654 32538
##	CPB1	137802 128
##	COL1A2	320551 65786
##	COL3A1	286157 72230
##	MT-CO1	320554 552252
##	TCGA-E2-A150-01A-11R-A115-07	TCGA-BH-A0DK-01A-21R-A056-07
##	COL1A1	100380 391327
##	FN1	115401 278552
##	CPB1	25 540824
##	COL1A2	69003 269974
##	COL3A1	53690 258143
##	MT-CO1	707940 539063
##	TCGA-B6-A0WW-01A-11R-A109-07	TCGA-E2-A2P5-01A-11R-A19W-07
##	COL1A1	347334 438838
##	FN1	201412 175007
##	CPB1	66744 162
##	COL1A2	254215 289374
##	COL3A1	174607 182707
##	MT-CO1	385220 767867
##	TCGA-E9-A3Q9-01A-11R-A21T-07	TCGA-E2-A158-01A-11R-A12D-07
##	COL1A1	37408 24374
##	FN1	29606 40149
##	CPB1	1016 12
##	COL1A2	33890 23595

##	COL3A1	31971	22696
##	MT-CO1	848492	762055
##	TCGA-A1-AOSD-01A-11R-A115-07	TCGA-E9-A5FL-01A-11R-A27Q-07	
##	COL1A1	343797	303647
##	FN1	406770	47452
##	CPB1	1287010	5
##	COL1A2	284174	139496
##	COL3A1	232080	132926
##	MT-CO1	764375	1340545
##	TCGA-BH-A1FU-11A-23R-A14D-07	TCGA-AR-A1AT-01A-11R-A12P-07	
##	COL1A1	69468	185022
##	FN1	16152	1502474
##	CPB1	125	12
##	COL1A2	71417	146120
##	COL3A1	61698	158010
##	MT-CO1	314524	199842
##	TCGA-A1-AOSE-01A-11R-A084-07	TCGA-AN-A0XN-01A-21R-A109-07	
##	COL1A1	735106	769442
##	FN1	313398	2465891
##	CPB1	254	8
##	COL1A2	573649	650553
##	COL3A1	596155	524279
##	MT-CO1	527240	281675
##	TCGA-A2-AOYG-01A-21R-A109-07	TCGA-E9-A1RF-01A-11R-A157-07	
##	COL1A1	167865	585681
##	FN1	285359	596473
##	CPB1	24202	11515
##	COL1A2	140876	441686
##	COL3A1	115061	328158
##	MT-CO1	513666	907993
##	TCGA-B6-AORQ-01A-11R-A115-07	TCGA-A7-A6VY-01A-12R-A33J-07	
##	COL1A1	74662	564923
##	FN1	19749	347877
##	CPB1	1547	5
##	COL1A2	49247	375432
##	COL3A1	62349	286934
##	MT-CO1	656624	219266
##	TCGA-A7-A13D-01B-04R-A277-07	TCGA-BH-A203-11A-42R-A169-07	
##	COL1A1	56083	8091
##	FN1	297180	15015
##	CPB1	85	244
##	COL1A2	68924	18535
##	COL3A1	69019	17562
##	MT-CO1	76637	881427
##	TCGA-GM-A2DD-01A-11R-A180-07	TCGA-E9-A2JT-01A-22R-A18M-07	
##	COL1A1	483800	51286
##	FN1	120282	21637
##	CPB1	33	4
##	COL1A2	248560	41869
##	COL3A1	220078	38489
##	MT-CO1	1174623	241449
##	TCGA-EW-A10W-01A-21R-A144-07	TCGA-AN-A03X-01A-21R-A00Z-07	
##	COL1A1	241903	1316898
##	FN1	294044	657036

##	CPB1	1	762
##	COL1A2	263458	806236
##	COL3A1	162733	639257
##	MT-CO1	212190	461253
##	TCGA-EW-A1PB-01A-11R-A144-07	TCGA-AC-A2FO-01A-11R-A180-07	
##	COL1A1	48852	269273
##	FN1	238282	96702
##	CPB1	59	169
##	COL1A2	33170	257869
##	COL3A1	29237	272784
##	MT-CO1	95290	410206
##	TCGA-BH-A18M-01A-11R-A12D-07	TCGA-A8-A090-01A-11R-A00Z-07	
##	COL1A1	712610	299684
##	FN1	562516	329276
##	CPB1	25012	4
##	COL1A2	590702	240507
##	COL3A1	524136	198690
##	MT-CO1	320733	275250
##	TCGA-AC-A2FM-01A-11R-A19W-07	TCGA-BH-A1FJ-11B-42R-A13Q-07	
##	COL1A1	113560	74637
##	FN1	54851	68842
##	CPB1	115	40556
##	COL1A2	78320	177904
##	COL3A1	101885	144504
##	MT-CO1	314607	831517
##	TCGA-A8-A099-01A-11R-A00Z-07	TCGA-A0-A12B-01A-11R-A10J-07	
##	COL1A1	321060	39440
##	FN1	222617	29628
##	CPB1	19355	42
##	COL1A2	259967	34495
##	COL3A1	209409	32528
##	MT-CO1	276575	147822
##	TCGA-E2-A15F-01A-11R-A115-07	TCGA-A0-A0JF-01A-11R-A056-07	
##	COL1A1	129539	740918
##	FN1	133294	311983
##	CPB1	535957	94
##	COL1A2	96480	696729
##	COL3A1	87152	824289
##	MT-CO1	435828	351740
##	TCGA-E2-A576-01A-11R-A310-07	TCGA-AN-A0AR-01A-11R-A00Z-07	
##	COL1A1	147612	99299
##	FN1	32065	82585
##	CPB1	193	327
##	COL1A2	91286	113571
##	COL3A1	56920	44412
##	MT-CO1	298848	451053
##	TCGA-AC-A30D-01B-06R-A220-07	TCGA-LL-A5YN-01A-11R-A28M-07	
##	COL1A1	31849	185054
##	FN1	26310	51557
##	CPB1	151	1513
##	COL1A2	20113	103420
##	COL3A1	26233	62028
##	MT-CO1	283062	214169
##	TCGA-E9-A1R0-01A-22R-A16F-07	TCGA-AN-A03Y-01A-21R-A00Z-07	

## COL1A1	465385	207618
## FN1	220839	128215
## CPB1	61	234
## COL1A2	362773	147724
## COL3A1	315576	112577
## MT-CO1	630278	1036009
## TCGA-A7-A4SC-01A-12R-A266-07	TCGA-OL-A5D6-01A-21R-A27Q-07	
## COL1A1	497724	651523
## FN1	110129	256715
## CPB1	118	52
## COL1A2	264221	352053
## COL3A1	274708	305336
## MT-CO1	417578	641391
## TCGA-EW-A1IZ-01A-11R-A13Q-07	TCGA-BH-A0EB-01A-11R-A034-07	
## COL1A1	44827	504010
## FN1	43655	418783
## CPB1	60	1673
## COL1A2	34464	484860
## COL3A1	37708	427896
## MT-CO1	455372	614270
## TCGA-C8-A3M8-01A-11R-A213-07	TCGA-A0-A129-01A-21R-A10J-07	
## COL1A1	627274	185606
## FN1	382201	141095
## CPB1	152878	3
## COL1A2	341146	104017
## COL3A1	287093	88002
## MT-CO1	1298295	597235
## TCGA-D8-A27P-01A-11R-A16F-07	TCGA-E2-A152-01A-11R-A12D-07	
## COL1A1	313320	717016
## FN1	229325	845840
## CPB1	13	5
## COL1A2	243324	596759
## COL3A1	187552	523175
## MT-CO1	274324	470502
## TCGA-BH-A1FU-01A-11R-A14D-07	TCGA-E9-A1NI-01A-11R-A14D-07	
## COL1A1	394783	418771
## FN1	551900	344747
## CPB1	6	6869
## COL1A2	336814	371188
## COL3A1	352193	305474
## MT-CO1	438578	449741
## TCGA-D8-A27M-01A-11R-A16F-07	TCGA-A2-A0YF-01A-21R-A109-07	
## COL1A1	495272	66755
## FN1	709976	138590
## CPB1	170	3646
## COL1A2	431532	62174
## COL3A1	375903	51481
## MT-CO1	372131	754175
## TCGA-E9-A1RG-01A-11R-A14D-07	TCGA-B6-A0X4-01A-11R-A109-07	
## COL1A1	389907	28209
## FN1	175673	69005
## CPB1	108	2749714
## COL1A2	242514	20880
## COL3A1	172553	10775

##	MT-C01	456489	399498
##	TCGA-A0-A0J4-01A-11R-A034-07	TCGA-A7-A425-01A-11R-A24H-07	
##	COL1A1	232383	520995
##	FN1	443659	123752
##	CPB1	3	14
##	COL1A2	173938	237403
##	COL3A1	169518	204455
##	MT-C01	637478	428274
##	TCGA-A8-A08X-01A-21R-A00Z-07	TCGA-OL-A66N-01A-12R-A310-07	
##	COL1A1	507303	872634
##	FN1	370233	68187
##	CPB1	6	6107
##	COL1A2	338191	487870
##	COL3A1	344534	351175
##	MT-C01	441980	131139
##	TCGA-B6-A1KF-01A-11R-A13Q-07	TCGA-AC-A2FE-01A-11R-A19W-07	
##	COL1A1	123693	753154
##	FN1	95994	246511
##	CPB1	5	950
##	COL1A2	115930	467835
##	COL3A1	67233	486572
##	MT-C01	565783	532513
##	TCGA-A2-AOCM-01A-31R-A034-07	TCGA-GM-A5PX-01A-12R-A28M-07	
##	COL1A1	260815	455839
##	FN1	172277	161500
##	CPB1	40	199
##	COL1A2	194790	405963
##	COL3A1	163734	331675
##	MT-C01	262915	432439
##	TCGA-AR-A1AR-01A-31R-A137-07	TCGA-A0-A0J6-01A-11R-A034-07	
##	COL1A1	405926	101818
##	FN1	586735	215206
##	CPB1	10	3
##	COL1A2	335767	70078
##	COL3A1	383468	70272
##	MT-C01	570704	510980
##	TCGA-E2-A156-01A-11R-A12D-07	TCGA-D8-A73W-01A-22R-A352-07	
##	COL1A1	19135	311365
##	FN1	57741	334070
##	CPB1	246	132922
##	COL1A2	16698	201026
##	COL3A1	10886	177159
##	MT-C01	135118	388951
##	TCGA-E2-A107-01A-11R-A10J-07	TCGA-E2-A158-11A-22R-A12D-07	
##	COL1A1	180005	66753
##	FN1	36703	86705
##	CPB1	1137	432
##	COL1A2	90198	111197
##	COL3A1	82757	102706
##	MT-C01	798785	588556
##	TCGA-AC-A23G-01A-11R-A213-07	TCGA-A2-A1G4-01A-11R-A13Q-07	
##	COL1A1	1114698	64049
##	FN1	379598	123173
##	CPB1	7	121

##	COL1A2	656222	53897
##	COL3A1	569264	49475
##	MT-CO1	181150	809914
##	TCGA-A0-A1K0-01A-31R-A13Q-07	TCGA-A2-A3XS-01A-11R-A22U-07	
##	COL1A1	66943	265450
##	FN1	34715	113494
##	CPB1	2760	2
##	COL1A2	68105	163791
##	COL3A1	79952	158861
##	MT-CO1	455001	684217
##	TCGA-B6-AORH-01A-21R-A115-07	TCGA-D8-A1JH-01A-11R-A13Q-07	
##	COL1A1	738276	1002149
##	FN1	481944	618746
##	CPB1	13	478479
##	COL1A2	469298	799485
##	COL3A1	266732	707097
##	MT-CO1	633136	562547
##	TCGA-A0-A03P-01A-11R-A00Z-07	TCGA-OL-A5DA-01A-11R-A27Q-07	
##	COL1A1	161292	288130
##	FN1	180503	129893
##	CPB1	16806	304
##	COL1A2	132562	158429
##	COL3A1	95412	119691
##	MT-CO1	1357363	428025
##	TCGA-A8-A07B-01A-11R-A00Z-07	TCGA-A2-A25A-01A-12R-A16F-07	
##	COL1A1	150241	589146
##	FN1	143475	223507
##	CPB1	11	228
##	COL1A2	150078	586254
##	COL3A1	126660	726788
##	MT-CO1	348880	857939
##	TCGA-A2-A04R-01A-41R-A109-07	TCGA-A2-A04T-01A-21R-A034-07	
##	COL1A1	345765	56325
##	FN1	398400	74581
##	CPB1	2954651	5
##	COL1A2	261457	43587
##	COL3A1	208108	48442
##	MT-CO1	417778	753627
##	TCGA-AR-A1AX-01A-11R-A12P-07	TCGA-C8-A26V-01A-11R-A16F-07	
##	COL1A1	354491	437164
##	FN1	291716	468145
##	CPB1	329	252
##	COL1A2	277388	380432
##	COL3A1	266203	318378
##	MT-CO1	498572	775164
##	TCGA-BH-A18R-11A-42R-A12D-07	TCGA-AN-A0XR-01A-11R-A109-07	
##	COL1A1	34098	306982
##	FN1	23544	164119
##	CPB1	92	40
##	COL1A2	65286	193359
##	COL3A1	37994	112693
##	MT-CO1	814485	352320
##	TCGA-BH-A0HQ-01A-11R-A034-07	TCGA-LL-A5YM-01A-11R-A28M-07	
##	COL1A1	677813	181503

##	FN1	315730	46856
##	CPB1	182	1870722
##	COL1A2	448043	94916
##	COL3A1	392710	55851
##	MT-CO1	462608	278350
##	TCGA-D8-A1X5-01A-11R-A14D-07	TCGA-A7-AODB-01A-11R-A00Z-07	
##	COL1A1	418705	463271
##	FN1	545169	269554
##	CPB1	962859	486
##	COL1A2	330046	331577
##	COL3A1	258650	298847
##	MT-CO1	445613	946335
##	TCGA-EW-A1P6-01A-11R-A144-07	TCGA-A7-AODB-01C-02R-A277-07	
##	COL1A1	211188	84323
##	FN1	222468	160399
##	CPB1	56	11
##	COL1A2	165121	104375
##	COL3A1	118290	87723
##	MT-CO1	535296	123147
##	TCGA-LL-A73Y-01A-11R-A33J-07	TCGA-E2-A1L8-01A-11R-A13Q-07	
##	COL1A1	496222	520417
##	FN1	462030	346786
##	CPB1	11	88
##	COL1A2	364614	383558
##	COL3A1	319878	272758
##	MT-CO1	329489	840122
##	TCGA-AR-A1AM-01A-41R-A22K-07	TCGA-E2-A1LG-01A-21R-A14M-07	
##	COL1A1	449742	15250
##	FN1	121640	37140
##	CPB1	17026	171
##	COL1A2	306886	59666
##	COL3A1	330808	12307
##	MT-CO1	333808	1486024
##	TCGA-E2-A14W-01A-11R-A12D-07	TCGA-C8-A27B-01A-11R-A169-07	
##	COL1A1	143073	58244
##	FN1	117182	130999
##	CPB1	39	3
##	COL1A2	105168	60888
##	COL3A1	72813	77607
##	MT-CO1	553458	728308
##	TCGA-EW-A1P1-01A-31R-A14D-07	TCGA-AN-AOFN-01A-11R-A034-07	
##	COL1A1	416405	1651109
##	FN1	257999	1058599
##	CPB1	14	364
##	COL1A2	292273	1514898
##	COL3A1	277975	1297847
##	MT-CO1	415647	579612
##	TCGA-BH-AOAW-01A-11R-A056-07	TCGA-AR-AOTX-01A-11R-A084-07	
##	COL1A1	223537	491774
##	FN1	181795	324966
##	CPB1	81	101
##	COL1A2	155287	331828
##	COL3A1	158581	306922
##	MT-CO1	410319	525233

##	TCGA-E9-A3QA-01A-61R-A22K-07	TCGA-A2-A0SU-01A-11R-A084-07
##	COL1A1	157973 318148
##	FN1	162527 145997
##	CPB1	1 88889
##	COL1A2	87255 232165
##	COL3A1	95494 227684
##	MT-CO1	390350 383342
##	TCGA-GM-A2DA-01A-11R-A18M-07	TCGA-OL-A5D7-01A-11R-A27Q-07
##	COL1A1	260568 64209
##	FN1	65293 37758
##	CPB1	12 0
##	COL1A2	139988 30686
##	COL3A1	128423 28753
##	MT-CO1	993212 317682
##	TCGA-E9-A1RF-11A-32R-A157-07	TCGA-BH-A0EE-01A-11R-A034-07
##	COL1A1	28095 31624
##	FN1	87609 13517
##	CPB1	137 274
##	COL1A2	78022 24602
##	COL3A1	64280 24546
##	MT-CO1	1135789 1443026
##	TCGA-BH-A0H6-01A-21R-A056-07	TCGA-OL-A66L-01A-12R-A310-07
##	COL1A1	722441 519200
##	FN1	438220 56977
##	CPB1	1692 1285
##	COL1A2	664758 281050
##	COL3A1	433832 190316
##	MT-CO1	441924 129517
##	TCGA-A7-A13D-01A-13R-A12P-07	TCGA-A7-A13D-01A-13R-A277-07
##	COL1A1	332943 141525
##	FN1	881953 864318
##	CPB1	7 5
##	COL1A2	262031 184558
##	COL3A1	200111 176147
##	MT-CO1	278125 34712
##	TCGA-E9-A5FK-01A-11R-A27Q-07	TCGA-D8-A1JL-01A-11R-A13Q-07
##	COL1A1	199005 523923
##	FN1	45389 377806
##	CPB1	3392 24
##	COL1A2	131018 384348
##	COL3A1	107356 404388
##	MT-CO1	407064 295117
##	TCGA-BH-A203-01A-12R-A169-07	TCGA-EW-A1PE-01A-11R-A144-07
##	COL1A1	248383 434449
##	FN1	384427 436829
##	CPB1	6 3903
##	COL1A2	197383 341832
##	COL3A1	184964 256769
##	MT-CO1	569035 297624
##	TCGA-AR-A0TP-01A-11R-A084-07	TCGA-EW-A6SA-01A-21R-A32P-07
##	COL1A1	61259 37741
##	FN1	208006 79539
##	CPB1	4 25645
##	COL1A2	49942 33090

##	COL3A1	58264	35138
##	MT-CO1	752133	583351
##	TCGA-BH-A18M-11A-33R-A12D-07	TCGA-BH-A18R-01A-11R-A12D-07	
##	COL1A1	205745	231527
##	FN1	97768	200435
##	CPB1	531	71587
##	COL1A2	196922	197921
##	COL3A1	214539	176995
##	MT-CO1	367448	645253
##	TCGA-BH-AOBP-01A-11R-A115-07	TCGA-E9-A1ND-01A-11R-A144-07	
##	COL1A1	377538	225148
##	FN1	259395	536012
##	CPB1	1002	268
##	COL1A2	240222	151208
##	COL3A1	221744	145988
##	MT-CO1	521807	378017
##	TCGA-E9-A1ND-11A-43R-A144-07	TCGA-E2-A1IH-01A-11R-A13Q-07	
##	COL1A1	303846	826553
##	FN1	57077	559076
##	CPB1	205	224183
##	COL1A2	251303	693462
##	COL3A1	233246	514927
##	MT-CO1	581376	667472
##	TCGA-S3-A6ZH-01A-22R-A32P-07	TCGA-BH-AOC1-01B-11R-A12D-07	
##	COL1A1	116193	752689
##	FN1	63356	661272
##	CPB1	1071097	395
##	COL1A2	85504	523237
##	COL3A1	89837	450621
##	MT-CO1	619514	293560
##	TCGA-A2-A4RX-01A-11R-A266-07	TCGA-AC-A5EH-01A-11R-A28M-07	
##	COL1A1	519625	801325
##	FN1	519650	644122
##	CPB1	47	5
##	COL1A2	349061	566648
##	COL3A1	268124	427508
##	MT-CO1	127121	195952
##	TCGA-AR-AOTR-01A-11R-A084-07	TCGA-E2-A106-01A-11R-A10J-07	
##	COL1A1	19971	142183
##	FN1	12354	65733
##	CPB1	1438077	895491
##	COL1A2	19998	86431
##	COL3A1	28319	79604
##	MT-CO1	140417	532652
##	TCGA-AQ-A1H2-01A-11R-A13Q-07	TCGA-A2-AOCX-01A-21R-A00Z-07	
##	COL1A1	191135	498277
##	FN1	161179	599529
##	CPB1	122021	9
##	COL1A2	151766	384565
##	COL3A1	97612	319182
##	MT-CO1	431162	1038938
##	TCGA-AR-AOTZ-01A-12R-A084-07	TCGA-A2-A25F-01A-11R-A169-07	
##	COL1A1	562826	473864
##	FN1	482616	536464

##	CPB1	80	16
##	COL1A2	373514	423984
##	COL3A1	361933	536315
##	MT-CO1	448076	498842
##	TCGA-E2-A15D-01A-11R-A115-07	TCGA-E2-A14X-01A-11R-A115-07	
##	COL1A1	386695	114452
##	FN1	219665	122946
##	CPB1	18	14
##	COL1A2	261010	109247
##	COL3A1	227191	110428
##	MT-CO1	434492	706482
##	TCGA-D8-A1Y1-01A-21R-A14M-07	TCGA-A8-A092-01A-11R-A00Z-07	
##	COL1A1	1351199	387911
##	FN1	1210243	291954
##	CPB1	18	3723
##	COL1A2	1160713	298239
##	COL3A1	793488	275190
##	MT-CO1	539893	452046
##	TCGA-E2-A573-01A-11R-A29R-07	TCGA-EW-A1IY-01A-11R-A13Q-07	
##	COL1A1	119127	367584
##	FN1	85432	487065
##	CPB1	101	8
##	COL1A2	79842	431119
##	COL3A1	95263	440701
##	MT-CO1	319048	822942
##	TCGA-C8-A133-01A-32R-A12D-07	TCGA-BH-A1EV-11A-24R-A137-07	
##	COL1A1	9652	78142
##	FN1	13627	39688
##	CPB1	3066263	283
##	COL1A2	8802	166573
##	COL3A1	6905	101055
##	MT-CO1	282059	1091977
##	TCGA-B6-A2IU-01A-32R-A18M-07	TCGA-D8-A147-01A-11R-A115-07	
##	COL1A1	93949	123592
##	FN1	38861	97756
##	CPB1	63	5
##	COL1A2	64926	108127
##	COL3A1	52629	126380
##	MT-CO1	344071	241332
##	TCGA-AC-A3EH-01A-22R-A22K-07	TCGA-A8-A09Z-01A-11R-A00Z-07	
##	COL1A1	927007	103299
##	FN1	373758	63067
##	CPB1	359	6
##	COL1A2	501544	64253
##	COL3A1	379635	70408
##	MT-CO1	871900	529306
##	TCGA-BH-A1FR-01A-11R-A13Q-07	TCGA-E2-A10B-01A-11R-A10J-07	
##	COL1A1	87477	581252
##	FN1	57814	380429
##	CPB1	774612	16977
##	COL1A2	60963	439164
##	COL3A1	46782	404689
##	MT-CO1	517362	493125
##	TCGA-E2-A2P6-01A-11R-A19W-07	TCGA-E9-A227-01A-11R-A157-07	

## COL1A1	188977	458342
## FN1	65754	246908
## CPB1	24	280
## COL1A2	116239	349778
## COL3A1	86972	371250
## MT-CO1	1506289	980071
## TCGA-C8-A132-01A-31R-A115-07	TCGA-AQ-A1H3-01A-31R-A13Q-07	
## COL1A1	618397	465956
## FN1	708155	491925
## CPB1	6222	274
## COL1A2	450143	363439
## COL3A1	448310	312709
## MT-CO1	319032	492275
## TCGA-A2-AOYE-01A-11R-A109-07	TCGA-A8-A08L-01A-11R-A00Z-07	
## COL1A1	244532	341816
## FN1	253260	243130
## CPB1	172	7872
## COL1A2	176689	237600
## COL3A1	152497	232869
## MT-CO1	375279	399800
## TCGA-S3-AA17-01A-11R-A41B-07	TCGA-BH-A0BC-01A-22R-A084-07	
## COL1A1	129751	592392
## FN1	107479	259795
## CPB1	18	1496
## COL1A2	78690	421720
## COL3A1	72910	362016
## MT-CO1	728830	702479
## TCGA-AR-A2LK-01A-11R-A180-07	TCGA-A0-A03R-01A-21R-A034-07	
## COL1A1	330686	45111
## FN1	431204	26649
## CPB1	924	2469
## COL1A2	322267	22290
## COL3A1	288163	21605
## MT-CO1	410582	1136269
## TCGA-BH-A1ET-01A-11R-A137-07	TCGA-OL-A6V0-01A-12R-A33J-07	
## COL1A1	479884	84956
## FN1	388380	58313
## CPB1	128251	4
## COL1A2	528442	53914
## COL3A1	539139	61754
## MT-CO1	804491	221563
## TCGA-AR-A1AU-01A-11R-A12P-07	TCGA-A0-A1KP-01A-11R-A13Q-07	
## COL1A1	377746	334905
## FN1	345737	179393
## CPB1	3210	2
## COL1A2	302206	214790
## COL3A1	237628	169266
## MT-CO1	512056	1323552
## TCGA-A8-A08R-01A-11R-A034-07	TCGA-BH-A0E1-11A-13R-A089-07	
## COL1A1	240608	7341
## FN1	248045	3829
## CPB1	3	1006
## COL1A2	203909	12838
## COL3A1	204159	9937

##	MT-C01	436259	255315
##	TCGA-AR-AOTV-01A-21R-A084-07	TCGA-E9-A1RD-01A-11R-A157-07	
##	COL1A1	55175	514240
##	FN1	69157	381181
##	CPB1	53787	11231
##	COL1A2	40324	409301
##	COL3A1	44180	325577
##	MT-C01	563902	549719
##	TCGA-A8-A09G-01A-21R-A00Z-07	TCGA-AR-A1AI-01A-11R-A12P-07	
##	COL1A1	246980	55064
##	FN1	203183	259836
##	CPB1	6	0
##	COL1A2	177184	43628
##	COL3A1	171344	40778
##	MT-C01	756469	208796
##	TCGA-BH-A1FC-11A-32R-A13Q-07	TCGA-BH-A1FC-01A-11R-A13Q-07	
##	COL1A1	34298	143594
##	FN1	23834	83223
##	CPB1	623	0
##	COL1A2	44668	110469
##	COL3A1	33570	108857
##	MT-C01	318344	218901
##	TCGA-C8-A12N-01A-11R-A115-07	TCGA-B6-A40C-01A-11R-A239-07	
##	COL1A1	1050845	925148
##	FN1	938621	412505
##	CPB1	4091	9
##	COL1A2	783546	508826
##	COL3A1	583890	399683
##	MT-C01	376877	1196309
##	TCGA-AC-A3TN-01A-11R-A22K-07	TCGA-A2-A0CP-01A-11R-A034-07	
##	COL1A1	560804	800857
##	FN1	107005	280638
##	CPB1	51	1797
##	COL1A2	283686	383946
##	COL3A1	174286	289884
##	MT-C01	497534	421230
##	TCGA-E2-A140-01A-31R-A115-07	TCGA-AQ-A04J-01A-02R-A034-07	
##	COL1A1	387890	175129
##	FN1	128069	252843
##	CPB1	366	36
##	COL1A2	197251	185364
##	COL3A1	129931	147478
##	MT-C01	745097	1500268
##	TCGA-S3-A6ZF-01A-32R-A32P-07	TCGA-A2-A0EU-01A-22R-A056-07	
##	COL1A1	319628	297849
##	FN1	299625	199147
##	CPB1	558	2220
##	COL1A2	255274	297224
##	COL3A1	234973	263249
##	MT-C01	397793	396829
##	TCGA-C8-A120-01A-11R-A115-07	TCGA-A2-A0CW-01A-21R-A115-07	
##	COL1A1	841649	175886
##	FN1	917168	133716
##	CPB1	7	185

##	COL1A2	670822	124688
##	COL3A1	601580	110760
##	MT-CO1	421208	375734
##	TCGA-E9-A1RA-01A-11R-A14D-07	TCGA-A1-A0SF-01A-11R-A144-07	
##	COL1A1	393267	250584
##	FN1	261490	64545
##	CPB1	69	256
##	COL1A2	306336	177497
##	COL3A1	274333	238584
##	MT-CO1	401775	590224
##	TCGA-AC-A3W5-01A-11R-A22K-07	TCGA-EW-A1PA-01A-11R-A144-07	
##	COL1A1	1151986	120257
##	FN1	631166	99143
##	CPB1	23	59847
##	COL1A2	764378	92283
##	COL3A1	607478	87528
##	MT-CO1	582865	842909
##	TCGA-D8-A73X-01A-11R-A32P-07	TCGA-C8-A8HR-01A-11R-A36F-07	
##	COL1A1	871197	824814
##	FN1	319611	271683
##	CPB1	8	131
##	COL1A2	543219	510473
##	COL3A1	471342	518768
##	MT-CO1	312394	545959
##	TCGA-AO-AOJ3-01A-11R-A034-07	TCGA-D8-A27L-01A-11R-A16F-07	
##	COL1A1	54265	833170
##	FN1	50342	730490
##	CPB1	12570	215
##	COL1A2	34837	781390
##	COL3A1	25437	733385
##	MT-CO1	1552635	362593
##	TCGA-A8-A09D-01A-11R-A00Z-07	TCGA-EW-A10X-01A-11R-A144-07	
##	COL1A1	479065	23218
##	FN1	552435	28426
##	CPB1	98815	1536704
##	COL1A2	386905	18459
##	COL3A1	313867	18938
##	MT-CO1	848299	313564
##	TCGA-AR-A1AY-01A-21R-A12P-07	TCGA-LD-A66U-01A-11R-A310-07	
##	COL1A1	175925	700404
##	FN1	322496	82096
##	CPB1	26	438
##	COL1A2	140646	402274
##	COL3A1	171839	353539
##	MT-CO1	289492	178661
##	TCGA-C8-A12Z-01A-11R-A115-07	TCGA-LD-A7W6-01A-81R-A352-07	
##	COL1A1	406545	205250
##	FN1	361435	40814
##	CPB1	160	169
##	COL1A2	293244	125998
##	COL3A1	285050	139706
##	MT-CO1	350682	292692
##	TCGA-E9-A1N9-11A-71R-A14D-07	TCGA-E9-A228-01A-31R-A157-07	
##	COL1A1	22736	195011

##	FN1	17193	104290
##	CPB1	492	1093358
##	COL1A2	43052	151763
##	COL3A1	31763	178136
##	MT-CO1	550509	638118
##	TCGA-A7-A6VW-01A-21R-A33J-07	TCGA-BH-AOHP-01A-12R-A084-07	
##	COL1A1	289585	397130
##	FN1	123750	139993
##	CPB1	256	198
##	COL1A2	155177	346316
##	COL3A1	114914	380033
##	MT-CO1	187827	771944
##	TCGA-EW-A1P0-01A-11R-A144-07	TCGA-A8-A094-01A-11R-A00Z-07	
##	COL1A1	154307	135468
##	FN1	142531	172038
##	CPB1	516	1
##	COL1A2	130253	118059
##	COL3A1	116877	113822
##	MT-CO1	437721	335596
##	TCGA-BH-A1FR-11B-42R-A13Q-07	TCGA-D8-A4Z1-01A-21R-A266-07	
##	COL1A1	287281	177423
##	FN1	80458	49928
##	CPB1	11349	9733
##	COL1A2	237685	129674
##	COL3A1	297429	139006
##	MT-CO1	417491	214268
##	TCGA-A2-A1FZ-01A-51R-A14D-07	TCGA-BH-A18F-01A-11R-A12D-07	
##	COL1A1	424393	426983
##	FN1	379633	298947
##	CPB1	46797	33
##	COL1A2	340656	350974
##	COL3A1	285889	385276
##	MT-CO1	343632	365788
##	TCGA-BH-AOHO-01A-11R-A034-07	TCGA-B6-AORI-01A-11R-A056-07	
##	COL1A1	222445	869250
##	FN1	172293	1105417
##	CPB1	4694659	81
##	COL1A2	190602	577650
##	COL3A1	188636	526817
##	MT-CO1	370149	499109
##	TCGA-S3-AA15-01A-11R-A41B-07	TCGA-E9-A245-01A-22R-A16F-07	
##	COL1A1	513009	86236
##	FN1	138517	41420
##	CPB1	14	3116281
##	COL1A2	264231	62797
##	COL3A1	284981	62327
##	MT-CO1	218332	284738
##	TCGA-AN-A041-01A-11R-A034-07	TCGA-A0-A125-01A-11R-A10J-07	
##	COL1A1	1016633	16526
##	FN1	886976	38140
##	CPB1	5484051	97
##	COL1A2	860953	10620
##	COL3A1	676404	9425
##	MT-CO1	581097	1141552

##	TCGA-BH-AOBA-11A-22R-A19E-07	TCGA-AN-AOFV-01A-11R-A00Z-07
##	COL1A1	47430 589881
##	FN1	30897 614976
##	CPB1	28900 331
##	COL1A2	82502 502672
##	COL3A1	58357 447047
##	MT-CO1	342382 692762
##	TCGA-AN-AOFK-01A-11R-A034-07	TCGA-B6-A0I2-01A-11R-A034-07
##	COL1A1	367169 31080
##	FN1	152765 27538
##	CPB1	320 1
##	COL1A2	228403 34085
##	COL3A1	161396 13443
##	MT-CO1	129915 640089
##	TCGA-BH-AODV-01A-21R-A12P-07	TCGA-OL-A66J-01A-11R-A29R-07
##	COL1A1	1781902 308190
##	FN1	746736 121189
##	CPB1	443 100
##	COL1A2	1312083 198319
##	COL3A1	1331310 151776
##	MT-CO1	722033 392696
##	TCGA-D8-A73U-01A-11R-A33J-07	TCGA-D8-A1XQ-01A-11R-A14M-07
##	COL1A1	677905 171910
##	FN1	199510 264287
##	CPB1	4 4
##	COL1A2	438123 165359
##	COL3A1	318412 172793
##	MT-CO1	261673 181276
##	TCGA-A2-AOT5-01A-21R-A084-07	TCGA-AR-A2LQ-01A-22R-A18M-07
##	COL1A1	333435 2014322
##	FN1	482854 425141
##	CPB1	247 279
##	COL1A2	240532 1337000
##	COL3A1	189426 1598661
##	MT-CO1	606240 354498
##	TCGA-HN-A2OB-01A-21R-A27Q-07	TCGA-AR-A1AO-01A-11R-A12P-07
##	COL1A1	1025240 1116827
##	FN1	227470 312427
##	CPB1	11 41
##	COL1A2	724174 775739
##	COL3A1	854758 1076008
##	MT-CO1	467739 656422
##	TCGA-BH-AOBR-01A-21R-A115-07	TCGA-A8-A0A1-01A-11R-A00Z-07
##	COL1A1	802267 441658
##	FN1	696202 295941
##	CPB1	48 69113
##	COL1A2	567935 302263
##	COL3A1	530155 225527
##	MT-CO1	291302 330221
##	TCGA-BH-AODG-11A-43R-A12P-07	TCGA-GM-A4EO-01A-12R-A266-07
##	COL1A1	42225 205987
##	FN1	18357 55969
##	CPB1	429 26
##	COL1A2	57333 102632

##	COL3A1	61518	85605
##	MT-CO1	350347	724206
##	TCGA-E9-A3H0-01A-11R-A213-07	TCGA-E9-A1N5-01A-11R-A14D-07	
##	COL1A1	145636	555933
##	FN1	49976	288577
##	CPB1	343	3990
##	COL1A2	80158	392918
##	COL3A1	71677	364948
##	MT-CO1	1540064	327258
##	TCGA-AR-A1AQ-01A-11R-A12P-07	TCGA-A8-A086-01A-11R-A00Z-07	
##	COL1A1	143548	861726
##	FN1	81831	731529
##	CPB1	18	10863
##	COL1A2	112521	765725
##	COL3A1	118752	650595
##	MT-CO1	135802	638211
##	TCGA-BH-A0DS-01A-11R-A056-07	TCGA-E9-A295-01A-11R-A16F-07	
##	COL1A1	399524	337344
##	FN1	234346	131623
##	CPB1	1972	182
##	COL1A2	284523	244257
##	COL3A1	199698	252121
##	MT-CO1	728256	703331
##	TCGA-BH-A42U-01A-12R-A24H-07	TCGA-A0-A0JJ-01A-11R-A056-07	
##	COL1A1	236570	391286
##	FN1	64144	170418
##	CPB1	201	4265
##	COL1A2	131489	309183
##	COL3A1	106446	419146
##	MT-CO1	420030	434612
##	TCGA-AQ-A04H-01B-11R-A10J-07	TCGA-E9-A2JS-01A-11R-A180-07	
##	COL1A1	229673	450035
##	FN1	268446	333616
##	CPB1	1328	7
##	COL1A2	157834	305179
##	COL3A1	145394	236416
##	MT-CO1	652413	879102
##	TCGA-AN-A049-01A-21R-A00Z-07	TCGA-A8-A08T-01A-21R-A00Z-07	
##	COL1A1	558267	173375
##	FN1	415661	374073
##	CPB1	413	14853
##	COL1A2	384165	166305
##	COL3A1	298709	147477
##	MT-CO1	359335	647877
##	TCGA-A8-A079-01A-21R-A00Z-07	TCGA-GM-A2DC-01A-11R-A18M-07	
##	COL1A1	167382	109830
##	FN1	184936	35169
##	CPB1	118	114941
##	COL1A2	110753	59674
##	COL3A1	91599	47029
##	MT-CO1	934584	772377
##	TCGA-E9-A226-01A-21R-A157-07	TCGA-AR-A254-01A-21R-A169-07	
##	COL1A1	676838	723805
##	FN1	1177243	850550

##	CPB1	60	33
##	COL1A2	782205	587889
##	COL3A1	584476	579107
##	MT-CO1	434507	446003
##	TCGA-BH-AODG-01A-21R-A12P-07	TCGA-E9-A1RH-11A-34R-A169-07	
##	COL1A1	991361	89924
##	FN1	591372	56398
##	CPB1	85254	425
##	COL1A2	732445	163751
##	COL3A1	589787	104663
##	MT-CO1	533771	1541323
##	TCGA-D8-A1J8-01A-11R-A13Q-07	TCGA-E9-A1RH-01A-21R-A169-07	
##	COL1A1	227684	399031
##	FN1	666798	546327
##	CPB1	6	10
##	COL1A2	251686	407870
##	COL3A1	238615	418341
##	MT-CO1	592027	510695
##	TCGA-A8-AOA7-01A-11R-A00Z-07	TCGA-AC-A6IW-01A-12R-A33J-07	
##	COL1A1	480218	45325
##	FN1	385795	24878
##	CPB1	515	2
##	COL1A2	385443	30453
##	COL3A1	443288	24395
##	MT-CO1	429541	379070
##	TCGA-A2-AOSW-01A-11R-A084-07	TCGA-XX-A899-01A-11R-A36F-07	
##	COL1A1	333039	542330
##	FN1	274455	267858
##	CPB1	38	31
##	COL1A2	252767	340825
##	COL3A1	218828	352346
##	MT-CO1	855238	466090
##	TCGA-PE-A5DE-01A-11R-A27Q-07	TCGA-E2-A150-01A-11R-A12D-07	
##	COL1A1	166191	190872
##	FN1	66676	201677
##	CPB1	3	13
##	COL1A2	104256	145393
##	COL3A1	86176	144145
##	MT-CO1	314062	541743
##	TCGA-A8-AOA6-01A-12R-A056-07	TCGA-AN-A04D-01A-21R-A034-07	
##	COL1A1	75707	42745
##	FN1	126876	61762
##	CPB1	3182	9
##	COL1A2	78980	26555
##	COL3A1	70677	23390
##	MT-CO1	266208	1563387
##	TCGA-C8-A1HM-01A-12R-A137-07	TCGA-A7-A4SD-01A-11R-A266-07	
##	COL1A1	61413	407692
##	FN1	238136	194047
##	CPB1	1	4
##	COL1A2	43681	259200
##	COL3A1	37828	205476
##	MT-CO1	374153	296162
##	TCGA-BH-AOBA-01A-11R-A056-07	TCGA-BH-A0BC-11A-22R-A089-07	

## COL1A1	197329	90542
## FN1	102964	36866
## CPB1	1226	2037
## COL1A2	160212	114423
## COL3A1	169260	90493
## MT-CO1	568894	436436
## TCGA-BH-A1ET-11B-23R-A137-07	TCGA-A2-AOCT-01A-31R-A056-07	
## COL1A1	57096	144379
## FN1	44912	102630
## CPB1	840	219
## COL1A2	100168	137555
## COL3A1	68107	119352
## MT-CO1	752271	584975
## TCGA-D8-A1JF-01A-11R-A13Q-07	TCGA-BH-AOE1-01A-11R-A056-07	
## COL1A1	430386	287544
## FN1	373582	130425
## CPB1	6	1060
## COL1A2	442716	226355
## COL3A1	409299	233091
## MT-CO1	1005351	242921
## TCGA-A8-A097-01A-11R-A034-07	TCGA-LD-A74U-01A-13R-A33J-07	
## COL1A1	925895	930889
## FN1	1179602	176312
## CPB1	16	109
## COL1A2	728745	555995
## COL3A1	642936	685545
## MT-CO1	1002130	427438
## TCGA-AO-AOJE-01A-11R-A056-07	TCGA-A2-A4S3-01A-21R-A266-07	
## COL1A1	905506	195340
## FN1	750438	237686
## CPB1	110	13
## COL1A2	692863	106920
## COL3A1	547407	74109
## MT-CO1	401510	195205
## TCGA-E9-A1RD-11A-33R-A157-07	TCGA-AO-A03T-01A-21R-A034-07	
## COL1A1	103660	108404
## FN1	66687	28690
## CPB1	2064	34427
## COL1A2	177360	55836
## COL3A1	182412	43874
## MT-CO1	1088758	3563979
## TCGA-A8-A09M-01A-11R-A00Z-07	TCGA-E9-A22B-01A-11R-A157-07	
## COL1A1	167659	438223
## FN1	165174	555683
## CPB1	12403	4027597
## COL1A2	131385	379726
## COL3A1	89210	304004
## MT-CO1	995380	1626990
## TCGA-E9-A1N9-01A-11R-A14D-07	TCGA-BH-A1EV-01A-11R-A137-07	
## COL1A1	169853	383930
## FN1	192294	635687
## CPB1	2472	3
## COL1A2	170043	359187
## COL3A1	173036	263837

##	MT-C01	360685	386754
##	TCGA-A0-A1KR-01A-12R-A144-07	TCGA-OL-A5RU-01A-11R-A28M-07	
##	COL1A1	29928	109515
##	FN1	7043	37289
##	CPB1	14	145
##	COL1A2	16822	83926
##	COL3A1	10913	69768
##	MT-C01	841806	216108
##	TCGA-E9-A1N5-11A-41R-A14D-07	TCGA-C8-A12U-01A-11R-A115-07	
##	COL1A1	342998	34282
##	FN1	56308	39893
##	CPB1	7	1647
##	COL1A2	190057	22260
##	COL3A1	285040	17421
##	MT-C01	206151	710734
##	TCGA-LL-A5Y0-01A-21R-A28M-07	TCGA-A8-A08A-01A-11R-A32Y-07	
##	COL1A1	167039	681777
##	FN1	60560	343491
##	CPB1	11	4
##	COL1A2	108912	458254
##	COL3A1	96718	409534
##	MT-C01	236656	420221
##	TCGA-PL-A8LX-01A-11R-A41B-07	TCGA-E9-A249-01A-11R-A169-07	
##	COL1A1	75979	222431
##	FN1	51732	198591
##	CPB1	3235426	108
##	COL1A2	42415	173702
##	COL3A1	28532	151851
##	MT-C01	691507	633670
##	TCGA-A2-A3XZ-01A-42R-A239-07	TCGA-OL-A66P-01A-11R-A310-07	
##	COL1A1	188612	357327
##	FN1	105823	38361
##	CPB1	8	5
##	COL1A2	102462	192422
##	COL3A1	91551	149180
##	MT-C01	363554	182417
##	TCGA-C8-A26W-01A-11R-A16F-07	TCGA-A7-A13H-01A-11R-A22K-07	
##	COL1A1	572667	444613
##	FN1	287662	151945
##	CPB1	254	157
##	COL1A2	425197	241161
##	COL3A1	429539	203241
##	MT-C01	634515	520187
##	TCGA-A2-A4S1-01A-21R-A266-07	TCGA-BH-A0E2-01A-11R-A056-07	
##	COL1A1	2279237	376828
##	FN1	358481	235585
##	CPB1	17	43451
##	COL1A2	2380801	311705
##	COL3A1	1291259	253590
##	MT-C01	130698	173793
##	TCGA-BH-A0HY-01A-11R-A056-07	TCGA-EW-A1PF-01A-11R-A144-07	
##	COL1A1	655650	801839
##	FN1	378949	584337
##	CPB1	835	1917031

##	COL1A2	444802	568294
##	COL3A1	363318	511436
##	MT-CO1	504782	563489
##	TCGA-AC-A3W6-01A-12R-A22K-07	TCGA-B6-A40B-01A-11R-A239-07	
##	COL1A1	1545224	1229476
##	FN1	225740	344550
##	CPB1	33	6848
##	COL1A2	847580	624966
##	COL3A1	675910	456796
##	MT-CO1	482066	560930
##	TCGA-A2-AOD0-01A-11R-A00Z-07	TCGA-A2-AOD3-01A-11R-A115-07	
##	COL1A1	54143	135174
##	FN1	110886	128854
##	CPB1	0	8
##	COL1A2	27663	98057
##	COL3A1	22153	93802
##	MT-CO1	224878	434998
##	TCGA-BH-A42T-01A-11R-A24H-07	TCGA-A2-A04N-01A-11R-A115-07	
##	COL1A1	442022	939224
##	FN1	154404	493793
##	CPB1	3	72
##	COL1A2	235640	603494
##	COL3A1	171848	434163
##	MT-CO1	550624	542837
##	TCGA-GI-A2C9-01A-11R-A21T-07	TCGA-GI-A2C9-11A-22R-A21T-07	
##	COL1A1	433855	167045
##	FN1	266326	62852
##	CPB1	20	7248
##	COL1A2	311311	204323
##	COL3A1	213088	145421
##	MT-CO1	688398	373180
##	TCGA-E2-A15P-01A-11R-A115-07	TCGA-A0-A03L-01A-41R-A056-07	
##	COL1A1	383451	490954
##	FN1	138297	408176
##	CPB1	1462855	2086
##	COL1A2	314878	390794
##	COL3A1	347762	437864
##	MT-CO1	1208445	274794
##	TCGA-3C-AALI-01A-11R-A41B-07	TCGA-E9-A22E-01A-11R-A157-07	
##	COL1A1	347225	2000790
##	FN1	201191	1249916
##	CPB1	49	18164
##	COL1A2	199850	1984099
##	COL3A1	138651	1516539
##	MT-CO1	366230	888848
##	TCGA-A2-A04V-01A-21R-A034-07	TCGA-E9-A1NG-01A-21R-A14M-07	
##	COL1A1	607897	923309
##	FN1	387391	2268083
##	CPB1	758	7984
##	COL1A2	489527	998577
##	COL3A1	419377	744900
##	MT-CO1	1481117	334896
##	TCGA-D8-A1JK-01A-11R-A13Q-07	TCGA-BH-AOD0-01B-11R-A12D-07	
##	COL1A1	367057	187199

##	FN1	451016	64179
##	CPB1	9	1258
##	COL1A2	258104	126548
##	COL3A1	262982	141395
##	MT-CO1	279341	529305
##	TCGA-AR-A24P-01A-11R-A169-07	TCGA-AC-A2QJ-01A-12R-A19W-07	
##	COL1A1	839171	1744629
##	FN1	626635	106979
##	CPB1	5092	56
##	COL1A2	729911	653211
##	COL3A1	617497	578396
##	MT-CO1	465148	111886
##	TCGA-AO-AOJ2-01A-11R-A034-07	TCGA-BH-A1FG-11B-12R-A13Q-07	
##	COL1A1	258190	6208
##	FN1	178198	13440
##	CPB1	7	3802
##	COL1A2	156894	22574
##	COL3A1	138623	11422
##	MT-CO1	467838	1092154
##	TCGA-BH-AOAY-01A-21R-A00Z-07	TCGA-A2-A0T0-01A-22R-A084-07	
##	COL1A1	698967	600671
##	FN1	426696	385282
##	CPB1	2677	47
##	COL1A2	461988	429486
##	COL3A1	401358	326290
##	MT-CO1	801373	698269
##	TCGA-A2-A04W-01A-31R-A115-07	TCGA-B6-A0IG-01A-11R-A034-07	
##	COL1A1	708285	496741
##	FN1	295538	264758
##	CPB1	21	81
##	COL1A2	378848	304445
##	COL3A1	328870	256480
##	MT-CO1	719101	987151
##	TCGA-AN-AOXS-01A-22R-A109-07	TCGA-D8-A1X9-01A-12R-A157-07	
##	COL1A1	528813	528020
##	FN1	563663	1066849
##	CPB1	11	218
##	COL1A2	403967	510814
##	COL3A1	346964	456146
##	MT-CO1	568328	534738
##	TCGA-5T-A9QA-01A-11R-A41B-07	TCGA-E9-A1NE-01A-21R-A14M-07	
##	COL1A1	14292	206449
##	FN1	13015	309551
##	CPB1	2492	700108
##	COL1A2	7373	213747
##	COL3A1	5601	231535
##	MT-CO1	173073	729962
##	TCGA-A2-AOEM-01A-11R-A034-07	TCGA-AC-A80Q-01A-11R-A41B-07	
##	COL1A1	601468	267364
##	FN1	476277	219538
##	CPB1	1117	19
##	COL1A2	532419	149628
##	COL3A1	383214	142257
##	MT-CO1	632269	420683

##	TCGA-AC-A23C-01A-12R-A169-07	TCGA-AN-A0XV-01A-11R-A109-07
##	COL1A1	521672 317275
##	FN1	437457 305528
##	CPB1	12607 537612
##	COL1A2	485677 300560
##	COL3A1	355104 242749
##	MT-CO1	885193 833629
##	TCGA-PE-A5DD-01A-12R-A27Q-07	TCGA-BH-A0H5-01A-21R-A115-07
##	COL1A1	454747 143979
##	FN1	107237 126105
##	CPB1	220 1834
##	COL1A2	245740 115656
##	COL3A1	185012 111112
##	MT-CO1	695765 274887
##	TCGA-A2-A1FX-01A-11R-A13Q-07	TCGA-E2-A105-01A-11R-A10J-07
##	COL1A1	391654 159884
##	FN1	229375 142200
##	CPB1	9516 37794
##	COL1A2	273743 102819
##	COL3A1	268609 84095
##	MT-CO1	122106 634567
##	TCGA-A8-A09R-01A-11R-A00Z-07	TCGA-C8-A1HE-01A-11R-A13Q-07
##	COL1A1	914546 247775
##	FN1	1136535 245596
##	CPB1	9196 508
##	COL1A2	735064 190296
##	COL3A1	597022 162090
##	MT-CO1	380830 358453
##	TCGA-A8-A09E-01A-11R-A00Z-07	TCGA-A7-A0DC-11A-41R-A089-07
##	COL1A1	173022 94831
##	FN1	122852 39503
##	CPB1	121 31008
##	COL1A2	109465 124995
##	COL3A1	120164 117576
##	MT-CO1	424249 584547
##	TCGA-A7-A0DC-01A-11R-A00Z-07	TCGA-A7-A0DC-01B-04R-A220-07
##	COL1A1	20347 25396
##	FN1	102306 75738
##	CPB1	59901 223
##	COL1A2	32445 18551
##	COL3A1	34540 21243
##	MT-CO1	20639 101723
##	TCGA-A8-A08F-01A-11R-A00Z-07	TCGA-LL-A6FP-01A-11R-A310-07
##	COL1A1	531457 53880
##	FN1	526908 14745
##	CPB1	96045 831
##	COL1A2	320403 30339
##	COL3A1	283458 21552
##	MT-CO1	337330 297590
##	TCGA-5L-AAT1-01A-12R-A41B-07	TCGA-A8-A07E-01A-11R-A034-07
##	COL1A1	359036 895653
##	FN1	157058 399812
##	CPB1	61 92
##	COL1A2	224234 709803

##	COL3A1	194370	765744
##	MT-CO1	183543	539099
##	TCGA-GI-A2C8-11A-22R-A16F-07	TCGA-A8-A09K-01A-11R-A00Z-07	
##	COL1A1	34961	24032
##	FN1	35461	23860
##	CPB1	79	420
##	COL1A2	48372	21895
##	COL3A1	73401	24762
##	MT-CO1	2138615	735363
##	TCGA-A8-A09I-01A-22R-A034-07	TCGA-A0-A0JI-01A-21R-A056-07	
##	COL1A1	618085	34819
##	FN1	701214	20022
##	CPB1	361	1431384
##	COL1A2	553318	32508
##	COL3A1	453339	42111
##	MT-CO1	612248	401062
##	TCGA-B6-A1KC-01B-11R-A157-07	TCGA-C8-A1HO-01A-11R-A13Q-07	
##	COL1A1	59309	132960
##	FN1	58227	98595
##	CPB1	908	313225
##	COL1A2	55859	104274
##	COL3A1	69994	105451
##	MT-CO1	2201602	817850
##	TCGA-E2-A1L7-01A-11R-A144-07	TCGA-C8-A274-01A-11R-A16F-07	
##	COL1A1	161971	183318
##	FN1	126074	303849
##	CPB1	463	3467
##	COL1A2	118489	143573
##	COL3A1	103086	107532
##	MT-CO1	271511	1307320
##	TCGA-D8-A1XU-01A-11R-A14M-07	TCGA-AR-A24Q-01A-12R-A169-07	
##	COL1A1	1120310	409539
##	FN1	798207	675338
##	CPB1	7032374	97
##	COL1A2	933615	341466
##	COL3A1	820558	288110
##	MT-CO1	1260411	610168
##	TCGA-BH-A18L-01A-32R-A12D-07	TCGA-A2-A0YM-01A-11R-A109-07	
##	COL1A1	43672	157690
##	FN1	35146	205053
##	CPB1	61	5
##	COL1A2	38895	120300
##	COL3A1	33610	143687
##	MT-CO1	576774	208806
##	TCGA-BH-A0B2-01A-11R-A10J-07	TCGA-E9-A22A-01A-11R-A157-07	
##	COL1A1	865948	1736737
##	FN1	353810	1227321
##	CPB1	2845	248
##	COL1A2	618746	1453259
##	COL3A1	527190	1041500
##	MT-CO1	292281	1319598
##	TCGA-A8-A08B-01A-11R-A00Z-07	TCGA-BH-A0B0-01A-21R-A115-07	
##	COL1A1	42233	1399998
##	FN1	187554	476502

##	CPB1	43	28
##	COL1A2	38312	864433
##	COL3A1	34130	898001
##	MT-CO1	188626	432089
##	TCGA-BH-A1FE-01A-11R-A13Q-07	TCGA-A1-AOSH-01A-11R-A084-07	
##	COL1A1	659301	3257937
##	FN1	662638	1660236
##	CPB1	1169	45
##	COL1A2	615288	2505608
##	COL3A1	689060	2140089
##	MT-CO1	702631	319940
##	TCGA-A8-A095-01A-11R-A00Z-07	TCGA-BH-A0BW-01A-11R-A115-07	
##	COL1A1	358286	122863
##	FN1	277583	262427
##	CPB1	244	0
##	COL1A2	282996	132920
##	COL3A1	277895	66646
##	MT-CO1	359167	207356
##	TCGA-AN-AOFF-01A-11R-A034-07	TCGA-A2-A3XX-01A-21R-A239-07	
##	COL1A1	394806	213301
##	FN1	247388	137192
##	CPB1	284873	1
##	COL1A2	258078	130348
##	COL3A1	245349	97841
##	MT-CO1	490776	422499
##	TCGA-BH-A1EN-01A-11R-A13Q-07	TCGA-BH-A1EN-11A-23R-A13Q-07	
##	COL1A1	243256	170544
##	FN1	202544	85745
##	CPB1	10	630
##	COL1A2	195075	211372
##	COL3A1	175836	174875
##	MT-CO1	933525	1478669
##	TCGA-A2-A3XW-01A-11R-A239-07	TCGA-A8-A08S-01A-11R-A034-07	
##	COL1A1	238398	409919
##	FN1	60435	332589
##	CPB1	209	1175
##	COL1A2	120465	292824
##	COL3A1	90269	203700
##	MT-CO1	340003	461693
##	TCGA-E2-A14V-01A-11R-A12D-07	TCGA-A8-A09C-01A-11R-A00Z-07	
##	COL1A1	421163	428914
##	FN1	286678	285747
##	CPB1	861	924
##	COL1A2	304818	302372
##	COL3A1	279938	241331
##	MT-CO1	523775	1273821
##	TCGA-D8-A1Y2-01A-11R-A157-07	TCGA-GM-A5PV-01A-11R-A28M-07	
##	COL1A1	556476	14823
##	FN1	730814	12866
##	CPB1	4738	397
##	COL1A2	482493	12936
##	COL3A1	335071	13492
##	MT-CO1	596443	484881
##	TCGA-E2-A1B1-01A-21R-A12P-07	TCGA-B6-A0WX-01A-11R-A109-07	

## COL1A1	897978	1076768
## FN1	685933	273730
## CPB1	635	6
## COL1A2	1000701	748566
## COL3A1	1187406	700519
## MT-CO1	725879	416483
## TCGA-EW-A1J2-01A-21R-A13Q-07	TCGA-AR-A255-01A-11R-A169-07	
## COL1A1	875892	706751
## FN1	925907	637759
## CPB1	2395	169
## COL1A2	790268	628736
## COL3A1	697253	574136
## MT-CO1	531375	216036
## TCGA-UL-AAZ6-01A-11R-A41B-07	TCGA-A2-AOCL-01A-11R-A115-07	
## COL1A1	112275	307221
## FN1	52064	326567
## CPB1	195847	5
## COL1A2	63109	241880
## COL3A1	47648	249011
## MT-CO1	996587	297865
## TCGA-AC-A3HN-01A-11R-A213-07	TCGA-E2-A108-01A-13R-A10J-07	
## COL1A1	1574832	361334
## FN1	584232	344943
## CPB1	583	637
## COL1A2	1010492	246494
## COL3A1	821323	233897
## MT-CO1	429396	407328
## TCGA-BH-A1F8-11B-21R-A13Q-07	TCGA-BH-A1F8-01A-11R-A13Q-07	
## COL1A1	101354	300303
## FN1	29704	507163
## CPB1	469	68822
## COL1A2	110630	303597
## COL3A1	86992	254649
## MT-CO1	445159	785791
## TCGA-A8-A08I-01A-11R-A00Z-07	TCGA-A8-A08J-01A-11R-A00Z-07	
## COL1A1	61398	576856
## FN1	101487	566867
## CPB1	331	432937
## COL1A2	50730	478513
## COL3A1	41778	446452
## MT-CO1	453481	651939
## TCGA-A8-A08P-01A-11R-A00Z-07	TCGA-C8-A12M-01A-11R-A115-07	
## COL1A1	256657	48185
## FN1	282554	61248
## CPB1	66	0
## COL1A2	223591	42263
## COL3A1	250676	35106
## MT-CO1	506208	369039
## TCGA-A2-AOEQ-01A-11R-A034-07	TCGA-BH-A0W7-01A-11R-A115-07	
## COL1A1	200029	481728
## FN1	81814	323679
## CPB1	53	187
## COL1A2	123847	349255
## COL3A1	141863	297570

##	MT-C01	396263	401783
##	TCGA-D8-A27W-01A-11R-A16F-07	TCGA-BH-AOHN-01A-11R-A10U-07	
##	COL1A1	209537	96901
##	FN1	310204	26355
##	CPB1	803	1
##	COL1A2	203030	49693
##	COL3A1	202809	44194
##	MT-C01	599230	722425
##	TCGA-A2-A1FW-01A-11R-A13Q-07	TCGA-E9-A1RI-11A-41R-A169-07	
##	COL1A1	141423	17257
##	FN1	183721	24808
##	CPB1	4632	178
##	COL1A2	137631	53848
##	COL3A1	149037	45536
##	MT-C01	672166	1208635
##	TCGA-BH-AODP-01A-21R-A056-07	TCGA-A7-A26G-01A-21R-A169-07	
##	COL1A1	662775	999360
##	FN1	240301	1323140
##	CPB1	1536	32
##	COL1A2	485005	950776
##	COL3A1	404943	895578
##	MT-C01	299506	768259
##	TCGA-3C-AALK-01A-11R-A41B-07	TCGA-C8-A12Q-01A-11R-A115-07	
##	COL1A1	1104141	477566
##	FN1	336918	706162
##	CPB1	312	128
##	COL1A2	626179	391992
##	COL3A1	502387	398714
##	MT-C01	539992	449638
##	TCGA-BH-AOB5-11A-23R-A12P-07	TCGA-BH-A1F6-01A-11R-A13Q-07	
##	COL1A1	70372	201080
##	FN1	27974	190153
##	CPB1	328	19
##	COL1A2	112978	142443
##	COL3A1	144744	105106
##	MT-C01	1205115	269394
##	TCGA-AO-A03M-01B-11R-A10J-07	TCGA-C8-A131-01A-11R-A115-07	
##	COL1A1	308548	669525
##	FN1	221723	497349
##	CPB1	15	25
##	COL1A2	218828	389447
##	COL3A1	154036	410414
##	MT-C01	351173	1007096
##	TCGA-AC-A6IV-01A-12R-A33J-07	TCGA-A2-A0T2-01A-11R-A084-07	
##	COL1A1	631650	190084
##	FN1	347417	190328
##	CPB1	95255	6
##	COL1A2	467701	137092
##	COL3A1	434999	108124
##	MT-C01	659593	1115850
##	TCGA-E9-A1NC-01A-21R-A26B-07	TCGA-XX-A89A-01A-11R-A36F-07	
##	COL1A1	192633	1020690
##	FN1	284863	384796
##	CPB1	7	158

##	COL1A2	131396	565258
##	COL3A1	115893	584940
##	MT-CO1	550744	460208
##	TCGA-E9-A5U0-01A-11R-A28M-07	TCGA-LL-A740-01A-21R-A32P-07	
##	COL1A1	45006	539899
##	FN1	57022	379202
##	CPB1	1489101	6
##	COL1A2	30230	414000
##	COL3A1	36109	390067
##	MT-CO1	470706	318134
##	TCGA-E9-A1N6-11A-32R-A144-07	TCGA-AN-A046-01A-21R-A034-07	
##	COL1A1	24168	603291
##	FN1	19345	375161
##	CPB1	78	1374
##	COL1A2	42432	447298
##	COL3A1	31587	348921
##	MT-CO1	329595	788272
##	TCGA-E9-A1N6-01A-11R-A144-07	TCGA-BH-A18G-01A-11R-A12D-07	
##	COL1A1	874477	318217
##	FN1	620636	166036
##	CPB1	46	0
##	COL1A2	640860	191715
##	COL3A1	596635	141021
##	MT-CO1	258829	694556
##	TCGA-B6-A0R0-01A-22R-A084-07	TCGA-E9-A54Y-01A-11R-A466-07	
##	COL1A1	99035	18048
##	FN1	54785	12094
##	CPB1	282	7
##	COL1A2	105027	6427
##	COL3A1	125694	2251
##	MT-CO1	300298	729362
##	TCGA-BH-A1ES-01A-11R-A137-07	TCGA-BH-A1ES-06A-12R-A24H-07	
##	COL1A1	118434	58810
##	FN1	234548	64737
##	CPB1	600	7
##	COL1A2	125679	32297
##	COL3A1	104350	55327
##	MT-CO1	129374	41548
##	TCGA-E9-A1R5-01A-11R-A14M-07	TCGA-B6-A0IC-01A-11R-A034-07	
##	COL1A1	759428	20815
##	FN1	889913	40605
##	CPB1	8	1747
##	COL1A2	784768	15202
##	COL3A1	765255	10511
##	MT-CO1	821159	517895
##	TCGA-A2-A0SY-01A-31R-A084-07	TCGA-AC-A2BK-01A-11R-A21T-07	
##	COL1A1	401932	65105
##	FN1	216731	142395
##	CPB1	14	0
##	COL1A2	336357	93374
##	COL3A1	318113	5614
##	MT-CO1	431776	217367
##	TCGA-C8-A26X-01A-31R-A16F-07	TCGA-A8-A07U-01A-11R-A034-07	
##	COL1A1	242021	278203

##	FN1	289537	216004
##	CPB1	2	7
##	COL1A2	187613	196017
##	COL3A1	187139	154903
##	MT-CO1	855929	580102
##	TCGA-A8-A07J-01A-11R-A00Z-07	TCGA-BH-A18H-01A-11R-A12D-07	
##	COL1A1	522059	98047
##	FN1	406085	76592
##	CPB1	290	34
##	COL1A2	529835	75279
##	COL3A1	498648	76832
##	MT-CO1	363482	577904
##	TCGA-E2-A14P-01A-31R-A12D-07	TCGA-AN-A04A-01A-21R-A034-07	
##	COL1A1	77777	1120218
##	FN1	79944	1183303
##	CPB1	3	129358
##	COL1A2	57069	888179
##	COL3A1	49320	743693
##	MT-CO1	482783	828861
##	TCGA-BH-A202-01A-11R-A14M-07	TCGA-WT-AB41-01A-11R-A41B-07	
##	COL1A1	269505	99194
##	FN1	1888823	23460
##	CPB1	386	5146
##	COL1A2	262581	55478
##	COL3A1	220832	43531
##	MT-CO1	556849	81003
##	TCGA-BH-AORX-01A-21R-A084-07	TCGA-A8-A07L-01A-11R-A00Z-07	
##	COL1A1	185868	453698
##	FN1	143686	305267
##	CPB1	529	6
##	COL1A2	138950	325638
##	COL3A1	140969	283924
##	MT-CO1	753388	231655
##	TCGA-D8-A140-01A-11R-A115-07	TCGA-AR-A5QQ-01A-11R-A28M-07	
##	COL1A1	192325	211814
##	FN1	130463	169462
##	CPB1	886	19
##	COL1A2	160268	117756
##	COL3A1	141505	147465
##	MT-CO1	227043	105215
##	TCGA-BH-AOHX-01A-21R-A056-07	TCGA-C8-A27A-01A-11R-A169-07	
##	COL1A1	351587	138763
##	FN1	369857	275298
##	CPB1	305379	28991
##	COL1A2	295289	144600
##	COL3A1	230112	126609
##	MT-CO1	402280	990936
##	TCGA-A2-A25D-01A-12R-A16F-07	TCGA-BH-A201-01A-11R-A14M-07	
##	COL1A1	117581	838143
##	FN1	83288	481962
##	CPB1	128	39831
##	COL1A2	87071	631457
##	COL3A1	64806	591768
##	MT-CO1	1188852	756010

##	TCGA-A8-A085-01A-11R-A00Z-07	TCGA-AR-A5QP-01A-11R-A28M-07
##	COL1A1	982729 494952
##	FN1	558846 155860
##	CPB1	29402 20
##	COL1A2	654439 367843
##	COL3A1	562930 337909
##	MT-CO1	1023348 224370
##	TCGA-BH-AOW3-01A-11R-A109-07	TCGA-AR-A24V-01A-21R-A169-07
##	COL1A1	88802 1458646
##	FN1	63924 1792563
##	CPB1	1778 24
##	COL1A2	59851 1479686
##	COL3A1	48635 1131400
##	MT-CO1	223734 592878
##	TCGA-A8-A09V-01A-11R-A034-07	TCGA-BH-A0HK-01A-11R-A056-07
##	COL1A1	653584 182257
##	FN1	518498 69314
##	CPB1	208 23
##	COL1A2	636011 123141
##	COL3A1	519651 144689
##	MT-CO1	456926 1178177
##	TCGA-E2-A14Q-01A-11R-A12D-07	TCGA-D8-A1X8-01A-11R-A14M-07
##	COL1A1	275363 199244
##	FN1	109556 136858
##	CPB1	12629 218
##	COL1A2	197927 204841
##	COL3A1	175468 213313
##	MT-CO1	778735 366063
##	TCGA-BH-A0H5-11A-62R-A115-07	TCGA-A7-A0CD-01A-11R-A00Z-07
##	COL1A1	65524 252060
##	FN1	12571 197194
##	CPB1	6575 34258
##	COL1A2	54875 225350
##	COL3A1	86578 183081
##	MT-CO1	440403 310397
##	TCGA-A8-A06T-01A-11R-A00Z-07	TCGA-A8-A06U-01A-11R-A00Z-07
##	COL1A1	397867 137284
##	FN1	150756 165085
##	CPB1	52955 55631
##	COL1A2	312248 113231
##	COL3A1	282408 88303
##	MT-CO1	565929 1120363
##	TCGA-AQ-A0Y5-01A-11R-A14M-07	TCGA-AN-A0XW-01A-11R-A109-07
##	COL1A1	753902 981212
##	FN1	727952 619846
##	CPB1	9645 570
##	COL1A2	796362 834784
##	COL3A1	727552 627255
##	MT-CO1	562000 481916
##	TCGA-BH-A0B1-01A-12R-A056-07	TCGA-AC-A62V-01A-11R-A310-07
##	COL1A1	425866 516034
##	FN1	209003 160656
##	CPB1	169 123
##	COL1A2	376113 269479

##	COL3A1	387092	153039
##	MT-C01	607878	197344
##	TCGA-AC-A2QH-01B-04R-A220-07	TCGA-A7-A4SB-01A-21R-A266-07	
##	COL1A1	32252	138666
##	FN1	285236	30264
##	CPB1	5	207
##	COL1A2	16151	79279
##	COL3A1	87243	84126
##	MT-C01	311952	717400
##	TCGA-A2-AOSV-01A-11R-A084-07	TCGA-BH-A0BS-01A-11R-A12P-07	
##	COL1A1	907778	61576
##	FN1	630311	116225
##	CPB1	215	252
##	COL1A2	690391	61198
##	COL3A1	492698	56175
##	MT-C01	821677	345800
##	TCGA-PE-A5DC-01A-12R-A27Q-07	TCGA-BH-A0HA-01A-11R-A12P-07	
##	COL1A1	161629	784387
##	FN1	89999	305977
##	CPB1	281	1646425
##	COL1A2	85409	690917
##	COL3A1	67614	532187
##	MT-C01	553417	607368
##	TCGA-A8-A06P-01A-11R-A00Z-07	TCGA-BH-A0HA-11A-31R-A12P-07	
##	COL1A1	840444	4847
##	FN1	857966	3381
##	CPB1	56	41
##	COL1A2	682218	6171
##	COL3A1	614561	8210
##	MT-C01	523576	268304
##	TCGA-A2-AOCO-01A-13R-A22K-07	TCGA-A2-A259-01A-11R-A16F-07	
##	COL1A1	370735	217006
##	FN1	85377	172397
##	CPB1	98985	267
##	COL1A2	203780	187944
##	COL3A1	188804	171409
##	MT-C01	626922	432706
##	TCGA-A8-A09T-01A-11R-A00Z-07	TCGA-E9-A6HE-01A-11R-A310-07	
##	COL1A1	132658	517914
##	FN1	228387	179048
##	CPB1	11	53
##	COL1A2	152715	341034
##	COL3A1	119518	235910
##	MT-C01	189806	223957
##	TCGA-A2-AOCR-01A-11R-A22K-07	TCGA-AR-A24M-01A-11R-A169-07	
##	COL1A1	311117	577020
##	FN1	40249	650403
##	CPB1	367	1585
##	COL1A2	179768	602985
##	COL3A1	145011	553508
##	MT-C01	250077	454614
##	TCGA-AC-A4ZE-01A-11R-A41B-07	TCGA-A2-A25B-01A-11R-A169-07	
##	COL1A1	75260	321594
##	FN1	17497	589831

##	CPB1	21	355
##	COL1A2	59451	320505
##	COL3A1	68094	244479
##	MT-CO1	631990	314247
##	TCGA-BH-A1FE-11B-14R-A13Q-07	TCGA-BH-A0BM-01A-11R-A056-07	
##	COL1A1	9882	215794
##	FN1	13646	76390
##	CPB1	95	2750
##	COL1A2	20209	139138
##	COL3A1	23264	113599
##	MT-CO1	670076	843812
##	TCGA-AR-AOU0-01A-11R-A109-07	TCGA-AR-A2LL-01A-11R-A180-07	
##	COL1A1	97888	251815
##	FN1	113651	160634
##	CPB1	5	6
##	COL1A2	76988	187207
##	COL3A1	64571	151044
##	MT-CO1	162402	516359
##	TCGA-B6-AOIK-01A-12R-A056-07	TCGA-A2-A0D4-01A-11R-A00Z-07	
##	COL1A1	112277	101900
##	FN1	111531	65989
##	CPB1	5653	2816243
##	COL1A2	84410	79843
##	COL3A1	77990	95726
##	MT-CO1	218022	971745
##	TCGA-E9-A1R4-01A-21R-A14D-07	TCGA-AC-A2QH-01A-11R-A18M-07	
##	COL1A1	209591	144510
##	FN1	361151	2168248
##	CPB1	4	24
##	COL1A2	167814	141694
##	COL3A1	141466	997089
##	MT-CO1	533801	39479
##	TCGA-A0-A1KQ-01A-11R-A13Q-07	TCGA-A8-A06Q-01A-11R-A034-07	
##	COL1A1	245394	33506
##	FN1	93476	38239
##	CPB1	8	50
##	COL1A2	136249	31582
##	COL3A1	96437	29786
##	MT-CO1	492708	1502717
##	TCGA-AC-A8OP-01A-11R-A36F-07	TCGA-BH-A18L-11A-42R-A12D-07	
##	COL1A1	260216	105817
##	FN1	116928	37420
##	CPB1	198	52158
##	COL1A2	143977	144013
##	COL3A1	150428	114217
##	MT-CO1	481793	517736
##	TCGA-E2-A1IG-11A-22R-A144-07	TCGA-LL-A442-01A-11R-A24H-07	
##	COL1A1	41408	57331
##	FN1	18741	74206
##	CPB1	65	1117
##	COL1A2	63261	51580
##	COL3A1	54617	52734
##	MT-CO1	499995	419874
##	TCGA-E2-A155-01A-11R-A12D-07	TCGA-C8-A134-01A-11R-A115-07	

## COL1A1	61050	68344
## FN1	121002	102618
## CPB1	85	12
## COL1A2	62762	56535
## COL3A1	73611	65684
## MT-CO1	588532	565477
## TCGA-BH-AOBS-11A-11R-A12P-07	TCGA-AR-A2LE-01A-11R-A180-07	
## COL1A1	4739	115926
## FN1	4825	118831
## CPB1	2789	27
## COL1A2	10818	99550
## COL3A1	12099	91010
## MT-CO1	329764	910204
## TCGA-BH-A28Q-01A-11R-A16F-07	TCGA-AR-A252-01A-11R-A169-07	
## COL1A1	556615	141583
## FN1	412874	247466
## CPB1	237	1332474
## COL1A2	484818	143272
## COL3A1	525597	151179
## MT-CO1	566935	631400
## TCGA-E2-A572-01A-13R-A310-07	TCGA-BH-AODI-01A-21R-A12P-07	
## COL1A1	20718	1646622
## FN1	6598	784443
## CPB1	22354	180
## COL1A2	11784	1089575
## COL3A1	7432	1056986
## MT-CO1	124522	493860
## TCGA-E2-A1IO-01A-11R-A144-07	TCGA-AN-AOFD-01A-11R-A034-07	
## COL1A1	1407831	923455
## FN1	556272	411054
## CPB1	453	200523
## COL1A2	1024333	559547
## COL3A1	1064767	437488
## MT-CO1	293985	558804
## TCGA-E9-A1NA-01A-11R-A144-07	TCGA-A1-AOSN-01A-11R-A144-07	
## COL1A1	1023228	891046
## FN1	527514	436343
## CPB1	2482	11
## COL1A2	927337	680201
## COL3A1	745626	657024
## MT-CO1	280005	426655
## TCGA-D8-A1XL-01A-11R-A14M-07	TCGA-A0-A030-01A-11R-A00Z-07	
## COL1A1	203577	85658
## FN1	95891	63023
## CPB1	26	10
## COL1A2	161204	64996
## COL3A1	112540	61789
## MT-CO1	235603	126580
## TCGA-B6-AOIP-01A-11R-A034-07	TCGA-BH-AOW5-01A-11R-A109-07	
## COL1A1	321030	482618
## FN1	153666	353469
## CPB1	19897	30
## COL1A2	232447	376833
## COL3A1	162776	288261

##	MT-C01	731508	448506
##	TCGA-B6-AOIM-01A-11R-A034-07	TCGA-A0-A128-01A-11R-A10J-07	
##	COL1A1	696157	124098
##	FN1	595624	199775
##	CPB1	14215	29
##	COL1A2	582108	97382
##	COL3A1	538452	85979
##	MT-C01	709876	319691
##	TCGA-A0-A124-01A-11R-A10J-07	TCGA-BH-A0BZ-01A-31R-A12P-07	
##	COL1A1	69063	615769
##	FN1	59586	719554
##	CPB1	16	7
##	COL1A2	44950	514189
##	COL3A1	42118	459598
##	MT-C01	223565	422243
##	TCGA-OL-A66K-01A-11R-A29R-07	TCGA-A2-A0ET-01A-31R-A034-07	
##	COL1A1	669208	31055
##	FN1	158440	44093
##	CPB1	81	91582
##	COL1A2	443317	24051
##	COL3A1	358964	24109
##	MT-C01	255960	553421
##	TCGA-BH-A1F5-01A-12R-A13Q-07	TCGA-A2-A0YT-01A-11R-A109-07	
##	COL1A1	156573	79814
##	FN1	177314	173353
##	CPB1	24	928
##	COL1A2	150629	78039
##	COL3A1	124172	72125
##	MT-C01	1084707	786020
##	TCGA-A2-A0EO-01A-11R-A034-07	TCGA-D8-A27G-01A-11R-A16F-07	
##	COL1A1	618593	960246
##	FN1	469553	517470
##	CPB1	19	49
##	COL1A2	613049	781976
##	COL3A1	658578	837444
##	MT-C01	871798	550257
##	TCGA-E9-A5UP-01A-11R-A28M-07	TCGA-E9-A1NF-11A-73R-A14D-07	
##	COL1A1	16038	101035
##	FN1	16577	100965
##	CPB1	7	57
##	COL1A2	8689	144916
##	COL3A1	6165	162824
##	MT-C01	632412	429215
##	TCGA-E9-A1NF-01A-11R-A14D-07	TCGA-B6-A0RT-01A-21R-A084-07	
##	COL1A1	1362914	157404
##	FN1	811646	159373
##	CPB1	53	64
##	COL1A2	1122411	110836
##	COL3A1	846323	109610
##	MT-C01	425212	488170
##	TCGA-A1-A0SP-01A-11R-A084-07	TCGA-C8-A278-01A-11R-A169-07	
##	COL1A1	179867	456802
##	FN1	130353	497831
##	CPB1	1002	32

##	COL1A2	119001	438384
##	COL3A1	130989	368083
##	MT-CO1	362041	709590
##	TCGA-AC-A2BM-01A-11R-A21T-07	TCGA-A2-AOCQ-01A-21R-A034-07	
##	COL1A1	157314	193850
##	FN1	115784	62791
##	CPB1	3439	106
##	COL1A2	145264	105355
##	COL3A1	144769	67166
##	MT-CO1	434818	1249943
##	TCGA-E9-A1QZ-01A-21R-A169-07	TCGA-BH-A6R8-01A-21R-A33J-07	
##	COL1A1	140170	345925
##	FN1	192321	94363
##	CPB1	16455	25311
##	COL1A2	146653	219187
##	COL3A1	122921	165058
##	MT-CO1	688156	153719
##	TCGA-AR-A1AS-01A-11R-A12P-07	TCGA-D8-A27K-01A-11R-A16F-07	
##	COL1A1	293177	371048
##	FN1	382602	314783
##	CPB1	0	1393
##	COL1A2	210130	321380
##	COL3A1	168131	320252
##	MT-CO1	219217	439790
##	TCGA-LD-A9QF-01A-32R-A41B-07	TCGA-BH-A0H9-01A-11R-A056-07	
##	COL1A1	822984	220164
##	FN1	245208	214437
##	CPB1	145	54354
##	COL1A2	474456	174638
##	COL3A1	451092	168653
##	MT-CO1	489505	231053
##	TCGA-E9-A1RI-01A-11R-A169-07	TCGA-BH-A209-11A-42R-A157-07	
##	COL1A1	899195	58734
##	FN1	741633	68178
##	CPB1	657	234
##	COL1A2	965485	103178
##	COL3A1	854166	82010
##	MT-CO1	876236	711560
##	TCGA-BH-AODP-11A-12R-A089-07	TCGA-BH-AOBQ-11A-33R-A115-07	
##	COL1A1	9166	48047
##	FN1	17964	21048
##	CPB1	12703	665
##	COL1A2	28726	54401
##	COL3A1	14446	77058
##	MT-CO1	398858	433174
##	TCGA-W8-A86G-01A-21R-A36F-07	TCGA-E2-A574-01A-11R-A29R-07	
##	COL1A1	322623	81541
##	FN1	80114	43850
##	CPB1	60	8
##	COL1A2	179709	54409
##	COL3A1	141505	43672
##	MT-CO1	216078	114965
##	TCGA-A1-AOSG-01A-11R-A144-07	TCGA-A0-A0J9-01A-11R-A034-07	
##	COL1A1	482058	471704

##	FN1	368714	193891
##	CPB1	13	64
##	COL1A2	363479	220347
##	COL3A1	295185	179696
##	MT-CO1	525017	3169341
##	TCGA-A8-A06N-01A-11R-A00Z-07	TCGA-B6-A0WY-01A-11R-A109-07	
##	COL1A1	10854	290511
##	FN1	29854	141788
##	CPB1	932	42
##	COL1A2	10054	236827
##	COL3A1	7512	214755
##	MT-CO1	756126	429566
##	TCGA-EW-A424-01A-11R-A24H-07	TCGA-AR-A1AL-01A-21R-A12P-07	
##	COL1A1	824307	449853
##	FN1	193133	218982
##	CPB1	18	243
##	COL1A2	485761	396144
##	COL3A1	439375	455294
##	MT-CO1	203633	356081
##	TCGA-D8-A1X6-01A-11R-A14M-07	TCGA-E2-A1L6-01A-11R-A13Q-07	
##	COL1A1	117441	307940
##	FN1	132408	190557
##	CPB1	208	24291
##	COL1A2	97712	232201
##	COL3A1	85929	244473
##	MT-CO1	502386	323578
##	TCGA-E2-A15S-01A-11R-A115-07	TCGA-A2-A04Q-01A-21R-A034-07	
##	COL1A1	74433	226393
##	FN1	34032	219684
##	CPB1	4663	26
##	COL1A2	62503	131520
##	COL3A1	103711	130920
##	MT-CO1	797389	633085
##	TCGA-GM-A3XG-01A-31R-A24H-07	TCGA-A8-A0A9-01A-11R-A00Z-07	
##	COL1A1	74171	583858
##	FN1	21264	493005
##	CPB1	2568	23
##	COL1A2	46296	389457
##	COL3A1	42164	335781
##	MT-CO1	450329	555251
##	TCGA-Z7-A8R5-01A-42R-A41B-07	TCGA-C8-A12T-01A-11R-A115-07	
##	COL1A1	130799	1069622
##	FN1	23601	621337
##	CPB1	5463	35
##	COL1A2	90821	777208
##	COL3A1	76945	661255
##	MT-CO1	310692	203708
##	TCGA-OL-A97C-01A-32R-A41B-07	TCGA-A8-A06X-01A-21R-A00Z-07	
##	COL1A1	571893	147649
##	FN1	247538	100630
##	CPB1	121	1
##	COL1A2	814278	123699
##	COL3A1	540128	120257
##	MT-CO1	171448	324695

##	TCGA-E2-A1IF-01A-11R-A144-07	TCGA-AR-A0U4-01A-11R-A109-07
##	COL1A1	929383 190212
##	FN1	248926 317287
##	CPB1	350930 3
##	COL1A2	457229 135285
##	COL3A1	374498 156423
##	MT-CO1	571286 175813
##	TCGA-BH-AODT-01A-21R-A12D-07	TCGA-A7-A3RF-01A-11R-A22K-07
##	COL1A1	382759 189900
##	FN1	287884 142359
##	CPB1	114 14252
##	COL1A2	275596 117194
##	COL3A1	265663 92553
##	MT-CO1	434203 591286
##	TCGA-A2-AOER-01A-21R-A034-07	TCGA-A0-A12A-01A-21R-A115-07
##	COL1A1	102293 371206
##	FN1	149191 219429
##	CPB1	4281962 1559
##	COL1A2	110615 245945
##	COL3A1	123126 229023
##	MT-CO1	714498 418530
##	TCGA-AR-A5QM-01A-11R-A27Q-07	TCGA-B6-AORG-01A-11R-A056-07
##	COL1A1	478987 325079
##	FN1	111310 180588
##	CPB1	504 13124
##	COL1A2	305630 203034
##	COL3A1	346506 217865
##	MT-CO1	405581 541615
##	TCGA-PL-A8LY-01A-11R-A41B-07	TCGA-A2-A4S2-01A-12R-A266-07
##	COL1A1	3940 410366
##	FN1	1572 153073
##	CPB1	431 884
##	COL1A2	2525 226370
##	COL3A1	3941 211639
##	MT-CO1	288730 321657
##	TCGA-A7-AOCH-01A-21R-A00Z-07	TCGA-A7-AOCH-11A-32R-A089-07
##	COL1A1	433407 69591
##	FN1	312794 56797
##	CPB1	6826 2370
##	COL1A2	373804 120294
##	COL3A1	276155 77415
##	MT-CO1	417144 729104
##	TCGA-BH-A1EW-11B-33R-A137-07	TCGA-E9-A24A-01A-11R-A169-07
##	COL1A1	29700 497586
##	FN1	56435 661067
##	CPB1	29 527
##	COL1A2	57459 479934
##	COL3A1	64699 369894
##	MT-CO1	625757 550286
##	TCGA-AR-A24W-01A-11R-A169-07	TCGA-E9-A1N4-01A-11R-A14M-07
##	COL1A1	296267 690454
##	FN1	165917 513947
##	CPB1	25262 28
##	COL1A2	260850 622725

##	COL3A1	278534	631688
##	MT-C01	405425	498073
##	TCGA-A1-AOSO-01A-22R-A084-07	TCGA-BH-A18N-01A-11R-A12D-07	
##	COL1A1	55518	163658
##	FN1	66157	140519
##	CPB1	221	5132
##	COL1A2	41083	135293
##	COL3A1	46591	109615
##	MT-C01	625285	464354
##	TCGA-A7-AOCG-01A-12R-A056-07	TCGA-D8-A1XO-01A-11R-A14M-07	
##	COL1A1	542311	951248
##	FN1	209883	774879
##	CPB1	47	12
##	COL1A2	480709	911523
##	COL3A1	591449	671967
##	MT-C01	547341	281900
##	TCGA-D8-A27E-01A-11R-A16F-07	TCGA-BH-A18V-11A-52R-A12D-07	
##	COL1A1	65026	81552
##	FN1	121603	12404
##	CPB1	1075336	1913
##	COL1A2	55826	114372
##	COL3A1	37748	94766
##	MT-C01	774687	270574
##	TCGA-BH-AOE6-01A-11R-A034-07	TCGA-C8-A12W-01A-11R-A115-07	
##	COL1A1	169871	588973
##	FN1	37058	704770
##	CPB1	304	5
##	COL1A2	82971	430223
##	COL3A1	65902	425192
##	MT-C01	757749	236786
##	TCGA-BH-A18V-06A-11R-A213-07	TCGA-AR-A2LR-01A-12R-A18M-07	
##	COL1A1	204567	244976
##	FN1	484191	211049
##	CPB1	8	14
##	COL1A2	100596	165740
##	COL3A1	98225	139318
##	MT-C01	455317	225492
##	TCGA-A8-A091-01A-11R-A00Z-07	TCGA-GI-A2C8-01A-11R-A16F-07	
##	COL1A1	278223	1282922
##	FN1	283901	428087
##	CPB1	4200461	169484
##	COL1A2	215466	875215
##	COL3A1	205891	817900
##	MT-C01	899441	822980
##	TCGA-D8-A146-01A-31R-A115-07	TCGA-A8-A07I-01A-11R-A00Z-07	
##	COL1A1	691232	483278
##	FN1	566343	527766
##	CPB1	892634	8
##	COL1A2	537261	387219
##	COL3A1	437670	338237
##	MT-C01	378609	266029
##	TCGA-A7-AODA-01A-31R-A115-07	TCGA-EW-A6SD-01A-12R-A33J-07	
##	COL1A1	615205	329644
##	FN1	581194	537846

##	CPB1	15	124
##	COL1A2	442809	231615
##	COL3A1	324939	218557
##	MT-CO1	859195	348416
##	TCGA-D8-A1XR-01A-11R-A14M-07	TCGA-E2-A1LE-01A-12R-A19W-07	
##	COL1A1	350644	795194
##	FN1	354871	324421
##	CPB1	8	2
##	COL1A2	296669	462621
##	COL3A1	236547	436806
##	MT-CO1	588871	649260
##	TCGA-E2-A1IG-01A-11R-A144-07	TCGA-3C-AAAU-01A-11R-A41B-07	
##	COL1A1	616507	170142
##	FN1	730696	137676
##	CPB1	2077	222179
##	COL1A2	493057	110463
##	COL3A1	360662	96241
##	MT-CO1	473494	675231
##	TCGA-E9-A1R2-01A-11R-A14D-07	TCGA-D8-A1XB-01A-11R-A14D-07	
##	COL1A1	286562	908220
##	FN1	162801	636896
##	CPB1	11	764
##	COL1A2	211443	751837
##	COL3A1	198560	708439
##	MT-CO1	362979	341740
##	TCGA-OL-A5RY-01A-21R-A28M-07	TCGA-LL-A9Q3-01A-11R-A41B-07	
##	COL1A1	314880	302077
##	FN1	266831	71986
##	CPB1	335	114
##	COL1A2	218328	166683
##	COL3A1	255868	121956
##	MT-CO1	197224	387510
##	TCGA-D8-A27R-01A-11R-A16F-07	TCGA-BH-A1FD-01A-11R-A13Q-07	
##	COL1A1	269814	357857
##	FN1	205266	477577
##	CPB1	1012	67
##	COL1A2	225150	305051
##	COL3A1	234483	270228
##	MT-CO1	561484	589406
##	TCGA-BH-A1FD-11B-21R-A13Q-07	TCGA-E9-A22G-01A-11R-A157-07	
##	COL1A1	16008	88389
##	FN1	23155	49580
##	CPB1	1570	20
##	COL1A2	33294	71335
##	COL3A1	23906	63997
##	MT-CO1	953812	426884
##	TCGA-AN-AOAK-01A-21R-A00Z-07	TCGA-D8-A1JJ-01A-31R-A14M-07	
##	COL1A1	95575	575546
##	FN1	102022	996095
##	CPB1	748	230
##	COL1A2	70345	563906
##	COL3A1	54796	497360
##	MT-CO1	186144	459092
##	TCGA-A2-A4RW-01A-21R-A266-07	TCGA-A8-A07S-01A-11R-A034-07	

## COL1A1	493006	342446
## FN1	119462	243970
## CPB1	708	64
## COL1A2	262006	271814
## COL3A1	248127	209566
## MT-CO1	320527	359403
## TCGA-AR-AOTS-01A-11R-A115-07	TCGA-D8-A1JP-01A-11R-A13Q-07	
## COL1A1	246749	436718
## FN1	145997	293436
## CPB1	4	128
## COL1A2	175684	342511
## COL3A1	199404	238237
## MT-CO1	361209	321405
## TCGA-AR-A24T-01A-11R-A169-07	TCGA-B6-A1KI-01A-11R-A14M-07	
## COL1A1	971216	436041
## FN1	577136	255175
## CPB1	187	55
## COL1A2	865290	366728
## COL3A1	938357	322368
## MT-CO1	496193	942638
## TCGA-S3-AA11-01A-31R-A41B-07	TCGA-A1-AOSB-01A-11R-A144-07	
## COL1A1	206648	40431
## FN1	146482	24984
## CPB1	78	233
## COL1A2	125284	53384
## COL3A1	86388	97289
## MT-CO1	513869	231129
## TCGA-BH-AOEI-01A-11R-A115-07	TCGA-AQ-A7U7-01A-22R-A352-07	
## COL1A1	1075639	1002421
## FN1	942192	180881
## CPB1	143	55372
## COL1A2	896628	630495
## COL3A1	649439	601197
## MT-CO1	352999	242350
## TCGA-AC-A2FF-11A-13R-A17B-07	TCGA-AC-A2FG-01A-11R-A17B-07	
## COL1A1	328151	179808
## FN1	74705	47414
## CPB1	85	551
## COL1A2	250430	174249
## COL3A1	428498	214035
## MT-CO1	602920	475208
## TCGA-A8-AOA2-01A-11R-A034-07	TCGA-D8-A1JG-01B-11R-A13Q-07	
## COL1A1	634236	436356
## FN1	330788	417774
## CPB1	107	4
## COL1A2	526298	407483
## COL3A1	381111	393744
## MT-CO1	1006223	627708
## TCGA-BH-AOBZ-11A-61R-A12P-07	TCGA-GM-A2DF-01A-11R-A180-07	
## COL1A1	63099	143735
## FN1	46441	96870
## CPB1	6533	0
## COL1A2	108860	81366
## COL3A1	91839	65751

##	MT-C01	892763	1292695
##	TCGA-BH-A1FL-01A-11R-A13Q-07	TCGA-A8-A06Y-01A-21R-A00Z-07	
##	COL1A1	336198	79933
##	FN1	157748	115492
##	CPB1	67	20145
##	COL1A2	241264	59522
##	COL3A1	211621	40527
##	MT-C01	326049	760540
##	TCGA-BH-A0AZ-01A-21R-A12P-07	TCGA-AR-A0TT-01A-31R-A084-07	
##	COL1A1	1583600	245289
##	FN1	910362	429735
##	CPB1	272	294
##	COL1A2	1198077	209849
##	COL3A1	929433	181575
##	MT-C01	499538	247765
##	TCGA-C8-A137-01A-11R-A115-07	TCGA-E9-A22H-01A-11R-A157-07	
##	COL1A1	352846	129823
##	FN1	386479	136784
##	CPB1	0	30464
##	COL1A2	293815	111115
##	COL3A1	304202	104361
##	MT-C01	462639	456069
##	TCGA-A2-A3XV-01A-21R-A239-07	TCGA-B6-A0I6-01A-11R-A034-07	
##	COL1A1	242599	369961
##	FN1	48515	362562
##	CPB1	21	41
##	COL1A2	132664	273996
##	COL3A1	66860	209732
##	MT-C01	245218	278144
##	TCGA-A7-A0D9-11A-53R-A089-07	TCGA-B6-A0WS-01A-11R-A115-07	
##	COL1A1	42372	574463
##	FN1	40903	290028
##	CPB1	290	56
##	COL1A2	84347	444245
##	COL3A1	95260	380553
##	MT-C01	853154	363593
##	TCGA-B6-A0I5-01A-11R-A034-07	TCGA-AR-A250-01A-31R-A169-07	
##	COL1A1	249098	1072437
##	FN1	111291	1123562
##	CPB1	67	18
##	COL1A2	130596	934515
##	COL3A1	87984	752319
##	MT-C01	1300525	779840
##	TCGA-BH-A1EU-11A-23R-A137-07	TCGA-S3-A6ZG-01A-22R-A32P-07	
##	COL1A1	12698	188651
##	FN1	17688	179984
##	CPB1	75	46
##	COL1A2	29754	153019
##	COL3A1	19351	138085
##	MT-C01	451337	381263
##	TCGA-BH-A1EU-01A-11R-A137-07	TCGA-AR-A1AW-01A-21R-A12P-07	
##	COL1A1	631733	182027
##	FN1	422590	248870
##	CPB1	23995	4

##	COL1A2	692207	132847
##	COL3A1	818646	129219
##	MT-CO1	828981	315752
##	TCGA-EW-A1P5-01A-11R-A144-07	TCGA-AN-AOFJ-01A-11R-A00Z-07	
##	COL1A1	258859	91177
##	FN1	189935	54818
##	CPB1	6854	1
##	COL1A2	197504	71544
##	COL3A1	182162	62494
##	MT-CO1	315006	99255
##	TCGA-A2-AOEW-01A-21R-A115-07	TCGA-BH-AOB8-01A-21R-A056-07	
##	COL1A1	642958	186648
##	FN1	231105	226034
##	CPB1	96	64345
##	COL1A2	482763	138491
##	COL3A1	515797	112311
##	MT-CO1	385191	690030
##	TCGA-BH-AOB8-11A-41R-A089-07	TCGA-BH-A18P-01A-11R-A12D-07	
##	COL1A1	46309	338713
##	FN1	40107	362682
##	CPB1	82	15
##	COL1A2	80525	315987
##	COL3A1	82714	283087
##	MT-CO1	1083840	259082
##	TCGA-AC-A6N0-01A-12R-A33J-07	TCGA-E2-A15H-01A-11R-A12D-07	
##	COL1A1	314447	517307
##	FN1	134579	429611
##	CPB1	46	168746
##	COL1A2	193962	315084
##	COL3A1	173859	231230
##	MT-CO1	345257	532919
##	TCGA-BH-AOE7-01A-11R-A034-07	TCGA-E9-A1NG-11A-52R-A14M-07	
##	COL1A1	289472	51786
##	FN1	301104	64769
##	CPB1	45	3447
##	COL1A2	298234	94029
##	COL3A1	225098	89782
##	MT-CO1	548126	557779
##	TCGA-AO-AOJL-01A-11R-A056-07	TCGA-BH-AOD0-11A-22R-A12D-07	
##	COL1A1	134641	13505
##	FN1	69960	11459
##	CPB1	2571	11226
##	COL1A2	70879	24337
##	COL3A1	57174	19874
##	MT-CO1	1826610	459288
##	TCGA-AC-A80S-01A-12R-A41B-07	TCGA-E2-A14Z-01A-11R-A115-07	
##	COL1A1	843513	247249
##	FN1	114525	249550
##	CPB1	956	1125
##	COL1A2	500250	164710
##	COL3A1	535360	132036
##	MT-CO1	989070	846526
##	TCGA-HN-A2NL-01A-11R-A18M-07	TCGA-EW-A1PG-01A-11R-A144-07	
##	COL1A1	78424	308621

##	FN1	37211	301829
##	CPB1	22	62
##	COL1A2	48728	225908
##	COL3A1	58592	216493
##	MT-CO1	283837	355874
##	TCGA-D8-A3Z5-01A-41R-A24H-07	TCGA-AR-A1AH-01A-11R-A12D-07	
##	COL1A1	103655	145427
##	FN1	16305	136223
##	CPB1	468	2
##	COL1A2	64878	123646
##	COL3A1	73407	99876
##	MT-CO1	449192	341136
##	TCGA-A8-A075-01A-11R-A084-07	TCGA-BH-A0AV-01A-31R-A115-07	
##	COL1A1	499389	981279
##	FN1	577128	1230217
##	CPB1	1307	12
##	COL1A2	346524	775338
##	COL3A1	365668	798504
##	MT-CO1	536786	284370
##	TCGA-BH-A1FM-11B-23R-A13Q-07	TCGA-A2-A0YJ-01A-11R-A109-07	
##	COL1A1	33881	71589
##	FN1	33011	169254
##	CPB1	1017	1
##	COL1A2	70650	55050
##	COL3A1	44758	48113
##	MT-CO1	839946	328392
##	TCGA-B6-A402-01A-11R-A239-07	TCGA-BH-A0DE-01A-11R-A115-07	
##	COL1A1	125104	348394
##	FN1	98705	244480
##	CPB1	2	3647
##	COL1A2	77727	235337
##	COL3A1	64466	178146
##	MT-CO1	583053	986230
##	TCGA-A2-A1G1-01A-21R-A13Q-07	TCGA-OL-A5S0-01A-11R-A28M-07	
##	COL1A1	354370	206134
##	FN1	678084	102291
##	CPB1	4	4
##	COL1A2	339667	137591
##	COL3A1	309685	131345
##	MT-CO1	535135	185107
##	TCGA-BH-A0E9-01B-11R-A115-07	TCGA-BH-A0GY-01A-11R-A056-07	
##	COL1A1	126104	234455
##	FN1	68170	323734
##	CPB1	178	83
##	COL1A2	97871	189266
##	COL3A1	115297	190691
##	MT-CO1	495919	525559
##	TCGA-GM-A2DH-01A-11R-A180-07	TCGA-BH-A1EO-11A-31R-A137-07	
##	COL1A1	149541	646024
##	FN1	362516	147593
##	CPB1	1	1494
##	COL1A2	120572	734747
##	COL3A1	110637	989651
##	MT-CO1	333837	1182158

##	TCGA-A2-AOT7-01A-21R-A084-07	TCGA-B6-A0I1-01A-11R-A21T-07
##	COL1A1	859691 38865
##	FN1	588201 22113
##	CPB1	2606 23
##	COL1A2	626783 20555
##	COL3A1	632232 13630
##	MT-CO1	418573 1003302
##	TCGA-BH-AOB9-01A-11R-A056-07	TCGA-A2-AOCK-01A-11R-A22K-07
##	COL1A1	41420 244037
##	FN1	63903 50931
##	CPB1	51 794
##	COL1A2	35519 158790
##	COL3A1	38968 158922
##	MT-CO1	337617 767235
##	TCGA-AO-A12G-01A-11R-A10J-07	TCGA-E2-A1L9-01A-11R-A13Q-07
##	COL1A1	350266 1138569
##	FN1	198061 1261551
##	CPB1	32075 10
##	COL1A2	257869 876908
##	COL3A1	220655 754019
##	MT-CO1	1203113 596276
##	TCGA-WT-AB44-01A-11R-A41B-07	TCGA-D8-A1XF-01A-11R-A14D-07
##	COL1A1	357612 65460
##	FN1	92710 113517
##	CPB1	11 981143
##	COL1A2	169414 61706
##	COL3A1	90092 56479
##	MT-CO1	987670 384602
##	TCGA-BH-A8G0-01A-11R-A352-07	TCGA-AR-A251-01A-12R-A169-07
##	COL1A1	765735 201684
##	FN1	150664 438479
##	CPB1	287 15
##	COL1A2	479857 195998
##	COL3A1	460732 241351
##	MT-CO1	296266 624593
##	TCGA-D8-A13Z-01A-11R-A115-07	TCGA-AR-AOTQ-01A-11R-A084-07
##	COL1A1	560906 218362
##	FN1	566036 295557
##	CPB1	4 149
##	COL1A2	407090 168301
##	COL3A1	365297 175590
##	MT-CO1	370001 583303
##	TCGA-A2-A4RY-01A-31R-A266-07	TCGA-A7-A13E-01B-06R-A277-07
##	COL1A1	299215 28631
##	FN1	70570 78711
##	CPB1	132 10
##	COL1A2	175524 41638
##	COL3A1	194727 56006
##	MT-CO1	266359 682087
##	TCGA-A7-A13E-11A-61R-A12P-07	TCGA-D8-A1X7-01A-11R-A14M-07
##	COL1A1	143966 228525
##	FN1	64423 389746
##	CPB1	286 30
##	COL1A2	142106 241259

##	COL3A1	170386	215990
##	MT-CO1	715328	505062
##	TCGA-E9-A1R6-01A-11R-A14D-07	TCGA-A2-A04P-01A-31R-A034-07	
##	COL1A1	302803	204518
##	FN1	309865	86249
##	CPB1	144320	5
##	COL1A2	224192	109482
##	COL3A1	156353	82379
##	MT-CO1	352918	2814340
##	TCGA-E2-A570-01A-11R-A29R-07	TCGA-AC-A80R-01A-21R-A41B-07	
##	COL1A1	795438	23222
##	FN1	312723	37565
##	CPB1	18	176
##	COL1A2	542664	13513
##	COL3A1	375112	8740
##	MT-CO1	457527	268077
##	TCGA-A8-A09W-01A-11R-A00Z-07	TCGA-BH-A0DH-11A-31R-A089-07	
##	COL1A1	157404	105027
##	FN1	191523	22694
##	CPB1	29	28370
##	COL1A2	176643	103990
##	COL3A1	160287	96348
##	MT-CO1	363884	735992
##	TCGA-A2-A1FV-01A-11R-A13Q-07	TCGA-S3-AAOZ-01A-11R-A41B-07	
##	COL1A1	129400	114622
##	FN1	119021	72747
##	CPB1	34	22
##	COL1A2	123564	62504
##	COL3A1	124541	66410
##	MT-CO1	1438151	133344
##	TCGA-D8-A1JT-01A-31R-A13Q-07	TCGA-LL-A6FQ-01A-11R-A310-07	
##	COL1A1	43172	257966
##	FN1	101380	75006
##	CPB1	13	485
##	COL1A2	49549	121610
##	COL3A1	33729	79847
##	MT-CO1	933877	152973
##	TCGA-BH-A0AY-11A-23R-A089-07	TCGA-E2-A1IU-01A-11R-A14D-07	
##	COL1A1	33140	73445
##	FN1	31883	59239
##	CPB1	7673	267791
##	COL1A2	60981	64253
##	COL3A1	50724	49837
##	MT-CO1	383587	194154
##	TCGA-E9-A3X8-01A-31R-A22U-07	TCGA-AC-A3QQ-01B-06R-A220-07	
##	COL1A1	554859	143717
##	FN1	97869	111385
##	CPB1	21	2062
##	COL1A2	325524	43159
##	COL3A1	181701	42203
##	MT-CO1	213232	551901
##	TCGA-D8-A1XK-01A-21R-A14M-07	TCGA-BH-A0AU-01A-11R-A12P-07	
##	COL1A1	46922	659348
##	FN1	287194	522268

##	CPB1	2	44
##	COL1A2	40494	519855
##	COL3A1	54517	459441
##	MT-CO1	114791	435252
##	TCGA-BH-AOBD-01A-11R-A034-07	TCGA-A0-A0J5-01A-11R-A034-07	
##	COL1A1	463179	2223512
##	FN1	255371	975067
##	CPB1	9	353968
##	COL1A2	343880	1675217
##	COL3A1	354710	1581640
##	MT-CO1	579892	771741
##	TCGA-PL-A8LV-01A-21R-A41B-07	TCGA-EW-A2FW-01A-11R-A17B-07	
##	COL1A1	215270	219583
##	FN1	129057	145139
##	CPB1	7	12214
##	COL1A2	117410	152638
##	COL3A1	77513	147015
##	MT-CO1	1233358	266060
##	TCGA-C8-A1HI-01A-11R-A137-07	TCGA-E9-A247-01A-11R-A169-07	
##	COL1A1	217172	244548
##	FN1	108036	304489
##	CPB1	5353	334
##	COL1A2	193415	189412
##	COL3A1	193308	143968
##	MT-CO1	838366	805722
##	TCGA-BH-A280-01A-11R-A22K-07	TCGA-OL-A6VQ-01A-12R-A41B-07	
##	COL1A1	644736	234026
##	FN1	101249	81002
##	CPB1	394	6
##	COL1A2	392627	149370
##	COL3A1	450509	165667
##	MT-CO1	444251	112875
##	TCGA-B6-A0WV-01A-11R-A109-07	TCGA-AR-A2LM-01A-11R-A180-07	
##	COL1A1	158296	953522
##	FN1	178535	528928
##	CPB1	46871	1760
##	COL1A2	122235	843154
##	COL3A1	106039	881414
##	MT-CO1	249211	564316
##	TCGA-A2-A0EP-01A-52R-A22U-07	TCGA-AR-A0TW-01A-11R-A084-07	
##	COL1A1	1268147	166132
##	FN1	347610	191901
##	CPB1	24	667
##	COL1A2	672911	122774
##	COL3A1	808264	105707
##	MT-CO1	701482	712094
##	TCGA-3C-AALJ-01A-31R-A41B-07	TCGA-B6-A0RM-01A-11R-A084-07	
##	COL1A1	209578	97929
##	FN1	135398	366535
##	CPB1	2557	122
##	COL1A2	115337	71073
##	COL3A1	83529	72327
##	MT-CO1	278549	505133
##	TCGA-A0-A0JD-01A-11R-A056-07	TCGA-E2-A15L-01A-11R-A12D-07	

## COL1A1	60883	824519
## FN1	78168	1035281
## CPB1	1066	99
## COL1A2	51561	743294
## COL3A1	52244	501172
## MT-CO1	486209	258435
## TCGA-A7-A6VV-01A-22R-A33J-07	TCGA-B6-A0IA-01A-11R-A034-07	
## COL1A1	939390	118166
## FN1	732073	42570
## CPB1	32	22
## COL1A2	605780	60076
## COL3A1	441615	46735
## MT-CO1	277304	889880
## TCGA-BH-A1FE-06A-11R-A213-07	TCGA-A8-A07F-01A-11R-A00Z-07	
## COL1A1	1287151	389340
## FN1	269474	336382
## CPB1	164	90
## COL1A2	914827	366453
## COL3A1	681350	367074
## MT-CO1	511143	313533
## TCGA-E2-A15T-01A-11R-A115-07	TCGA-C8-A12Y-01A-11R-A12D-07	
## COL1A1	73811	531624
## FN1	99154	177827
## CPB1	2	12670
## COL1A2	58387	313519
## COL3A1	51351	237478
## MT-CO1	651229	540859
## TCGA-EW-A1PC-01B-11R-A21T-07	TCGA-A2-A3XT-01A-11R-A22U-07	
## COL1A1	512997	170682
## FN1	545103	132101
## CPB1	469	5
## COL1A2	437805	101454
## COL3A1	367722	115874
## MT-CO1	308464	661690
## TCGA-C8-A138-01A-11R-A115-07	TCGA-BH-A0B6-01A-11R-A19W-07	
## COL1A1	428822	567099
## FN1	198501	92127
## CPB1	7951	21251
## COL1A2	302110	322022
## COL3A1	249740	358182
## MT-CO1	1270480	259067
## TCGA-BH-A1FB-01A-11R-A13Q-07	TCGA-C8-A1HN-01A-11R-A137-07	
## COL1A1	603967	214800
## FN1	491631	259774
## CPB1	368	110
## COL1A2	544564	182281
## COL3A1	530620	254051
## MT-CO1	478586	525633
## TCGA-BH-A1FB-11A-33R-A13Q-07	TCGA-A8-A084-01A-21R-A00Z-07	
## COL1A1	9499	591484
## FN1	7604	457005
## CPB1	64	8
## COL1A2	20926	444814
## COL3A1	12655	311659

##	MT-C01	369452	511835
##	TCGA-D8-A1Y3-01A-11R-A157-07	TCGA-BH-A5J0-01A-11R-A27Q-07	
##	COL1A1	166655	600869
##	FN1	64911	244421
##	CPB1	11	14
##	COL1A2	119048	354713
##	COL3A1	99627	275087
##	MT-C01	319684	475954
##	TCGA-D8-A1XD-01A-11R-A14D-07	TCGA-BH-A0HL-01A-11R-A10U-07	
##	COL1A1	482354	133019
##	FN1	475002	167535
##	CPB1	0	1395737
##	COL1A2	438196	84957
##	COL3A1	370432	58630
##	MT-C01	534910	495434
##	TCGA-AR-AOU3-01A-11R-A109-07	TCGA-B6-A0IE-01A-11R-A034-07	
##	COL1A1	66971	230822
##	FN1	95567	36448
##	CPB1	228	50813
##	COL1A2	52115	132122
##	COL3A1	50290	145626
##	MT-C01	240941	997246
##	TCGA-A0-A0J8-01A-21R-A034-07	TCGA-A7-A6VX-01A-12R-A33J-07	
##	COL1A1	734673	836336
##	FN1	605566	710298
##	CPB1	46	1620
##	COL1A2	519278	590224
##	COL3A1	359671	490374
##	MT-C01	568158	145691
##	TCGA-AC-A23E-01A-11R-A157-07	TCGA-AC-A3W7-01A-11R-A22K-07	
##	COL1A1	1066492	1236620
##	FN1	2213779	317175
##	CPB1	818	473
##	COL1A2	1136987	644355
##	COL3A1	921561	535511
##	MT-C01	380299	423967
##	TCGA-BH-A0BF-01A-21R-A12P-07	TCGA-BH-A0H9-11A-22R-A466-07	
##	COL1A1	1428569	23056
##	FN1	1203005	24662
##	CPB1	15	136
##	COL1A2	1123874	42715
##	COL3A1	984293	29837
##	MT-C01	306147	437001
##	TCGA-A8-A09B-01A-11R-A00Z-07	TCGA-EW-A1P8-01A-11R-A144-07	
##	COL1A1	511688	164516
##	FN1	254808	123113
##	CPB1	682665	27
##	COL1A2	344942	133054
##	COL3A1	301854	104417
##	MT-C01	1040594	342001
##	TCGA-BH-A0GZ-01A-11R-A056-07	TCGA-D8-A1XC-01A-11R-A14D-07	
##	COL1A1	470231	15459
##	FN1	239266	46089
##	CPB1	287	213924

##	COL1A2	386666	16674
##	COL3A1	266750	12436
##	MT-CO1	1522743	477310
##	TCGA-A2-A3KD-01A-12R-A213-07	TCGA-C8-A1HK-01A-21R-A13Q-07	
##	COL1A1	197169	109458
##	FN1	89735	73379
##	CPB1	860	44
##	COL1A2	122058	70294
##	COL3A1	129224	63931
##	MT-CO1	577444	409097
##	TCGA-A0-A12H-01A-11R-A115-07	TCGA-AC-A2FK-01A-12R-A180-07	
##	COL1A1	80081	22214
##	FN1	94404	13267
##	CPB1	4463642	11
##	COL1A2	70053	33855
##	COL3A1	46617	37282
##	MT-CO1	1229208	333523
##	TCGA-A2-AOCV-01A-31R-A115-07	TCGA-EW-A1IX-01A-12R-A144-07	
##	COL1A1	541128	1451410
##	FN1	148867	636175
##	CPB1	81	92
##	COL1A2	404862	983677
##	COL3A1	494769	772563
##	MT-CO1	609210	254986
##	TCGA-AN-AOXP-01A-11R-A109-07	TCGA-EW-A6S9-01A-22R-A33J-07	
##	COL1A1	425287	936341
##	FN1	329379	549936
##	CPB1	8830	1
##	COL1A2	341791	670819
##	COL3A1	235477	633634
##	MT-CO1	413211	220837
##	TCGA-A7-A4SF-01A-11R-A266-07	TCGA-AR-A24H-01A-11R-A169-07	
##	COL1A1	156203	160773
##	FN1	68275	258709
##	CPB1	9	56
##	COL1A2	101026	148180
##	COL3A1	93442	142073
##	MT-CO1	782703	395504
##	TCGA-EW-A1J5-01A-11R-A13Q-07	TCGA-E2-A1LB-11A-22R-A144-07	
##	COL1A1	222409	71987
##	FN1	228730	8004
##	CPB1	36	1086
##	COL1A2	195539	46843
##	COL3A1	171456	81126
##	MT-CO1	728174	293333
##	TCGA-E2-A1LB-01A-11R-A144-07	TCGA-A8-A07R-01A-21R-A034-07	
##	COL1A1	394935	105741
##	FN1	146152	196990
##	CPB1	2	1327
##	COL1A2	286770	106239
##	COL3A1	246265	78554
##	MT-CO1	497593	595426
##	TCGA-EW-A6SB-01A-12R-A32P-07	TCGA-E2-A15I-11A-32R-A137-07	
##	COL1A1	15593	14802

##	FN1	5943	20308
##	CPB1	18	1855
##	COL1A2	9778	42033
##	COL3A1	14211	38263
##	MT-CO1	173142	1277404
##	TCGA-E2-A15I-01A-21R-A137-07	TCGA-EW-A3E8-01B-11R-A24H-07	
##	COL1A1	214474	403274
##	FN1	123436	138083
##	CPB1	6051	33
##	COL1A2	179783	283078
##	COL3A1	217830	259822
##	MT-CO1	356225	548661
##	TCGA-BH-AOBV-01A-11R-A00Z-07	TCGA-BH-AOBV-11A-31R-A089-07	
##	COL1A1	555943	381195
##	FN1	367764	65419
##	CPB1	2640	4224
##	COL1A2	412294	264153
##	COL3A1	359117	447539
##	MT-CO1	287123	407644
##	TCGA-AR-A24S-01A-11R-A169-07	TCGA-C8-A12X-01A-11R-A115-07	
##	COL1A1	473941	63180
##	FN1	969311	46396
##	CPB1	12	86
##	COL1A2	450767	46916
##	COL3A1	445931	29904
##	MT-CO1	516329	123978
##	TCGA-A2-AOEV-01A-11R-A034-07	TCGA-A0-A12C-01A-11R-A10J-07	
##	COL1A1	565928	1014145
##	FN1	544561	688239
##	CPB1	319	1771
##	COL1A2	449966	811800
##	COL3A1	336487	737468
##	MT-CO1	441839	654817
##	TCGA-A2-AOYI-01A-31R-A10J-07	TCGA-A2-A04X-01A-21R-A034-07	
##	COL1A1	396167	202674
##	FN1	217267	83738
##	CPB1	1904421	12
##	COL1A2	314709	136139
##	COL3A1	318943	83320
##	MT-CO1	567200	840107
##	TCGA-BH-AODT-11A-12R-A12D-07	TCGA-E2-A15A-01A-11R-A12D-07	
##	COL1A1	109843	348077
##	FN1	26121	493027
##	CPB1	2199	36
##	COL1A2	90947	279205
##	COL3A1	109725	217795
##	MT-CO1	499056	347065
##	TCGA-B6-A408-01A-12R-A24H-07	TCGA-E2-A1B0-01A-11R-A12P-07	
##	COL1A1	468859	656421
##	FN1	265639	1219132
##	CPB1	2	13
##	COL1A2	268239	558581
##	COL3A1	135694	530072
##	MT-CO1	299791	1284430

##	TCGA-A2-AOD2-01A-21R-A034-07	TCGA-BH-A1EW-01A-11R-A137-07
##	COL1A1	356704 87717
##	FN1	274373 151739
##	CPB1	32 17
##	COL1A2	226580 100188
##	COL3A1	222779 99087
##	MT-CO1	226052 430153
##	TCGA-AN-AO4C-01A-21R-A034-07	TCGA-A2-AOEX-01A-21R-A034-07
##	COL1A1	121373 567906
##	FN1	97067 250627
##	CPB1	1 89
##	COL1A2	80402 523084
##	COL3A1	67661 492855
##	MT-CO1	466300 348504
##	TCGA-4H-AAAK-01A-12R-A41B-07	TCGA-A7-AOCE-01A-11R-A00Z-07
##	COL1A1	1012726 102912
##	FN1	588624 282770
##	CPB1	35 894
##	COL1A2	764706 74796
##	COL3A1	597654 70789
##	MT-CO1	309710 607454
##	TCGA-AC-A3TM-01A-11R-A22K-07	TCGA-GM-A2DI-01A-31R-A18M-07
##	COL1A1	365523 169771
##	FN1	49151 140344
##	CPB1	61 37
##	COL1A2	172353 134433
##	COL3A1	122314 135045
##	MT-CO1	222149 157515
##	TCGA-BH-A1F2-11A-32R-A13Q-07	TCGA-C8-A8HQ-01A-11R-A36F-07
##	COL1A1	83071 1092274
##	FN1	32371 606587
##	CPB1	6977 4735
##	COL1A2	138032 661308
##	COL3A1	88954 556679
##	MT-CO1	1465887 533968
##	TCGA-A7-A3IY-01A-21R-A21T-07	TCGA-E2-A1II-01A-11R-A144-07
##	COL1A1	163379 19487
##	FN1	277273 45714
##	CPB1	1089 3
##	COL1A2	149162 19704
##	COL3A1	109545 17803
##	MT-CO1	394831 203576
##	TCGA-A7-A13G-11A-51R-A13Q-07	TCGA-D8-A27T-01A-11R-A16F-07
##	COL1A1	57911 1225260
##	FN1	54288 502529
##	CPB1	180 10805
##	COL1A2	83745 1148530
##	COL3A1	90349 1064151
##	MT-CO1	2029260 397893
##	TCGA-A2-AOYH-01A-11R-A109-07	TCGA-D8-A1JI-01A-11R-A13Q-07
##	COL1A1	523001 41385
##	FN1	314151 19255
##	CPB1	11039 696
##	COL1A2	415040 32935

##	COL3A1	326808	170405
##	MT-C01	629320	424235
##	TCGA-AN-AOFT-01A-11R-A034-07	TCGA-A0-A03V-01A-11R-A115-07	
##	COL1A1	364238	282440
##	FN1	161265	167124
##	CPB1	215	383
##	COL1A2	214065	192455
##	COL3A1	168608	192430
##	MT-C01	1839743	361051
##	TCGA-GM-A3NY-01A-11R-A21T-07	TCGA-E2-A10A-01A-21R-A115-07	
##	COL1A1	520264	385249
##	FN1	93357	513316
##	CPB1	3	588956
##	COL1A2	178697	289152
##	COL3A1	166790	214625
##	MT-C01	1568941	227541
##	TCGA-D8-A3Z6-01A-11R-A239-07	TCGA-A8-A08C-01A-11R-A00Z-07	
##	COL1A1	808108	270257
##	FN1	240563	230049
##	CPB1	11499	2
##	COL1A2	408625	211960
##	COL3A1	366443	183584
##	MT-C01	837057	269215
##	TCGA-BH-A1EX-01A-11R-A13Q-07	TCGA-BH-A0C3-01A-21R-A12P-07	
##	COL1A1	657074	586710
##	FN1	839516	215632
##	CPB1	10	232
##	COL1A2	638099	471944
##	COL3A1	579665	527540
##	MT-C01	1096297	595196
##	TCGA-AR-A1AN-01A-11R-A12P-07	TCGA-A0-A0JM-01A-21R-A056-07	
##	COL1A1	716026	535966
##	FN1	558585	400992
##	CPB1	167	3515
##	COL1A2	612812	395416
##	COL3A1	668603	325157
##	MT-C01	604962	272879
##	TCGA-UU-A93S-01A-21R-A41B-07	TCGA-OL-A5RZ-01A-11R-A28M-07	
##	COL1A1	23669	67379
##	FN1	72881	48647
##	CPB1	2	8
##	COL1A2	21727	42072
##	COL3A1	14934	36476
##	MT-C01	369192	266824
##	TCGA-GM-A2DN-01A-11R-A180-07	TCGA-A2-A0EN-01A-13R-A084-07	
##	COL1A1	157606	912682
##	FN1	134849	546266
##	CPB1	34	822
##	COL1A2	124389	604682
##	COL3A1	98922	519501
##	MT-C01	471673	1113979
##	TCGA-E2-A15M-01A-11R-A12D-07	TCGA-E9-A1RC-11A-33R-A157-07	
##	COL1A1	495699	162507
##	FN1	371761	100988

##	CPB1	3769	196
##	COL1A2	533433	204427
##	COL3A1	581404	269208
##	MT-CO1	824999	1324201
##	TCGA-A8-A080-01A-21R-A056-07	TCGA-AN-A0XL-01A-11R-A10J-07	
##	COL1A1	373690	406212
##	FN1	431084	212333
##	CPB1	992	18
##	COL1A2	356977	354339
##	COL3A1	339819	385446
##	MT-CO1	323334	722273
##	TCGA-A0-A12F-01A-11R-A115-07	TCGA-B6-A401-01A-11R-A239-07	
##	COL1A1	283610	719404
##	FN1	411271	513952
##	CPB1	20	5
##	COL1A2	221215	326106
##	COL3A1	168395	260515
##	MT-CO1	355037	813417
##	TCGA-BH-AODH-01A-11R-A084-07	TCGA-AN-A0FW-01A-11R-A034-07	
##	COL1A1	141788	386102
##	FN1	151646	192802
##	CPB1	74	741886
##	COL1A2	102073	280182
##	COL3A1	83118	277533
##	MT-CO1	538190	729544
##	TCGA-D8-A1XM-01A-21R-A14M-07	TCGA-E9-A1RB-01A-11R-A157-07	
##	COL1A1	685049	287202
##	FN1	810590	1072515
##	CPB1	225	923
##	COL1A2	596662	266515
##	COL3A1	566852	411897
##	MT-CO1	454091	1001250
##	TCGA-PL-A8LZ-01A-31R-A36F-07	TCGA-MS-A51U-01A-31R-A266-07	
##	COL1A1	278354	578804
##	FN1	182313	283100
##	CPB1	20	83
##	COL1A2	179294	366192
##	COL3A1	168431	334172
##	MT-CO1	305558	280880
##	TCGA-BH-AOC7-01B-11R-A115-07	TCGA-E2-A1L7-11A-33R-A144-07	
##	COL1A1	699663	80994
##	FN1	548314	60206
##	CPB1	9801	143
##	COL1A2	576223	86301
##	COL3A1	610539	80460
##	MT-CO1	368918	248311
##	TCGA-A8-A07G-01A-11R-A034-07	TCGA-GM-A2D0-01A-11R-A18M-07	
##	COL1A1	1173537	28712
##	FN1	995500	29353
##	CPB1	69	128
##	COL1A2	1066461	14775
##	COL3A1	936149	14042
##	MT-CO1	619800	290705
##	TCGA-BH-AOC0-11A-21R-A089-07	TCGA-BH-AOC0-01A-21R-A056-07	

## COL1A1	17139	77373
## FN1	14971	229827
## CPB1	3870	133
## COL1A2	28790	54958
## COL3A1	22143	49652
## MT-CO1	407057	282150
## TCGA-D8-A143-01A-11R-A115-07	TCGA-D8-A1XV-01A-11R-A14M-07	
## COL1A1	105118	19960
## FN1	104721	41273
## CPB1	8	41493
## COL1A2	67085	24212
## COL3A1	75097	19634
## MT-CO1	172702	488370
## TCGA-E2-A1LI-01A-12R-A157-07	TCGA-A0-A03N-01B-11R-A10J-07	
## COL1A1	51051	570825
## FN1	46070	210323
## CPB1	63	292
## COL1A2	37456	306933
## COL3A1	44391	259891
## MT-CO1	909803	3188676
## TCGA-B6-AOX5-01A-21R-A109-07	TCGA-E2-A15R-01A-11R-A115-07	
## COL1A1	107416	340559
## FN1	81872	181071
## CPB1	52	18210
## COL1A2	79287	286371
## COL3A1	56646	222397
## MT-CO1	354521	835530
## TCGA-E9-A1NA-11A-33R-A144-07	TCGA-BH-A0BM-11A-12R-A089-07	
## COL1A1	97441	4471
## FN1	44799	3257
## CPB1	1874	280
## COL1A2	117615	5440
## COL3A1	105172	5826
## MT-CO1	511481	598720
## TCGA-AN-AOFZ-01A-11R-A034-07	TCGA-AC-A2FF-01A-11R-A17B-07	
## COL1A1	580441	685831
## FN1	620422	155242
## CPB1	650	43
## COL1A2	511898	615895
## COL3A1	422216	867830
## MT-CO1	558231	536429
## TCGA-BH-AOBW-11A-12R-A115-07	TCGA-AN-AOFL-01A-11R-A034-07	
## COL1A1	130205	49998
## FN1	52137	60441
## CPB1	745	1
## COL1A2	103971	35308
## COL3A1	137178	29474
## MT-CO1	421592	459599
## TCGA-B6-AOWT-01A-11R-A109-07	TCGA-C8-A12V-01A-11R-A115-07	
## COL1A1	457469	77868
## FN1	330215	151132
## CPB1	21786	5
## COL1A2	375730	59608
## COL3A1	317542	50705

## MT-C01	197888	438936
## TCGA-A7-A26F-01B-04R-A220-07	TCGA-A7-A26F-01A-21R-A169-07	
## COL1A1	359176	464377
## FN1	733862	991437
## CPB1	229	207
## COL1A2	254390	708206
## COL3A1	299638	963049
## MT-C01	330728	22505
## TCGA-A8-A093-01A-11R-A00Z-07	TCGA-A2-A1G0-01A-11R-A13Q-07	
## COL1A1	1239587	440068
## FN1	437582	196729
## CPB1	3340	195
## COL1A2	854686	341850
## COL3A1	726969	359883
## MT-C01	388986	415471
## TCGA-BH-AOBQ-01A-21R-A115-07	TCGA-E2-A1IN-01A-11R-A13Q-07	
## COL1A1	632401	1604892
## FN1	376645	1095615
## CPB1	6505	3848
## COL1A2	447154	1294765
## COL3A1	421339	815547
## MT-C01	282619	1090870
## TCGA-AR-A24K-01A-11R-A169-07	TCGA-BH-A0HO-01A-11R-A056-07	
## COL1A1	339481	42804
## FN1	252629	29234
## CPB1	97237	522
## COL1A2	293573	37780
## COL3A1	266145	41736
## MT-C01	319144	1107973
## TCGA-EW-A1PH-01A-11R-A14M-07	TCGA-BH-A18T-01A-11R-A12D-07	
## COL1A1	107860	138140
## FN1	74585	120915
## CPB1	27	5
## COL1A2	119556	110569
## COL3A1	110734	104467
## MT-C01	363093	357121
## TCGA-AR-A0U2-01A-11R-A109-07	TCGA-BH-A209-01A-11R-A157-07	
## COL1A1	155411	145685
## FN1	146949	415159
## CPB1	3397	10
## COL1A2	108942	137491
## COL3A1	109947	157945
## MT-C01	457263	449911
## TCGA-AR-A2LN-01A-21R-A18M-07	TCGA-E2-A14S-01A-11R-A12D-07	
## COL1A1	500912	57171
## FN1	248275	62464
## CPB1	19	264
## COL1A2	421867	46064
## COL3A1	405444	42158
## MT-C01	192924	864935
## TCGA-A0-A1KS-01A-11R-A13Q-07	TCGA-AC-A62X-01A-11R-A29R-07	
## COL1A1	106224	30708
## FN1	272409	10258
## CPB1	730	20

##	COL1A2	100327	25255
##	COL3A1	93086	20204
##	MT-CO1	593374	347073
##	TCGA-A8-A083-01A-21R-A00Z-07	TCGA-B6-A0IH-01A-11R-A115-07	
##	COL1A1	67161	519546
##	FN1	51681	101285
##	CPB1	40	1640
##	COL1A2	57510	374169
##	COL3A1	46479	435951
##	MT-CO1	410763	289279
##	TCGA-E2-A1IJ-01A-11R-A144-07	TCGA-A1-A0SK-01A-12R-A084-07	
##	COL1A1	720595	85967
##	FN1	383458	184512
##	CPB1	88	28
##	COL1A2	562024	69052
##	COL3A1	414459	62299
##	MT-CO1	331668	100843
##	TCGA-A8-A081-01A-11R-A00Z-07	TCGA-AR-A2LJ-01A-12R-A19W-07	
##	COL1A1	49996	676001
##	FN1	150016	107852
##	CPB1	7	94
##	COL1A2	47350	383372
##	COL3A1	45587	400836
##	MT-CO1	369685	296033
##	TCGA-BH-AOB5-01A-11R-A12P-07	TCGA-BH-A1F6-11B-94R-A13Q-07	
##	COL1A1	566403	90918
##	FN1	363593	107488
##	CPB1	53	5488
##	COL1A2	429992	141592
##	COL3A1	462800	127756
##	MT-CO1	353419	1235972
##	TCGA-BH-AOB4-01A-11R-A00Z-07	TCGA-BH-AOBL-01A-11R-A115-07	
##	COL1A1	374086	728232
##	FN1	231341	744705
##	CPB1	33	23
##	COL1A2	268320	576805
##	COL3A1	273940	613456
##	MT-CO1	766533	517796
##	TCGA-A2-AOEY-01A-11R-A034-07	TCGA-GM-A2DK-01A-21R-A180-07	
##	COL1A1	224587	184875
##	FN1	538297	117466
##	CPB1	181	21634
##	COL1A2	161650	86929
##	COL3A1	160174	80359
##	MT-CO1	523910	1467424
##	TCGA-D8-A27I-01A-11R-A16F-07	TCGA-A8-A08G-01A-11R-A00Z-07	
##	COL1A1	1187332	482857
##	FN1	324642	302940
##	CPB1	674	386060
##	COL1A2	888650	380721
##	COL3A1	952369	317043
##	MT-CO1	560514	494738
##	TCGA-D8-A1XA-01A-11R-A14D-07	TCGA-E2-A1B5-01A-21R-A12P-07	
##	COL1A1	299051	89016

##	FN1	282921	33478
##	CPB1	6994	37
##	COL1A2	215070	78966
##	COL3A1	160480	89881
##	MT-CO1	148069	902758
##	TCGA-E2-A15A-06A-11R-A12D-07	TCGA-E2-A14N-01A-31R-A137-07	
##	COL1A1	206620	37536
##	FN1	272299	90083
##	CPB1	2	3
##	COL1A2	175086	29907
##	COL3A1	187435	21730
##	MT-CO1	347966	729918
##	TCGA-BH-A18I-01A-11R-A12D-07	TCGA-AC-A6IX-06A-11R-A32P-07	
##	COL1A1	1043002	122344
##	FN1	858212	22180
##	CPB1	199	45
##	COL1A2	948219	96393
##	COL3A1	860967	107318
##	MT-CO1	564101	931226
##	TCGA-AC-A6IX-01A-12R-A32P-07	TCGA-E2-A10C-01A-21R-A10J-07	
##	COL1A1	799217	143068
##	FN1	190862	91583
##	CPB1	1167	78
##	COL1A2	511628	76449
##	COL3A1	527742	65833
##	MT-CO1	376593	795237
##	TCGA-A2-A04Y-01A-21R-A034-07	TCGA-D8-A1JA-01A-11R-A13Q-07	
##	COL1A1	132990	489052
##	FN1	100607	672036
##	CPB1	17	119
##	COL1A2	90302	477963
##	COL3A1	79604	507300
##	MT-CO1	2443709	566003
##	TCGA-E9-A1N4-11A-33R-A14M-07	TCGA-A7-A0CE-11A-21R-A089-07	
##	COL1A1	14509	188566
##	FN1	10945	33135
##	CPB1	884	19849
##	COL1A2	25551	156456
##	COL3A1	22639	138253
##	MT-CO1	624719	318475
##	TCGA-LL-A441-01A-11R-A24H-07	TCGA-BH-A18N-11A-43R-A12D-07	
##	COL1A1	304139	39622
##	FN1	90780	31927
##	CPB1	846	6529
##	COL1A2	155268	77259
##	COL3A1	157858	41929
##	MT-CO1	527612	666321
##	TCGA-BH-A1F2-01A-31R-A13Q-07	TCGA-A8-A06D-01A-11R-A00Z-07	
##	COL1A1	581484	344462
##	FN1	896405	170147
##	CPB1	66	4048
##	COL1A2	564695	236850
##	COL3A1	621250	234914
##	MT-CO1	1036563	1054025

##	TCGA-A7-A26J-01B-02R-A277-07	TCGA-B6-A0RU-01A-11R-A084-07
##	COL1A1	40275 116109
##	FN1	58842 85598
##	CPB1	311 416
##	COL1A2	65078 111643
##	COL3A1	82792 115626
##	MT-CO1	69507 454602
##	TCGA-A7-A26J-01A-11R-A277-07	TCGA-A7-A26J-01A-11R-A169-07
##	COL1A1	44391 488644
##	FN1	164440 492733
##	CPB1	4134 6314
##	COL1A2	125376 504972
##	COL3A1	256751 481338
##	MT-CO1	28916 510699
##	TCGA-C8-A12K-01A-21R-A115-07	TCGA-BH-A18V-01A-11R-A12D-07
##	COL1A1	147100 287671
##	FN1	160175 241410
##	CPB1	6 1
##	COL1A2	109249 218367
##	COL3A1	102519 243886
##	MT-CO1	209437 189775
##	TCGA-A7-A13G-01B-04R-A220-07	TCGA-D8-A1XS-01A-11R-A14M-07
##	COL1A1	178217 496788
##	FN1	240162 622147
##	CPB1	429 214
##	COL1A2	119606 565409
##	COL3A1	112250 477582
##	MT-CO1	357779 572853
##	TCGA-A7-A3J0-01A-11R-A213-07	TCGA-E2-A15C-01A-31R-A12D-07
##	COL1A1	67481 388887
##	FN1	123037 443111
##	CPB1	40163 59
##	COL1A2	38815 288219
##	COL3A1	34237 214004
##	MT-CO1	429628 453468
##	TCGA-AR-A1AV-01A-21R-A12P-07	TCGA-A8-A08H-01A-21R-A00Z-07
##	COL1A1	476410 894173
##	FN1	545169 436676
##	CPB1	426690 212
##	COL1A2	375240 627545
##	COL3A1	341176 601998
##	MT-CO1	451970 317667
##	TCGA-OL-A6VR-01A-32R-A33J-07	TCGA-BH-A0DD-01A-31R-A12P-07
##	COL1A1	30013 198057
##	FN1	23586 440755
##	CPB1	145 7431
##	COL1A2	23696 146935
##	COL3A1	20758 150048
##	MT-CO1	239083 270774
##	TCGA-BH-A18Q-01A-12R-A12D-07	TCGA-A7-A4SE-01A-11R-A266-07
##	COL1A1	50023 562884
##	FN1	59354 320827
##	CPB1	1569 30
##	COL1A2	40685 323031

##	COL3A1	35838	300602
##	MT-CO1	438783	138166
##	TCGA-BH-AOHB-01A-11R-A056-07	TCGA-BH-A1FN-01A-11R-A13Q-07	
##	COL1A1	248237	303431
##	FN1	116522	368165
##	CPB1	40964	1530
##	COL1A2	136113	247749
##	COL3A1	116541	222885
##	MT-CO1	718185	238270
##	TCGA-BH-AOBO-01A-23R-A12D-07	TCGA-D8-A1JS-01A-11R-A13Q-07	
##	COL1A1	473968	15021
##	FN1	182381	24235
##	CPB1	1424	1
##	COL1A2	370128	14775
##	COL3A1	361625	14473
##	MT-CO1	306809	97809
##	TCGA-E9-A1R7-01A-11R-A14M-07	TCGA-D8-A27H-01A-11R-A16F-07	
##	COL1A1	93714	560444
##	FN1	135304	735004
##	CPB1	2344	176
##	COL1A2	100945	476090
##	COL3A1	121622	457522
##	MT-CO1	912017	405722
##	TCGA-E9-A1R7-11A-42R-A14M-07	TCGA-B6-A400-01A-11R-A239-07	
##	COL1A1	30085	499235
##	FN1	71848	255151
##	CPB1	552	2
##	COL1A2	101487	306883
##	COL3A1	61805	220873
##	MT-CO1	898663	379054
##	TCGA-B6-A1KN-01A-11R-A13Q-07	TCGA-E2-A15G-01A-11R-A12D-07	
##	COL1A1	298175	159480
##	FN1	335456	85690
##	CPB1	40	252
##	COL1A2	242243	106620
##	COL3A1	217509	69071
##	MT-CO1	553217	283165
##	TCGA-A2-A1G6-01A-11R-A13Q-07	TCGA-A2-A25E-01A-11R-A169-07	
##	COL1A1	38655	186657
##	FN1	15406	274326
##	CPB1	1383	46065
##	COL1A2	49626	170202
##	COL3A1	54403	131927
##	MT-CO1	342762	579967
##	TCGA-A7-A26H-01A-11R-A169-07	TCGA-D8-A1JB-01A-11R-A13Q-07	
##	COL1A1	914350	139760
##	FN1	932015	58608
##	CPB1	30	619522
##	COL1A2	862089	111213
##	COL3A1	743744	108283
##	MT-CO1	762221	371151
##	TCGA-BH-A208-01A-11R-A157-07	TCGA-A2-A4S0-01A-21R-A266-07	
##	COL1A1	2462371	46435
##	FN1	1366367	37679

##	CPB1	39	391
##	COL1A2	1921097	29424
##	COL3A1	1896259	20313
##	MT-CO1	483027	310516
##	TCGA-A8-A08Z-01A-21R-A00Z-07	TCGA-OL-A66H-01A-11R-A29R-07	
##	COL1A1	590929	192215
##	FN1	1023296	54424
##	CPB1	355	2159427
##	COL1A2	579802	118388
##	COL3A1	530383	98681
##	MT-CO1	392175	272225
##	TCGA-EW-A1J3-01A-11R-A13Q-07	TCGA-AQ-A04L-01B-21R-A10J-07	
##	COL1A1	294656	415819
##	FN1	239326	381823
##	CPB1	4821	218
##	COL1A2	221229	313708
##	COL3A1	170351	236014
##	MT-CO1	1457204	521447
##	TCGA-EW-A423-01A-11R-A24H-07	TCGA-GM-A2DM-01A-11R-A180-07	
##	COL1A1	164576	127402
##	FN1	44551	205234
##	CPB1	139	3523763
##	COL1A2	72535	74790
##	COL3A1	43770	59431
##	MT-CO1	936751	322868
##	TCGA-E2-A153-01A-12R-A12D-07	TCGA-BH-AOWA-01A-11R-A109-07	
##	COL1A1	918301	176661
##	FN1	926909	288322
##	CPB1	118	3
##	COL1A2	634613	168425
##	COL3A1	374632	164222
##	MT-CO1	408207	316744
##	TCGA-E2-A1IK-01A-11R-A144-07	TCGA-BH-AODL-01A-11R-A115-07	
##	COL1A1	30151	126965
##	FN1	17801	231139
##	CPB1	630830	0
##	COL1A2	24999	87408
##	COL3A1	25065	96871
##	MT-CO1	191202	567572
##	TCGA-A2-A25C-01A-11R-A169-07	TCGA-A2-AOCY-01A-12R-A034-07	
##	COL1A1	591690	30803
##	FN1	517337	293784
##	CPB1	91	506
##	COL1A2	589180	18826
##	COL3A1	655895	12579
##	MT-CO1	617021	960019
##	TCGA-D8-A141-01A-11R-A115-07	TCGA-AN-AOAS-01A-11R-A00Z-07	
##	COL1A1	1489160	1073091
##	FN1	1156027	677134
##	CPB1	14	30
##	COL1A2	1106014	775824
##	COL3A1	879556	731454
##	MT-CO1	429501	354959
##	TCGA-BH-A18S-01A-11R-A12D-07	TCGA-BH-A18S-11A-43R-A12D-07	

## COL1A1	46740	11509
## FN1	71896	13574
## CPB1	30802	5194
## COL1A2	53518	21401
## COL3A1	34137	26657
## MT-CO1	384589	463940
## TCGA-BH-A6R9-01A-21R-A32P-07	TCGA-BH-A1EO-01A-11R-A137-07	
## COL1A1	2092802	580295
## FN1	1081925	532439
## CPB1	23	302
## COL1A2	1616776	516277
## COL3A1	1765090	499913
## MT-CO1	314629	228162
## TCGA-BH-A42V-01A-11R-A24H-07	TCGA-AQ-A54N-01A-11R-A266-07	
## COL1A1	277306	1329
## FN1	112965	3617
## CPB1	193	1
## COL1A2	235842	487
## COL3A1	231840	356
## MT-CO1	489642	459377
## TCGA-E9-A1RC-01A-11R-A157-07	TCGA-E2-A9RU-01A-11R-A41B-07	
## COL1A1	725907	278284
## FN1	959735	120829
## CPB1	69	93
## COL1A2	588094	177394
## COL3A1	557776	92620
## MT-CO1	468669	131458
## TCGA-AC-A3QP-01A-11R-A22U-07	TCGA-D8-A145-01A-11R-A115-07	
## COL1A1	522938	389879
## FN1	123616	101165
## CPB1	228	1317
## COL1A2	320788	312560
## COL3A1	304794	349076
## MT-CO1	504228	375502
## TCGA-A7-A13E-01A-11R-A12P-07	TCGA-OK-A5Q2-01A-11R-A27Q-07	
## COL1A1	183445	347423
## FN1	200650	116649
## CPB1	0	71
## COL1A2	123645	203295
## COL3A1	104193	198528
## MT-CO1	509265	344160
## TCGA-A7-A13E-01A-11R-A277-07	TCGA-BH-A0DZ-01A-11R-A00Z-07	
## COL1A1	105393	365823
## FN1	298788	371931
## CPB1	2	6
## COL1A2	125977	300231
## COL3A1	133488	290747
## MT-CO1	67573	210208
## TCGA-A8-A07P-01A-11R-A00Z-07	TCGA-C8-A275-01A-21R-A16F-07	
## COL1A1	512783	253901
## FN1	215536	353850
## CPB1	8538	9
## COL1A2	395135	192057
## COL3A1	346678	170789

##	MT-C01	421723	375530
##	TCGA-AR-A24X-01A-11R-A169-07	TCGA-A8-A070-01A-11R-A00Z-07	
##	COL1A1	297098	282674
##	FN1	319077	169270
##	CPB1	1	26211
##	COL1A2	277510	183261
##	COL3A1	220622	229628
##	MT-C01	206891	794587
##	TCGA-B6-AORS-01A-11R-A084-07	TCGA-E2-A1LS-11A-32R-A157-07	
##	COL1A1	230728	46112
##	FN1	175113	35202
##	CPB1	44	291
##	COL1A2	177492	80540
##	COL3A1	175409	96043
##	MT-C01	978246	1957867
##	TCGA-E2-A10F-01A-11R-A10J-07	TCGA-A7-A2KD-01A-31R-A21T-07	
##	COL1A1	795962	657665
##	FN1	695736	564370
##	CPB1	26	4
##	COL1A2	597046	480105
##	COL3A1	460920	424498
##	MT-C01	333489	566638
##	TCGA-BH-AOHI-01A-11R-A084-07	TCGA-EW-A10Z-01A-11R-A144-07	
##	COL1A1	381346	661488
##	FN1	282946	696914
##	CPB1	170821	4
##	COL1A2	300672	493278
##	COL3A1	231140	385298
##	MT-C01	516645	596607
##	TCGA-E2-A14T-01A-11R-A115-07	TCGA-AR-A1AK-01A-21R-A12P-07	
##	COL1A1	293197	117920
##	FN1	206946	103931
##	CPB1	1444026	21
##	COL1A2	170042	103197
##	COL3A1	154744	102964
##	MT-C01	1001020	432408
##	TCGA-A7-A3J1-01A-11R-A213-07	TCGA-E2-A1B6-01A-31R-A12P-07	
##	COL1A1	619488	55366
##	FN1	168433	37220
##	CPB1	52	10
##	COL1A2	407349	39174
##	COL3A1	386385	33139
##	MT-C01	496526	987808
##	TCGA-GM-A3XL-01A-11R-A22U-07	TCGA-C8-A3M7-01A-12R-A21T-07	
##	COL1A1	50449	115143
##	FN1	82235	51713
##	CPB1	3	16
##	COL1A2	30597	107527
##	COL3A1	30680	126017
##	MT-C01	329187	1053684
##	TCGA-A2-AOCU-01A-12R-A034-07	TCGA-OL-A66I-01A-21R-A29R-07	
##	COL1A1	209312	387368
##	FN1	126599	106429
##	CPB1	2	0

##	COL1A2	130655	270083
##	COL3A1	123370	209478
##	MT-CO1	935118	135672
##	TCGA-AN-AOG0-01A-11R-A034-07	TCGA-LL-A440-01A-11R-A24H-07	
##	COL1A1	300167	2482119
##	FN1	290165	356919
##	CPB1	0	3279
##	COL1A2	211647	1262501
##	COL3A1	222018	1738013
##	MT-CO1	445808	688325
##	TCGA-A8-A09N-01A-11R-A00Z-07	TCGA-BH-A0DV-11A-22R-A12P-07	
##	COL1A1	374701	7992
##	FN1	157900	14751
##	CPB1	3737	677
##	COL1A2	254553	21217
##	COL3A1	230611	11989
##	MT-CO1	273328	508409
##	TCGA-AC-A3BB-01A-21R-A19W-07	TCGA-B6-A0IO-01A-11R-A034-07	
##	COL1A1	1054868	194986
##	FN1	273706	86682
##	CPB1	139	12188
##	COL1A2	623582	128458
##	COL3A1	534098	95655
##	MT-CO1	288339	1350613
##	TCGA-C8-A1HF-01A-11R-A137-07	TCGA-A8-A07Z-01A-11R-A00Z-07	
##	COL1A1	908573	173501
##	FN1	639165	71336
##	CPB1	4	4397
##	COL1A2	638415	98122
##	COL3A1	590827	82415
##	MT-CO1	479433	293306
##	TCGA-A2-AOES-01A-11R-A115-07	TCGA-BH-A0DQ-11A-12R-A089-07	
##	COL1A1	167202	286498
##	FN1	106170	51172
##	CPB1	455	6175
##	COL1A2	155151	235098
##	COL3A1	179981	429733
##	MT-CO1	505715	368283
##	TCGA-BH-A0DQ-01A-11R-A084-07	TCGA-A7-A26E-01A-11R-A169-07	
##	COL1A1	757734	675067
##	FN1	540097	552715
##	CPB1	9950	1715
##	COL1A2	623354	596165
##	COL3A1	583930	515113
##	MT-CO1	680951	484282
##	TCGA-A7-A26E-01B-06R-A277-07	TCGA-A7-A26E-01A-11R-A277-07	
##	COL1A1	84956	71214
##	FN1	114305	223183
##	CPB1	5	1019
##	COL1A2	102371	166482
##	COL3A1	100303	282884
##	MT-CO1	36568	13239
##	TCGA-B6-A0IQ-01A-11R-A034-07	TCGA-LL-A73Z-01A-11R-A32P-07	
##	COL1A1	137425	190871

##	FN1	43573	75265
##	CPB1	11	2882
##	COL1A2	127128	145951
##	COL3A1	37145	129533
##	MT-CO1	682179	746106
##	TCGA-AN-AOAL-01A-11R-A00Z-07	TCGA-GM-A2DB-01A-31R-A18M-07	
##	COL1A1	581463	73094
##	FN1	341840	26549
##	CPB1	624	4
##	COL1A2	385493	35561
##	COL3A1	379332	30355
##	MT-CO1	462374	1173237
##	TCGA-A2-AOT3-01A-21R-A115-07	TCGA-A0-A03U-01B-21R-A10J-07	
##	COL1A1	232872	459916
##	FN1	391595	552949
##	CPB1	202	227
##	COL1A2	181071	347489
##	COL3A1	164647	284080
##	MT-CO1	490645	487914
##	TCGA-BH-AOW4-01A-11R-A109-07	TCGA-AR-A5QN-01A-12R-A28M-07	
##	COL1A1	617368	457127
##	FN1	481449	109047
##	CPB1	486	979
##	COL1A2	436138	333665
##	COL3A1	431049	343978
##	MT-CO1	444701	293005
##	TCGA-A8-AOA4-01A-11R-A00Z-07	TCGA-A0-A12E-01A-11R-A10J-07	
##	COL1A1	157057	171654
##	FN1	60099	175140
##	CPB1	6299	375
##	COL1A2	116469	122018
##	COL3A1	121415	126597
##	MT-CO1	386502	1262785
##	TCGA-E2-A1LL-01A-11R-A144-07	TCGA-LL-A7SZ-01A-32R-A352-07	
##	COL1A1	59795	119360
##	FN1	148217	63720
##	CPB1	8	1
##	COL1A2	64109	68282
##	COL3A1	32854	45591
##	MT-CO1	93565	202014
##	TCGA-E2-A1IE-01A-11R-A13Q-07	TCGA-AR-A24L-01A-11R-A169-07	
##	COL1A1	245828	731125
##	FN1	179497	1033660
##	CPB1	2072134	1881
##	COL1A2	182801	774333
##	COL3A1	164214	737700
##	MT-CO1	970746	680053
##	TCGA-S3-AA10-01A-21R-A41B-07	TCGA-A2-A3XY-01A-11R-A239-07	
##	COL1A1	153428	699111
##	FN1	65697	481629
##	CPB1	5	40
##	COL1A2	88573	411164
##	COL3A1	104822	367675
##	MT-CO1	147464	353221

##	TCGA-B6-AORN-01A-12R-A084-07	TCGA-BH-A1FO-01A-11R-A137-07
##	COL1A1	1026955 269821
##	FN1	431000 329379
##	CPB1	12 24
##	COL1A2	784099 256001
##	COL3A1	648669 256225
##	MT-CO1	423812 677642
##	TCGA-EW-A1P3-01A-11R-A144-07	TCGA-A1-AOSJ-01A-11R-A084-07
##	COL1A1	199347 215351
##	FN1	91044 208561
##	CPB1	194 73
##	COL1A2	118998 177651
##	COL3A1	110221 191557
##	MT-CO1	848289 665930
##	TCGA-A7-A4SA-01A-11R-A266-07	TCGA-BH-A0EA-01A-11R-A115-07
##	COL1A1	359433 386468
##	FN1	99203 144870
##	CPB1	151178 116
##	COL1A2	220909 257276
##	COL3A1	227244 220699
##	MT-CO1	662936 417537
##	TCGA-AO-A1KT-01A-11R-A13Q-07	TCGA-BH-A0BG-01A-11R-A115-07
##	COL1A1	509857 163117
##	FN1	519045 144743
##	CPB1	268 1
##	COL1A2	408794 122696
##	COL3A1	256207 98622
##	MT-CO1	632407 636286
##	TCGA-BH-A8FZ-01A-11R-A352-07	TCGA-E2-A1LK-01A-21R-A14D-07
##	COL1A1	273356 20361
##	FN1	189448 30162
##	CPB1	182 0
##	COL1A2	178092 13626
##	COL3A1	206340 10100
##	MT-CO1	277825 503319
##	TCGA-OL-A5RV-01A-12R-A28M-07	TCGA-B6-A409-01A-11R-A24H-07
##	COL1A1	161003 171367
##	FN1	143578 82830
##	CPB1	187 41
##	COL1A2	96062 98428
##	COL3A1	104781 85002
##	MT-CO1	175871 722203
##	TCGA-AC-A2FB-01A-11R-A17B-07	TCGA-B6-A0X0-01A-21R-A115-07
##	COL1A1	161263 64541
##	FN1	146792 49858
##	CPB1	457 10
##	COL1A2	125397 40628
##	COL3A1	145094 35800
##	MT-CO1	660748 232239
##	TCGA-LL-A6FR-01A-12R-A310-07	TCGA-D8-A1JC-01A-11R-A13Q-07
##	COL1A1	427370 609913
##	FN1	126773 611806
##	CPB1	25 701
##	COL1A2	276150 513289

##	COL3A1	141481	446202
##	MT-CO1	117359	689936
##	TCGA-BH-AODX-01A-11R-A115-07	TCGA-A1-AOSI-01A-11R-A144-07	
##	COL1A1	1352075	2297178
##	FN1	555669	1756810
##	CPB1	35	443
##	COL1A2	797023	1755756
##	COL3A1	678788	1682021
##	MT-CO1	657053	382638
##	TCGA-A7-AOD9-01A-31R-A056-07	TCGA-E2-A15K-06A-11R-A12P-07	
##	COL1A1	294299	11256
##	FN1	330715	17511
##	CPB1	356	187
##	COL1A2	220340	9351
##	COL3A1	178770	8841
##	MT-CO1	431157	626238
##	TCGA-AR-A24Z-01A-11R-A169-07	TCGA-EW-A10Y-01A-11R-A144-07	
##	COL1A1	172895	133704
##	FN1	352639	154982
##	CPB1	14661	4313
##	COL1A2	174875	90147
##	COL3A1	160646	59027
##	MT-CO1	450079	324481
##	TCGA-C8-A130-01A-31R-A115-07	TCGA-E2-A14Y-01A-21R-A12D-07	
##	COL1A1	348090	87626
##	FN1	243408	108745
##	CPB1	7	1
##	COL1A2	224930	57136
##	COL3A1	172006	53541
##	MT-CO1	735912	415191
##	TCGA-D8-A27N-01A-11R-A16F-07	TCGA-AR-A0TY-01A-12R-A115-07	
##	COL1A1	850333	145574
##	FN1	769923	104401
##	CPB1	428	290
##	COL1A2	708557	114826
##	COL3A1	637011	141065
##	MT-CO1	482021	500161
##	TCGA-E2-A1BC-01A-11R-A12P-07	TCGA-E9-A243-01A-21R-A169-07	
##	COL1A1	311238	422067
##	FN1	237206	853079
##	CPB1	25	38
##	COL1A2	272315	477846
##	COL3A1	254565	494632
##	MT-CO1	275820	1416415
##	TCGA-BH-AOBJ-11A-23R-A089-07	TCGA-BH-AOBJ-01A-11R-A056-07	
##	COL1A1	184236	483755
##	FN1	48548	303738
##	CPB1	2221	5931
##	COL1A2	170136	413374
##	COL3A1	225332	452584
##	MT-CO1	289337	468350
##	TCGA-AR-A240-01A-11R-A169-07	TCGA-AR-A24R-01A-11R-A169-07	
##	COL1A1	771351	305680
##	FN1	457712	361268

##	CPB1	120	594
##	COL1A2	616389	296278
##	COL3A1	662820	253865
##	MT-CO1	451763	841367
##	TCGA-LL-A7T0-01A-31R-A352-07	TCGA-AC-A5XS-01A-11R-A29R-07	
##	COL1A1	207338	285390
##	FN1	139276	41608
##	CPB1	33	723
##	COL1A2	106331	199352
##	COL3A1	73429	171993
##	MT-CO1	357525	226531
##	TCGA-BH-AODD-11A-23R-A12P-07	TCGA-BH-A18Q-11A-34R-A12D-07	
##	COL1A1	74303	27789
##	FN1	74539	23128
##	CPB1	303	630
##	COL1A2	136311	49274
##	COL3A1	154245	34916
##	MT-CO1	1045582	542170
##	TCGA-E2-A1B4-01A-11R-A12P-07	TCGA-C8-A26Z-01A-11R-A16F-07	
##	COL1A1	521340	509624
##	FN1	165641	341140
##	CPB1	77	416343
##	COL1A2	261143	389873
##	COL3A1	164851	351339
##	MT-CO1	683310	542312
##	TCGA-E2-A10E-01A-21R-A10J-07	TCGA-BH-A0HU-01A-11R-A034-07	
##	COL1A1	410516	236675
##	FN1	439877	120436
##	CPB1	10224	18045
##	COL1A2	322250	152252
##	COL3A1	263348	113192
##	MT-CO1	358960	932553
##	TCGA-OL-A5D8-01A-11R-A27Q-07	TCGA-C8-A1HG-01A-11R-A137-07	
##	COL1A1	220343	124713
##	FN1	123229	76695
##	CPB1	100	0
##	COL1A2	116803	86579
##	COL3A1	89176	69445
##	MT-CO1	351933	160341
##	TCGA-BH-A1FN-11A-34R-A13Q-07	TCGA-GM-A3NW-01A-21R-A22K-07	
##	COL1A1	95018	152478
##	FN1	37445	36942
##	CPB1	38	843
##	COL1A2	135728	93254
##	COL3A1	89719	98843
##	MT-CO1	641790	759100
##	TCGA-A8-A09X-01A-11R-A00Z-07	TCGA-E2-A1BD-01A-11R-A12P-07	
##	COL1A1	360375	535523
##	FN1	302035	225982
##	CPB1	9	117
##	COL1A2	258752	316028
##	COL3A1	204958	240429
##	MT-CO1	730827	1001022
##	TCGA-D8-A1JN-01A-11R-A13Q-07	TCGA-A7-A5ZV-01A-11R-A28M-07	

## COL1A1	20732	14522
## FN1	15233	81275
## CPB1	3	25
## COL1A2	15361	141656
## COL3A1	11397	5251
## MT-CO1	421318	209786
## TCGA-BH-AOHW-01A-11R-A034-07	TCGA-BH-A2L8-01A-11R-A18M-07	
## COL1A1	75043	567458
## FN1	94589	206720
## CPB1	8954	68
## COL1A2	48644	376046
## COL3A1	37570	317296
## MT-CO1	712641	277582
## TCGA-BH-A208-11A-51R-A157-07	TCGA-C8-A26Y-01A-11R-A16F-07	
## COL1A1	46028	150539
## FN1	16247	84827
## CPB1	14653	0
## COL1A2	65996	87284
## COL3A1	87330	71995
## MT-CO1	492637	1166848
## TCGA-A1-AOSM-01A-11R-A084-07	TCGA-BH-A0DL-11A-13R-A115-07	
## COL1A1	1055100	151309
## FN1	713532	57211
## CPB1	56	1820
## COL1A2	808924	166470
## COL3A1	665843	168551
## MT-CO1	332421	718898
## TCGA-E2-A153-11A-31R-A12D-07	TCGA-A7-A5ZX-01A-12R-A29R-07	
## COL1A1	196406	509567
## FN1	57466	141569
## CPB1	157	1792
## COL1A2	141910	373578
## COL3A1	135617	403766
## MT-CO1	451841	308362
## TCGA-BH-AOC3-11A-23R-A12P-07	TCGA-C8-A12L-01A-11R-A115-07	
## COL1A1	58478	286940
## FN1	14869	647315
## CPB1	376	4
## COL1A2	59917	183922
## COL3A1	68100	174313
## MT-CO1	422313	314020
## TCGA-B6-A0I9-01A-11R-A034-07	TCGA-JL-A3YX-01A-11R-A22U-07	
## COL1A1	143868	506162
## FN1	193498	217340
## CPB1	31	1143
## COL1A2	115662	253399
## COL3A1	103633	204140
## MT-CO1	485948	508173
## TCGA-AC-A23H-01A-11R-A157-07	TCGA-OL-A5RX-01A-11R-A28M-07	
## COL1A1	634452	173958
## FN1	722419	96690
## CPB1	42	36
## COL1A2	481751	109839
## COL3A1	419478	118976

##	MT-C01	344975	249383
##	TCGA-AC-A23H-11A-12R-A157-07	TCGA-BH-A1FM-01A-11R-A13Q-07	
##	COL1A1	126855	399591
##	FN1	62782	926931
##	CPB1	414	3021
##	COL1A2	223696	462770
##	COL3A1	146089	388909
##	MT-C01	1142085	1283062
##	TCGA-B6-AORP-01A-21R-A084-07	TCGA-EW-A1IW-01A-11R-A13Q-07	
##	COL1A1	903009	364006
##	FN1	598832	315476
##	CPB1	1156203	661
##	COL1A2	675157	294140
##	COL3A1	495886	256235
##	MT-C01	438346	427720
##	TCGA-A0-AOJG-01A-31R-A084-07	TCGA-C8-A273-01A-11R-A16F-07	
##	COL1A1	388196	567054
##	FN1	110985	503075
##	CPB1	728634	18438
##	COL1A2	312691	427910
##	COL3A1	395994	431846
##	MT-C01	704055	653975
##	TCGA-A8-A082-01A-11R-A00Z-07	TCGA-B6-A0X7-01A-11R-A10J-07	
##	COL1A1	258654	1361181
##	FN1	363384	212943
##	CPB1	0	321
##	COL1A2	185547	797595
##	COL3A1	154388	610381
##	MT-C01	446434	708118
##	TCGA-E2-A15M-11A-22R-A12D-07	TCGA-BH-A8FY-01A-11R-A36F-07	
##	COL1A1	106179	13573
##	FN1	56275	11569
##	CPB1	1378	159
##	COL1A2	123974	8739
##	COL3A1	91185	7255
##	MT-C01	547322	700342
##	TCGA-E2-A56Z-01A-12R-A29R-07	TCGA-D8-A1XJ-01A-11R-A14M-07	
##	COL1A1	128719	436186
##	FN1	85517	395240
##	CPB1	18063	137
##	COL1A2	94102	398206
##	COL3A1	59760	357027
##	MT-C01	507266	388356
##	TCGA-D8-A142-01A-11R-A115-07	TCGA-BH-A0DZ-11A-22R-A089-07	
##	COL1A1	832557	95746
##	FN1	1204200	7734
##	CPB1	24	793
##	COL1A2	634706	57620
##	COL3A1	577173	83523
##	MT-C01	443053	324339
##	TCGA-E2-A1LA-01A-11R-A144-07	TCGA-E9-A229-01A-31R-A157-07	
##	COL1A1	347793	1323321
##	FN1	173889	877386
##	CPB1	38351	1702

##	COL1A2	194477	1383640
##	COL3A1	125538	1443258
##	MT-CO1	646360	430608
##	TCGA-E9-A22D-01A-11R-A157-07	TCGA-E9-A1R3-01A-31R-A14M-07	
##	COL1A1	277043	1161113
##	FN1	182675	1356548
##	CPB1	283888	27699
##	COL1A2	209364	1066053
##	COL3A1	186714	1062706
##	MT-CO1	680291	531867
##	TCGA-D8-A1Y0-01A-11R-A14M-07	TCGA-B6-A0RV-01A-11R-A084-07	
##	COL1A1	335595	84912
##	FN1	298661	49682
##	CPB1	118	3463
##	COL1A2	302377	73613
##	COL3A1	341154	101542
##	MT-CO1	641718	242757
##	TCGA-E2-A1LS-01A-12R-A157-07	TCGA-LL-A50Y-01A-11R-A266-07	
##	COL1A1	28725	107815
##	FN1	28925	55206
##	CPB1	501	72
##	COL1A2	24523	69616
##	COL3A1	22267	37613
##	MT-CO1	3563523	132860
##	TCGA-LL-A8F5-01A-11R-A36F-07	TCGA-A2-A0CS-01A-11R-A115-07	
##	COL1A1	143554	1276251
##	FN1	688795	295005
##	CPB1	37	10467
##	COL1A2	82214	702148
##	COL3A1	61269	483317
##	MT-CO1	274438	192648
##	TCGA-A2-A0YK-01A-22R-A109-07	TCGA-B6-A3ZX-01A-11R-A239-07	
##	COL1A1	280314	104110
##	FN1	107006	33443
##	CPB1	805	2
##	COL1A2	207532	63096
##	COL3A1	237555	56804
##	MT-CO1	702860	312928
##	TCGA-AR-A0TU-01A-31R-A109-07	TCGA-AR-A1AP-01A-11R-A12P-07	
##	COL1A1	34686	288146
##	FN1	50478	402422
##	CPB1	6	44
##	COL1A2	22349	209985
##	COL3A1	20303	177360
##	MT-CO1	119642	204073
##	TCGA-S3-AA14-01A-11R-A41B-07	TCGA-E9-A1RB-11A-33R-A157-07	
##	COL1A1	818436	40889
##	FN1	288337	119060
##	CPB1	8	1640
##	COL1A2	502994	118314
##	COL3A1	475283	88502
##	MT-CO1	395046	1145355
##	TCGA-A0-A0J7-01A-11R-A034-07	TCGA-AC-A5XU-01A-11R-A28M-07	
##	COL1A1	292526	504652

##	FN1	145713	271457
##	CPB1	409176	2671
##	COL1A2	214576	277315
##	COL3A1	178255	239500
##	MT-CO1	479797	255998
##	TCGA-EW-A6SC-01A-12R-A32P-07	TCGA-AN-AOAM-01A-11R-A034-07	
##	COL1A1	637050	426867
##	FN1	430339	1356246
##	CPB1	6445	7767
##	COL1A2	457116	266980
##	COL3A1	444323	215172
##	MT-CO1	458728	606803
##	TCGA-AQ-A540-01A-11R-A266-07	TCGA-AN-AOAJ-01A-11R-A00Z-07	
##	COL1A1	215175	321179
##	FN1	75102	227867
##	CPB1	2408	80
##	COL1A2	118799	270611
##	COL3A1	111488	240154
##	MT-CO1	742127	108627
##	TCGA-C8-A135-01A-11R-A115-07	TCGA-BH-AOAU-11A-11R-A12P-07	
##	COL1A1	335000	19375
##	FN1	303348	6503
##	CPB1	16	7951
##	COL1A2	216910	25239
##	COL3A1	192508	22640
##	MT-CO1	570659	309191
##	TCGA-AR-A2LH-01A-31R-A18M-07	TCGA-OL-A5RW-01A-11R-A28M-07	
##	COL1A1	106710	155221
##	FN1	48885	69351
##	CPB1	541	671
##	COL1A2	76173	96367
##	COL3A1	74045	70009
##	MT-CO1	310466	185019
##	TCGA-C8-A1HJ-01A-11R-A13Q-07	TCGA-C8-A1HL-01A-11R-A137-07	
##	COL1A1	81700	207879
##	FN1	112878	215336
##	CPB1	6	2289
##	COL1A2	68872	178502
##	COL3A1	85728	167366
##	MT-CO1	1537747	884782
##	TCGA-A8-A07W-01A-11R-A00Z-07	TCGA-D8-A1XY-01A-11R-A14M-07	
##	COL1A1	260304	981641
##	FN1	1239486	656150
##	CPB1	32	903
##	COL1A2	203224	959720
##	COL3A1	295345	837077
##	MT-CO1	381941	510548
##	TCGA-BH-A5IZ-01A-11R-A27Q-07	TCGA-A2-A3XU-01A-12R-A22U-07	
##	COL1A1	177225	33026
##	FN1	71289	11516
##	CPB1	4	1
##	COL1A2	115128	17027
##	COL3A1	123925	12319
##	MT-CO1	165940	290593

##	TCGA-EW-A1P7-01A-21R-A144-07	TCGA-E9-A54X-01A-11R-A266-07
##	COL1A1	212574 8881
##	FN1	347524 9470
##	CPB1	350 27154
##	COL1A2	151838 5588
##	COL3A1	166818 3954
##	MT-CO1	363171 267719
##	TCGA-AC-A2FB-11A-13R-A17B-07	TCGA-AN-AOXT-01A-11R-A109-07
##	COL1A1	69367 76024
##	FN1	59544 59802
##	CPB1	7146 2232
##	COL1A2	159248 61757
##	COL3A1	77368 61837
##	MT-CO1	589102 598884
##	TCGA-AO-AOJB-01A-11R-A32Y-07	TCGA-D8-A1JM-01A-11R-A13Q-07
##	COL1A1	50497 17378
##	FN1	11546 97959
##	CPB1	3 0
##	COL1A2	21610 13425
##	COL3A1	13507 7663
##	MT-CO1	1963769 412485
##	TCGA-AC-A2B8-01A-11R-A17B-07	TCGA-B6-A0I8-01A-11R-A034-07
##	COL1A1	725543 475219
##	FN1	729462 256150
##	CPB1	16 510
##	COL1A2	628374 406914
##	COL3A1	559392 396015
##	MT-CO1	720122 1151538
##	TCGA-AN-AOAT-01A-11R-A034-07	TCGA-A2-A0YC-01A-11R-A109-07
##	COL1A1	147856 405145
##	FN1	190342 367743
##	CPB1	12 281
##	COL1A2	91459 301733
##	COL3A1	144444 227639
##	MT-CO1	592631 758540
##	TCGA-BH-A0H3-01A-11R-A12P-07	TCGA-BH-A204-11A-53R-A157-07
##	COL1A1	722963 166896
##	FN1	324569 108231
##	CPB1	2066 3505
##	COL1A2	639214 252539
##	COL3A1	834309 185814
##	MT-CO1	508003 1752052
##	TCGA-A8-A09Q-01A-11R-A00Z-07	TCGA-A2-A0T6-01A-11R-A084-07
##	COL1A1	265926 1107485
##	FN1	586177 1058456
##	CPB1	3067 15
##	COL1A2	211505 957691
##	COL3A1	200754 755152
##	MT-CO1	525701 438187
##	TCGA-A2-A0D1-01A-11R-A034-07	TCGA-B6-A0WZ-01A-11R-A109-07
##	COL1A1	305222 314843
##	FN1	148388 166776
##	CPB1	1373 16
##	COL1A2	175887 202305

##	COL3A1	136977	149431
##	MT-CO1	779671	563160
##	TCGA-B6-AORE-01A-11R-A056-07	TCGA-A7-A13F-01A-11R-A12P-07	
##	COL1A1	293822	181537
##	FN1	168541	195754
##	CPB1	23	277
##	COL1A2	194659	129525
##	COL3A1	146469	81549
##	MT-CO1	263131	652480
##	TCGA-D8-A1XG-01A-11R-A14D-07	TCGA-A2-A0YD-01A-11R-A109-07	
##	COL1A1	172909	927362
##	FN1	428093	680605
##	CPB1	1586256	34
##	COL1A2	168393	843053
##	COL3A1	113052	752709
##	MT-CO1	312640	307565
##	TCGA-E2-A14U-01A-11R-A22K-07	TCGA-A8-A076-01A-21R-A00Z-07	
##	COL1A1	16743	942015
##	FN1	9106	764639
##	CPB1	1459	2950
##	COL1A2	16025	759696
##	COL3A1	12103	690241
##	MT-CO1	440258	631639
##	TCGA-BH-AOB7-11A-34R-A115-07	TCGA-BH-A18U-01A-21R-A12D-07	
##	COL1A1	38283	273835
##	FN1	11259	647949
##	CPB1	472	1977
##	COL1A2	42146	264712
##	COL3A1	60374	231317
##	MT-CO1	411106	313439
##	TCGA-A8-A06Z-01A-11R-A00Z-07	TCGA-BH-A0BT-11A-21R-A12P-07	
##	COL1A1	128457	15070
##	FN1	92418	9399
##	CPB1	1957420	3927
##	COL1A2	84035	18022
##	COL3A1	73460	14955
##	MT-CO1	1282913	247562
##	TCGA-BH-AOH7-01A-13R-A056-07	TCGA-BH-A0EO-01A-11R-A056-07	
##	COL1A1	2429818	38671
##	FN1	947229	21884
##	CPB1	227	3445
##	COL1A2	1768444	28713
##	COL3A1	1484947	27953
##	MT-CO1	821895	828757
##	TCGA-A7-A5ZW-01A-12R-A29R-07	TCGA-E9-A244-01A-11R-A169-07	
##	COL1A1	660393	163088
##	FN1	205266	345762
##	CPB1	101	56
##	COL1A2	713066	143670
##	COL3A1	821490	112852
##	MT-CO1	163444	624366
##	TCGA-D8-A1JD-01A-11R-A13Q-07	TCGA-BH-A18K-11A-13R-A12D-07	
##	COL1A1	1000865	11047
##	FN1	1094587	10381

##	CPB1	1258	27
##	COL1A2	949997	20146
##	COL3A1	783973	25333
##	MT-CO1	306599	669020
##	TCGA-E9-A248-01A-11R-A169-07	TCGA-A2-A0T1-01A-21R-A084-07	
##	COL1A1	74590	684334
##	FN1	175378	451706
##	CPB1	21	23
##	COL1A2	76937	444490
##	COL3A1	88009	451797
##	MT-CO1	422999	721688
##	TCGA-BH-AOAZ-11A-22R-A12P-07	TCGA-AN-AOFX-01A-11R-A034-07	
##	COL1A1	269879	398462
##	FN1	76605	365472
##	CPB1	1543	466
##	COL1A2	213928	273174
##	COL3A1	356657	250860
##	MT-CO1	381159	620873
##	TCGA-BH-A1FH-01A-12R-A13Q-07	TCGA-A7-A3IZ-01A-11R-A213-07	
##	COL1A1	1982148	108395
##	FN1	533315	93664
##	CPB1	168179	213659
##	COL1A2	1553243	71005
##	COL3A1	1729356	55386
##	MT-CO1	488800	521330
##	TCGA-D8-A13Y-01A-11R-A115-07	TCGA-A2-A0SX-01A-12R-A084-07	
##	COL1A1	205208	571074
##	FN1	130712	382880
##	CPB1	100	50
##	COL1A2	145272	401514
##	COL3A1	119435	419641
##	MT-CO1	458332	274911
##	TCGA-B6-A0IN-01A-11R-A034-07	TCGA-A2-A3Y0-01A-11R-A239-07	
##	COL1A1	217399	217554
##	FN1	98687	33106
##	CPB1	153435	22
##	COL1A2	144422	113124
##	COL3A1	130203	119840
##	MT-CO1	2964382	241651
##	TCGA-BH-AOB3-01A-11R-A056-07	TCGA-AN-AOX0-01A-11R-A109-07	
##	COL1A1	312466	627842
##	FN1	946733	680795
##	CPB1	2	482
##	COL1A2	205466	492466
##	COL3A1	176502	386540
##	MT-CO1	323252	484166
##	TCGA-E2-A15K-11A-13R-A12P-07	TCGA-BH-AOHF-01A-11R-A056-07	
##	COL1A1	55128	749375
##	FN1	23773	328704
##	CPB1	187	9263
##	COL1A2	92218	453063
##	COL3A1	46991	457792
##	MT-CO1	895346	420101
##	TCGA-GM-A2D9-01A-11R-A18M-07	TCGA-A2-AOCZ-01A-11R-A034-07	

## COL1A1	259289	1302865
## FN1	105056	688774
## CPB1	3421	8621
## COL1A2	139844	879081
## COL3A1	108833	707088
## MT-CO1	626610	860059
## TCGA-D8-A1JE-01A-11R-A13Q-07	TCGA-D8-A1XT-01A-11R-A14M-07	
## COL1A1	1349606	966554
## FN1	861727	903608
## CPB1	1016828	3
## COL1A2	1281081	822692
## COL3A1	1324347	770837
## MT-CO1	490675	371318
## TCGA-EW-A2FS-01A-11R-A17B-07	TCGA-A7-AOCJ-01A-21R-A00Z-07	
## COL1A1	93522	78994
## FN1	74106	156651
## CPB1	265530	21
## COL1A2	56781	70663
## COL3A1	57379	54906
## MT-CO1	1631346	721784
## TCGA-BH-A1FG-01A-11R-A13Q-07	TCGA-E2-A159-01A-11R-A115-07	
## COL1A1	119839	218770
## FN1	63136	186319
## CPB1	797	137
## COL1A2	91908	182723
## COL3A1	77160	191895
## MT-CO1	967316	454950
## TCGA-AC-A62Y-01A-11R-A29R-07	TCGA-A7-A13G-01A-11R-A13Q-07	
## COL1A1	203650	120258
## FN1	118243	230011
## CPB1	13401	1195
## COL1A2	168009	241841
## COL3A1	156013	278928
## MT-CO1	520799	23409
## TCGA-AN-AOXU-01A-11R-A109-07	TCGA-JL-A3YW-01A-12R-A239-07	
## COL1A1	97579	1479674
## FN1	210006	822334
## CPB1	6	8
## COL1A2	61327	811919
## COL3A1	56881	834452
## MT-CO1	348893	619273
## TCGA-BH-A1EY-01A-11R-A13Q-07	TCGA-BH-A204-01A-11R-A157-07	
## COL1A1	657439	190104
## FN1	247807	544222
## CPB1	15	509264
## COL1A2	558225	209439
## COL3A1	634336	160243
## MT-CO1	312683	472054
## TCGA-LL-A5YP-01A-21R-A28M-07	TCGA-BH-AB28-01A-31R-A41B-07	
## COL1A1	564015	260134
## FN1	229443	153384
## CPB1	16	65
## COL1A2	346520	180312
## COL3A1	281957	164655

##	MT-C01	179934	339106
##	TCGA-B6-A0IJ-01A-11R-A034-07	TCGA-E2-A15E-01A-11R-A12D-07	
##	COL1A1	56440	229544
##	FN1	290461	185800
##	CPB1	28	12
##	COL1A2	38689	176447
##	COL3A1	46644	152258
##	MT-C01	441930	429916
##	TCGA-D8-A1XZ-01A-11R-A14M-07	TCGA-E2-A15E-06A-11R-A12D-07	
##	COL1A1	338310	38034
##	FN1	595818	36321
##	CPB1	236	51
##	COL1A2	361272	35380
##	COL3A1	306221	38130
##	MT-C01	408345	397148
##	TCGA-A1-AOSQ-01A-21R-A144-07	TCGA-A7-A13F-11A-42R-A12P-07	
##	COL1A1	55898	47901
##	FN1	53860	26152
##	CPB1	848852	148
##	COL1A2	50881	76644
##	COL3A1	48989	82425
##	MT-C01	481389	797026
##	TCGA-GM-A3XN-01A-12R-A22U-07	TCGA-AC-A7VC-01A-11R-A352-07	
##	COL1A1	966623	935189
##	FN1	199112	770856
##	CPB1	12	7
##	COL1A2	510621	477920
##	COL3A1	536313	751787
##	MT-C01	716260	207906
##	TCGA-S3-AA12-01A-11R-A41B-07	TCGA-A2-A3KC-01A-11R-A213-07	
##	COL1A1	164284	574810
##	FN1	95561	160390
##	CPB1	5664	6254
##	COL1A2	91234	365545
##	COL3A1	50702	377124
##	MT-C01	197245	316201
##	TCGA-AN-AOFY-01A-11R-A034-07	TCGA-BH-A0B7-01A-12R-A115-07	
##	COL1A1	790728	332227
##	FN1	646893	494995
##	CPB1	224	21
##	COL1A2	604888	275276
##	COL3A1	511273	318332
##	MT-C01	654566	386576
##	TCGA-E2-A15J-01A-11R-A12P-07	TCGA-BH-A1F0-11B-23R-A137-07	
##	COL1A1	265333	84197
##	FN1	249278	32914
##	CPB1	27	1213
##	COL1A2	178822	114068
##	COL3A1	134424	77211
##	MT-C01	573314	782907
##	TCGA-AR-A1AJ-01A-21R-A12P-07	TCGA-A8-A0AB-01A-11R-A034-07	
##	COL1A1	220552	339807
##	FN1	205962	386269
##	CPB1	25	692

##	COL1A2	160936	281966
##	COL3A1	175875	252558
##	MT-CO1	311993	628512
##	TCGA-E9-A1RE-01A-11R-A157-07	TCGA-BH-A18U-11A-23R-A12D-07	
##	COL1A1	179999	27776
##	FN1	271118	18346
##	CPB1	7	11216
##	COL1A2	134504	50599
##	COL3A1	129519	27795
##	MT-CO1	951376	372646
##	TCGA-BH-AOH7-11A-13R-A089-07	TCGA-BH-AOBT-01A-11R-A12P-07	
##	COL1A1	37092	357964
##	FN1	23247	149718
##	CPB1	537	12
##	COL1A2	46161	224376
##	COL3A1	28745	189549
##	MT-CO1	491879	746324
##	TCGA-OL-A660-01A-11R-A310-07	TCGA-AC-A3YI-01A-21R-A239-07	
##	COL1A1	361101	878704
##	FN1	73336	197141
##	CPB1	1708565	50
##	COL1A2	212639	536354
##	COL3A1	156822	494152
##	MT-CO1	149846	529237
##	TCGA-BH-AOE0-11A-13R-A089-07	TCGA-E2-A14R-01A-11R-A115-07	
##	COL1A1	34809	45438
##	FN1	11451	176241
##	CPB1	386	1
##	COL1A2	31265	30216
##	COL3A1	46979	28813
##	MT-CO1	555071	659903
##	TCGA-LL-A5YL-01A-12R-A29R-07	TCGA-BH-A18K-01A-11R-A12D-07	
##	COL1A1	248905	814942
##	FN1	108552	422116
##	CPB1	420	391
##	COL1A2	191406	545465
##	COL3A1	126957	481798
##	MT-CO1	173202	759223
##	TCGA-A2-AOST-01A-12R-A084-07	TCGA-D8-A27F-01A-11R-A16F-07	
##	COL1A1	58700	452488
##	FN1	57460	745456
##	CPB1	801	10
##	COL1A2	43791	391781
##	COL3A1	49590	361272
##	MT-CO1	361739	294101
##	TCGA-EW-A1P4-01A-21R-A144-07	TCGA-B6-A0IB-01A-11R-A034-07	
##	COL1A1	305400	245152
##	FN1	198429	85968
##	CPB1	8	15031
##	COL1A2	215572	128202
##	COL3A1	244856	91839
##	MT-CO1	275104	929607
##	TCGA-AO-AOJC-01A-11R-A056-07	TCGA-BH-A1FH-11B-42R-A13Q-07	
##	COL1A1	26345	82530

##	FN1	31697	24480
##	CPB1	24	1437
##	COL1A2	28328	118277
##	COL3A1	23536	78747
##	MT-CO1	709998	327050
##	TCGA-B6-AOX1-01A-11R-A109-07	TCGA-E9-A1N3-01A-12R-A157-07	
##	COL1A1	23699	74286
##	FN1	51690	88795
##	CPB1	1	2790666
##	COL1A2	19443	67923
##	COL3A1	13470	59698
##	MT-CO1	126528	1098144
##	TCGA-BH-AOB3-11B-21R-A089-07	TCGA-AN-AOFS-01A-11R-A034-07	
##	COL1A1	36018	851583
##	FN1	16464	267545
##	CPB1	6493	433
##	COL1A2	56302	677206
##	COL3A1	59411	844531
##	MT-CO1	487125	648490
##	TCGA-E2-A15K-01A-11R-A12P-07	TCGA-AC-A7VB-01A-11R-A352-07	
##	COL1A1	300428	85407
##	FN1	177607	43434
##	CPB1	43943	5
##	COL1A2	201665	51784
##	COL3A1	168047	50440
##	MT-CO1	414281	376731
##	TCGA-LD-A7W5-01A-22R-A352-07	TCGA-B6-AORL-01A-11R-A084-07	
##	COL1A1	487388	9420
##	FN1	153108	9631
##	CPB1	24	259
##	COL1A2	274586	4139
##	COL3A1	250474	2722
##	MT-CO1	520989	507956
##	TCGA-BH-A18P-11A-43R-A12D-07	TCGA-C8-A8HP-01A-11R-A36F-07	
##	COL1A1	19385	166652
##	FN1	8507	73934
##	CPB1	480	5
##	COL1A2	37262	87470
##	COL3A1	34000	69748
##	MT-CO1	1149132	395721
##	TCGA-EW-A3U0-01A-11R-A22K-07	TCGA-D8-A1J9-01A-11R-A13Q-07	
##	COL1A1	394432	644803
##	FN1	227783	838730
##	CPB1	8	1392678
##	COL1A2	190620	541380
##	COL3A1	172024	391428
##	MT-CO1	356824	384389
##	TCGA-V7-A7HQ-01A-11R-A33J-07		
##	COL1A1	371846	
##	FN1	54053	
##	CPB1	77	
##	COL1A2	169718	
##	COL3A1	80616	
##	MT-CO1	88402	

```
# Heatmap Analysis
library (ComplexHeatmap)
```

```
## Loading required package: grid
```

```
## =====
## ComplexHeatmap version 2.24.1
## Bioconductor page: http://bioconductor.org/packages/ComplexHeatmap/
## Github page: https://github.com/jokergoo/ComplexHeatmap
## Documentation: http://jokergoo.github.io/ComplexHeatmap-reference
##
## If you use it in published research, please cite either one:
## - Gu, Z. Complex Heatmap Visualization. iMeta 2022.
## - Gu, Z. Complex heatmaps reveal patterns and correlations in multidimensional
##   genomic data. Bioinformatics 2016.
##
##
## The new InteractiveComplexHeatmap package can directly export static
## complex heatmaps into an interactive Shiny app with zero effort. Have a try!
##
## This message can be suppressed by:
##   suppressPackageStartupMessages(library(ComplexHeatmap))
## =====
```

```
## Heatmap annotations
column_ha <- HeatmapAnnotation(
  # change annotation track names
  "Tissue Type" = meta_data$tissue_type,
  "Gender"      = meta_data$gender,
  # set colors
  col = list(
    "Tissue Type" = c(Tumor = "red", Normal = "blue"),
    "Gender"      = c(male = "green", female = "purple")
  ),
  # change legend titles and labels
  annotation_legend_param = list(
    "Tissue Type" = list(title = "Tissue Type", labels = c("Normal Tissue", "Tumor Tissue")),
    "Gender"      = list(title = "Gender", labels = c("Female", "Male"))
  ),
  # position annotations on the right
  annotation_name_side = "right",
  show_annotation_name = TRUE
)
```

```
# Create heatmap
p <- Heatmap(
  subset_merged,
  name = "Counts",
  top_annotation = column_ha,
  show_row_names = TRUE,
  show_column_names = FALSE,
  cluster_columns = TRUE,
```



```

cluster_rows = TRUE,
column_title = "Samples",
row_title = "Genes",
heatmap_legend_param = list(title = "Count")
)

```

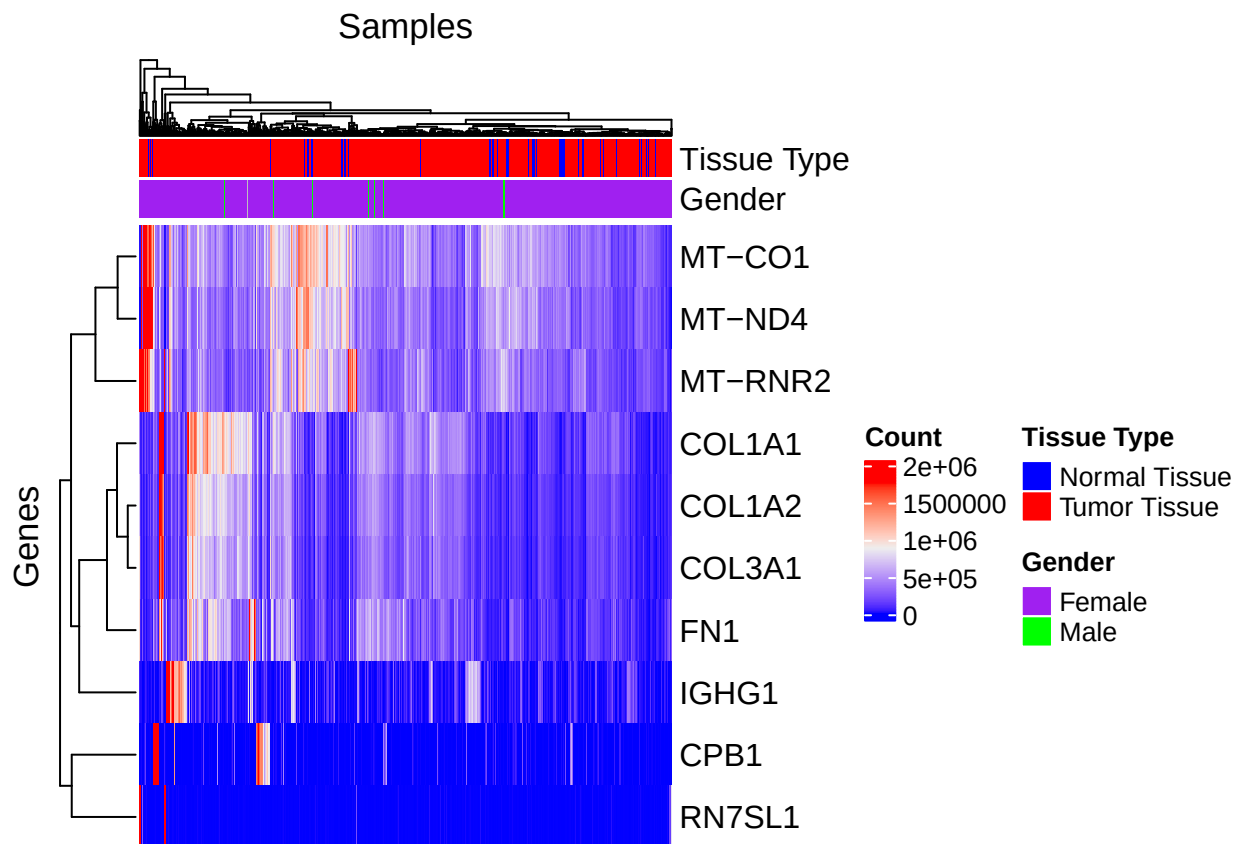
```
## Warning: The input is a data frame-like object, convert it to a matrix.
```

```
## The automatically generated colors map from the 1st and 99th of the
## values in the matrix. There are outliers in the matrix whose patterns
## might be hidden by this color mapping. You can manually set the color
## to 'col' argument.
```

```
##
```

```
## Use 'suppressMessages()' to turn off this message.
```

```
print(p)
```



```

png("/Users/jmelb/Projects/HDS-Foundations_of_Data_Science/Final_Project/figures_tables/heatmap.png", w
print(p)
dev.off()

```

```
## pdf
```

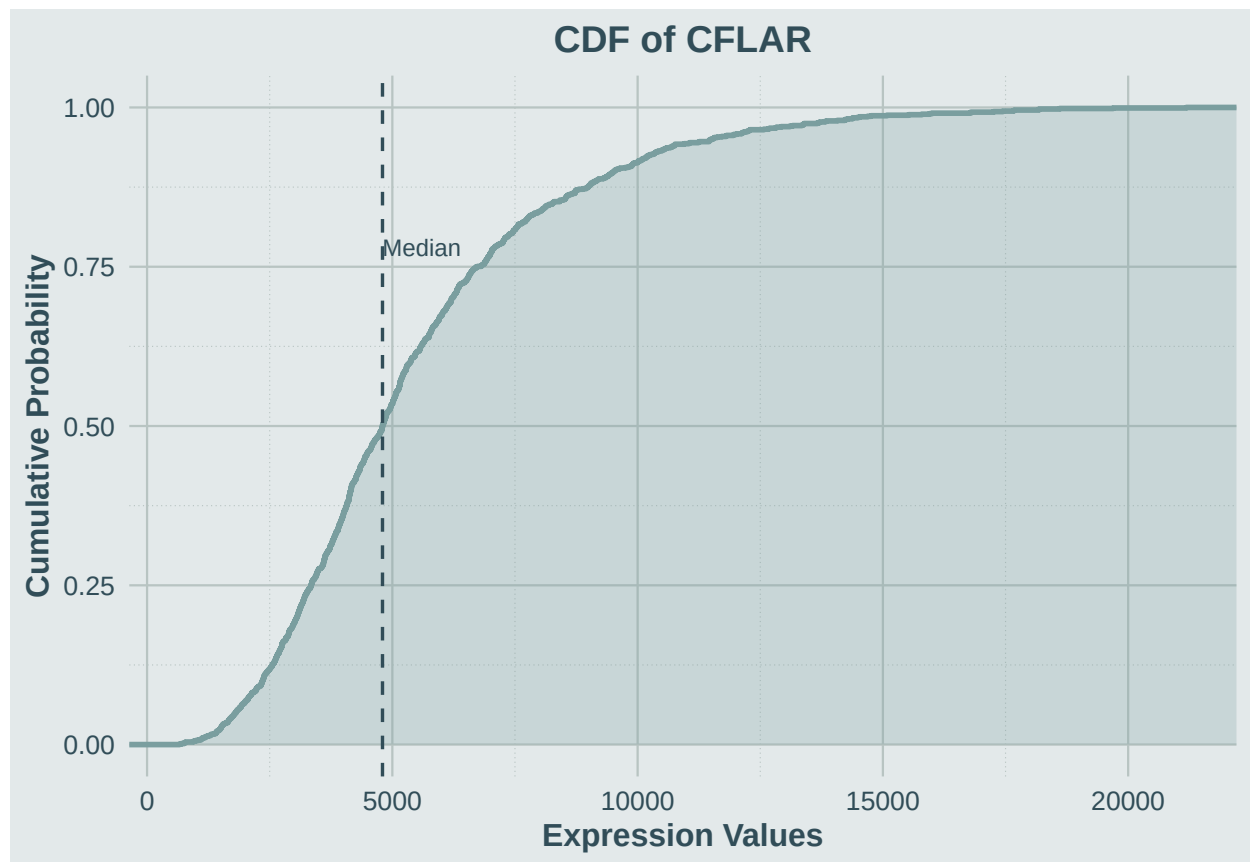
```
## 2
```

```

# make a new plot! I chose a cumulative distribution function
# Cumulative Distribution Function
p <- ggplot(gene_data, aes(x = gene1)) +
  # shaded area under ECDF
  stat_ecdf(
    geom = "area",
    fill = coolor_code[3],
    alpha = 0.25
  ) +
  # ECDF line
  stat_ecdf(
    geom = "step",
    color = coolor_code[3],
    linewidth = 1
  ) +
  # median line
  geom_vline(
    xintercept = median(gene_data$gene1, na.rm = TRUE),
    linetype = "dashed",
    color = coolor_code[5],
    linewidth = 0.6
  ) +
  # add median line text
  annotate("text", x = 5600, y = 0.75, label = "Median",
    vjust = -0.6, color = coolor_code[5], size = 3) +
  labs(
    title = "CDF of CFLAR",
    x = "Expression Values",
    y = "Cumulative Probability"
  )

print(p)

```



```
png("/Users/jmelb/Projects/HDS-Foundations_of_Data_Science/Final_Project/figures_tables/CDF.png", width
print(p)
dev.off()
```

```
## pdf
## 2
```